ENERGY INFRASTRUCTURE

The visual engineering handbook

First principles. Real measurements.
Reliable control. A company built on evidence.



Conceptual opening artwork. Technical teaching figures inside are original vector diagrams.

96 LESSONS

8 PROJECTS

WORKED ANSWERS

A self-contained foundation for Chris and Lupine’s energy direction.
Hobart, Tasmania / October 2026

This book teaches directly

You do not need to return to the two background books to follow these lessons. The definitions, explanations, calculations and answers are here.

What this edition covers

Electricity and electronics; measurement; AC machines and networks; Tasmania and hydro; storage; heat and cooling; AI data centres; feedback control; device communication; reliable execution; and technical problem discovery.

How a lesson works

Read the opening explanation. Follow the figure. Work the example yourself, then answer the question before checking the answer section. Close the page and redraw the mechanism. If you cannot explain a term, use the glossary rather than pressing ahead.

Your learning rule

Use AI to question and check your reasoning. First make your own prediction, sketch or calculation. Keep mistakes with the correction: that record becomes a reference you understand.

Your evidence rule

Every numerical example here is hypothetical unless explicitly identified otherwise. No bench measurement, field validation or current-repository behaviour is claimed by this book.

Scope: a standalone foundation and prototype reference, not a full engineering degree or an installation manual. Real equipment requires its exact documentation and an appropriate competent review. Specialist study remains necessary for power-system design and protective functions.

Original text, diagrams and exercises. AI-assisted edition; no independent professional technical sign-off. References are for provenance and optional verification, not assigned reading.

Your complete route

Twelve connected subjects, followed by projects, runnable examples, answers and a working reference. Page numbers match the PDF viewer.

Part	Focus	Page
A Think in systems	Energy, units, maths and evidence	7
B Understand circuits	From charge to stored energy	16
C Meet the electronics	Small signals controlling real loads	25
D Make measurements	Numbers you can reason from	34
E Understand power systems	AC, machines, networks and protection	43
F Ground it in Tasmania	Hydro, renewable generation and the grid	52
G Storage and continuity	Batteries, converters and resilient supply	61
H Follow the heat	Thermal systems and data-centre cooling	70
I Understand the AI factory	Compute, power paths and workloads	79
J Control the system	Feedback, dynamics and limits	88
K Connect the equipment	Data, code, protocols and security	97
L Build Lupine with evidence	Execution, reconciliation and discovery	106

Use while learning	Pages
Eight practical projects / Python and Modbus code	115-126
Answers / Formulas / Glossary	127-143
Recording sheets / Sources / Subject index	144-150

A pace you can sustain

Read one connected section at a time. This is a reference you revisit, not a test you finish and forget.

First pass	Read and explain	Make or test
Stage 1	A-B: units, energy and circuits	Project 1
Stage 2	C-D: electronics and measurement	Projects 2-3
Stage 3	E-G: power, Tasmania and storage	System sketches
Stage 4	H-I: heat and the AI factory	Thermal balance
Stage 5	J-K: control, data and protocols	Projects 4 and 7
Stage 6	L: execution and validation	Projects 5-6, then 8

Start today: lessons 01 and 02, pages 8-9

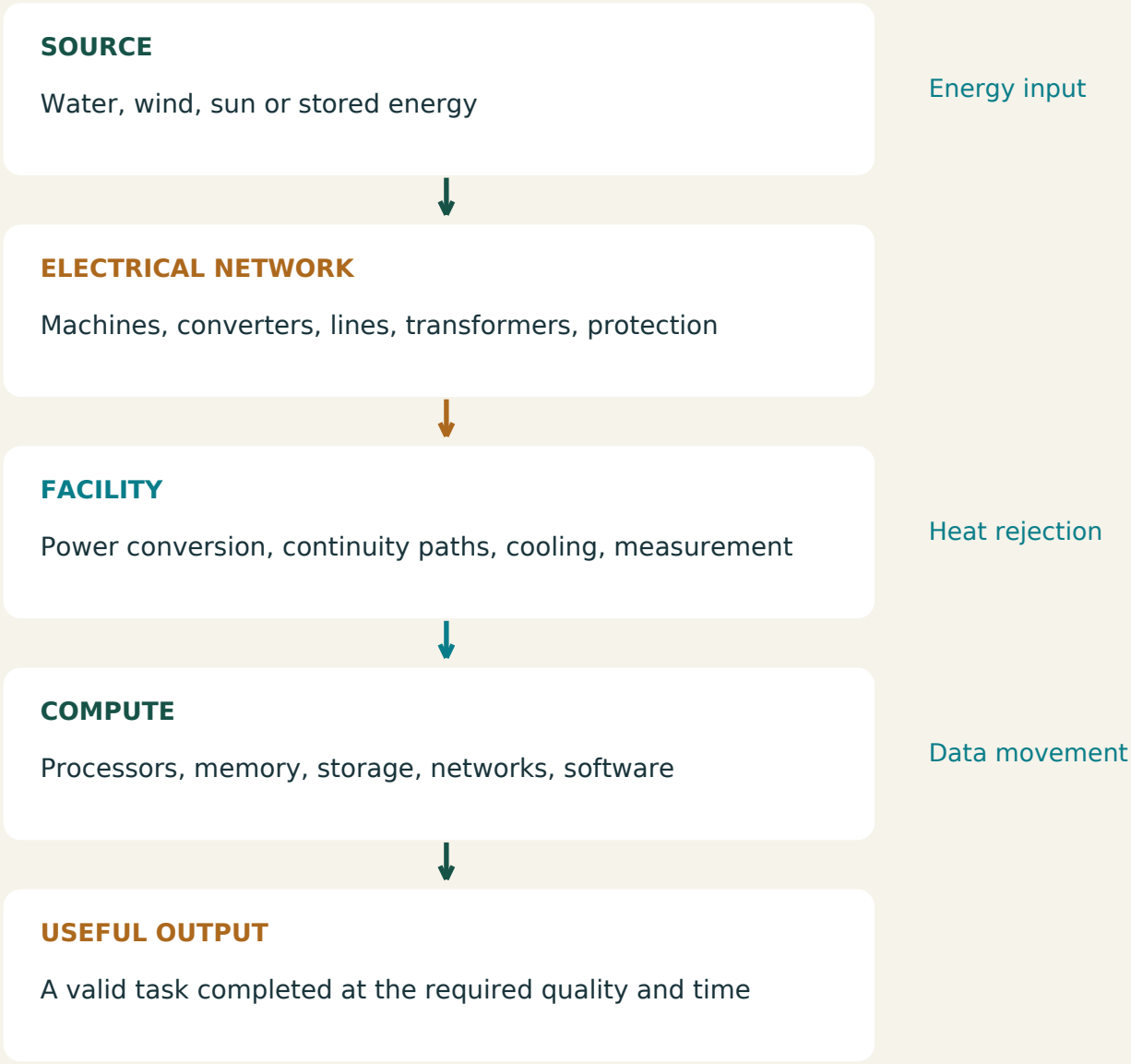
Draw one energy path and a separate information path. Convert 250 mA to A and 0.8 kWh to Wh. Explain why a lost network message does not prove a load stopped. That is enough for the first session.

A study block: 15 minutes reading and inspecting the figure; 15 minutes working the example; 15 minutes recalling, drawing and answering. Spend extra time where the model or units do not yet make sense.

At roughly 8-12 hours a week, allow about 4-6 months for a careful first pass and projects. This is a planning estimate, not an expertise deadline. Long-term depth comes from repeated measurement, device access, review and field experience.

From water to useful computation

Keep this overview beside you. Each layer has a different physical quantity, time scale and failure boundary.



LUPINE'S FIRST INTERFACE

Aiva: choose only feasible actions. Rail: manage execution and unresolved outcomes. Cloked: preserve records within a defined trust model. All three depend on real equipment semantics and valid measurements.

Solve, then challenge the result

A repeatable method makes unfamiliar engineering problems manageable.

Step	Ask yourself
1 / Boundary	What system and measurement point does this question concern?
2 / Quantity	Is the answer a rate, an amount, a temperature, a delay or a probability?
3 / Model	Which relationship applies, and what does it ignore?
4 / Units	Have I converted time, prefixes and reference quantities correctly?
5 / Estimate	What order of magnitude should I expect?
6 / Calculate	Keep the units and enough intermediate precision.
7 / Challenge	What evidence could contradict this result? What remains uncertain?

Example: a 5 V, 100 Ω resistor

Boundary: resistor terminals. Model: steady ideal source and ohmic resistor. Current: 5/100 = 0.050 A. Power: 5 × 0.050 = 0.25 W. Two-hour energy: 0.50 Wh. Limits: source variation, resistance tolerance, heating and meter error. A real result needs actual readings at that boundary.

Diagram key: gold = electrical/work input; teal = data, fluid or controlled path; red = heat, conflict or fault. Follow the labels: colour alone never defines the physics.

Think in systems

Start here. Learn to define a boundary, check a unit and explain what a result can support.

A Energy, units, maths and evidence

EIGHT IDEAS TO UNDERSTAND

01 The physical system comes first 8

02 Units are part of the answer 9

03 Use algebra to ask better questions 10

04 Power is a rate; energy accumulates 11

05 Efficiency needs a boundary 12

06 Read a graph before trusting it 13

07 Keep claims at the level of evidence 14

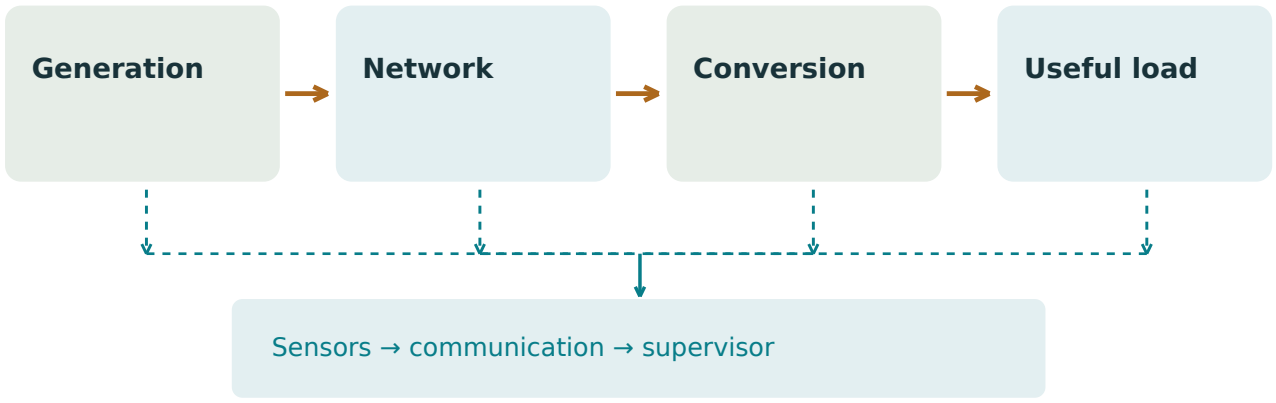
08 Set a sensible home-lab boundary 15

Before you move on

Explain one mechanism without looking, solve one example with units and identify one limit of the model. Use the answer section to correct your reasoning, then keep your corrected sketch.

The physical system comes first

An energy company starts with physical constraints. Software helps observe and influence equipment; it does not replace the physics.



Energy flow and information flow must be distinguished.

HOW IT WORKS

A **system** is the set of things you choose to study together. Water, a turbine, a generator, a network and a server form an energy-conversion chain. Draw a boundary around your question. Energy crossing that boundary must be accounted for; a change in stored energy matters too.

USE IT CORRECTLY

A sensor reports information about the system. A controller uses that information to request a change. The electrical path and the information path can fail independently. If Wi-Fi drops, the motor may keep running. If a sensor freezes, a fresh network message may still contain an old value.

WORKED EXAMPLE / HYPOTHETICAL

For a fan experiment, put the boundary at its 5 V input. Measure voltage and current there. This describes electrical input, including its internal electronics. It does not by itself describe shaft speed or cooling delivered.

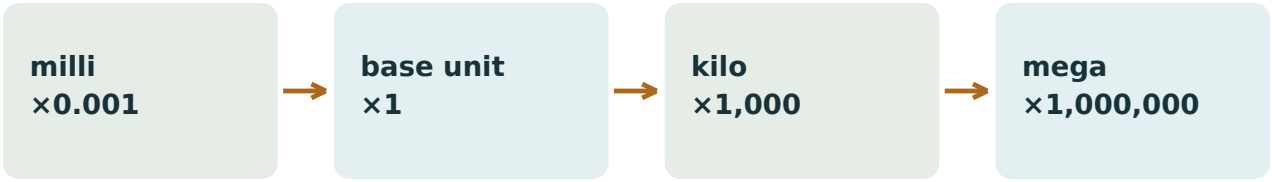
CHECK YOUR UNDERSTANDING

If a network link disappears, what do you know about the fan’s physical state?

Answer: 127 | Sources: B1, B2

Units are part of the answer

A number without a unit is incomplete. Prefixes help describe systems from a tiny sensor to an electricity network.



250 mA = 0.250 A 0.8 kWh = 800 Wh

Prefixes change scale. They do not change the underlying quantity.

HOW IT WORKS

Use **watts (W)** for power, **joules (J)** or **watt-hours (Wh)** for energy, **volts (V)** for potential difference and **amperes (A)** for current. The prefix kilo means 1,000; mega means 1,000,000; milli means one thousandth. A unit conversion changes the representation, not the physical quantity.

USE IT CORRECTLY

Write the unit at every calculation step. Multiply and divide units as you do numbers. A watt is a joule per second; multiplying watts by seconds produces joules. Multiplying watts by hours produces watt-hours. Celsius differences and kelvin differences have the same size, but absolute Celsius and kelvin temperatures differ.

WORKED EXAMPLE / HYPOTHETICAL

250 mA = 0.250 A. At 5.0 V, P = 5.0 × 0.250 = 1.25 W. Over 2 h, E = 2.50 Wh = 9,000 J. The last conversion uses 1 Wh = 3,600 J.

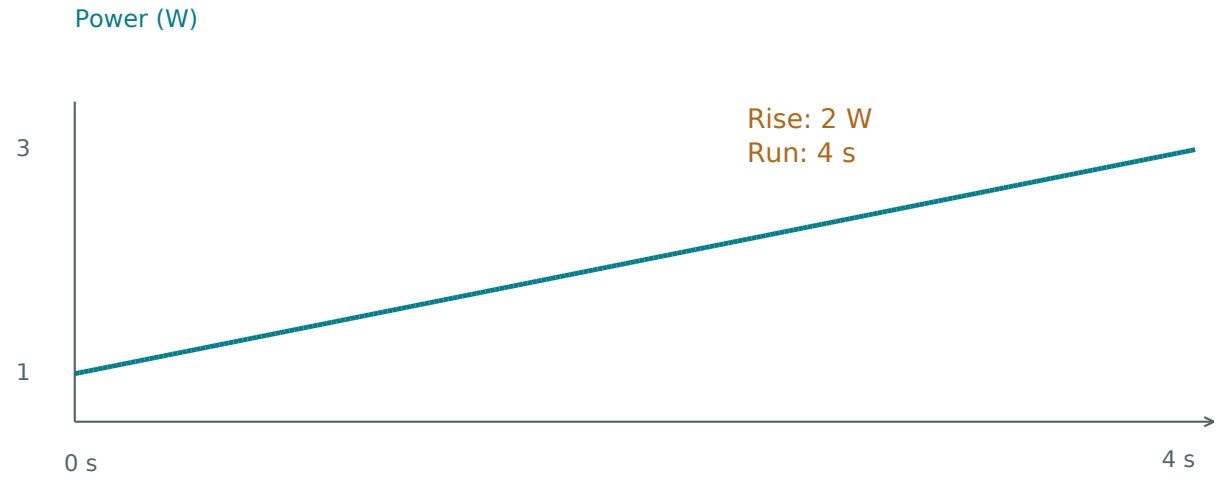
CHECK YOUR UNDERSTANDING

Convert 0.8 kWh to Wh and joules.

Answer: 127 | Sources: B1, B2

Use algebra to ask better questions

You need a small, useful maths toolkit: rearranging formulas, ratios, percentages and a sense of scale.



Slope is a rate of change; area under a power-time curve is energy.

HOW IT WORKS

An equation connects quantities. If P = VI and you know P and V, divide both sides by V to obtain I = P/V. Keep parentheses around grouped terms. A ratio compares two like quantities; a percentage is a ratio multiplied by 100. Estimate the expected order of magnitude before using a calculator.

USE IT CORRECTLY

Scientific notation keeps very large and small values readable: 2.5×10^3 means 2,500. For a graph, **slope** is change in the vertical quantity divided by change in the horizontal quantity. You can understand an initial rate of change without mastering calculus. Always define the interval over which you calculated it.

WORKED EXAMPLE / HYPOTHETICAL

A power reading rises from 1 W to 3 W over 4 s. Average slope = (3 - 1)/4 = 0.5 W/s. This is a ramp rate, not energy. A straight-line ramp over those 4 s transfers (1 + 3)/2 × 4 = 8 J.

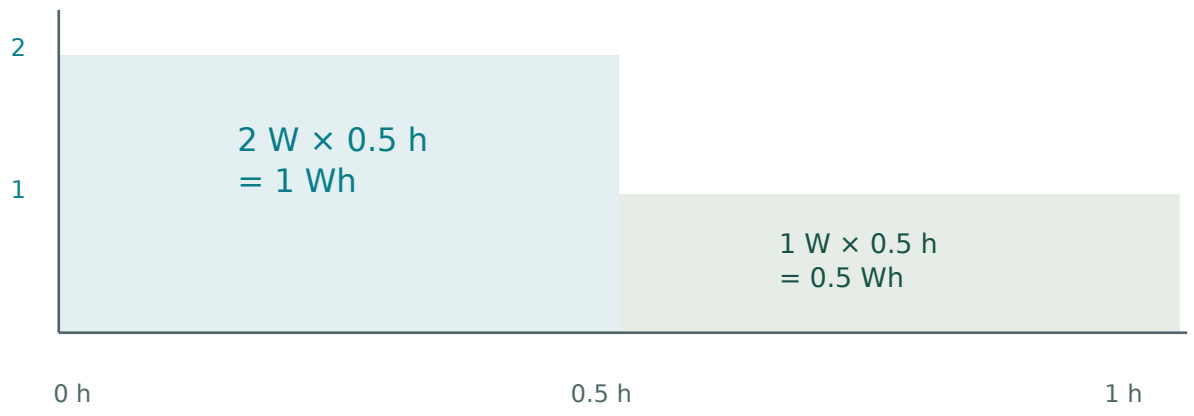
CHECK YOUR UNDERSTANDING

A device is 80% efficient. What input power is required for 40 W of useful output?

Answer: 127 | Sources: B1, B2

Power is a rate; energy accumulates

This distinction connects a USB load, a battery and an entire power station.



Each rectangle contributes power × duration.

HOW IT WORKS

Power is the rate of energy transfer. A steady DC circuit has P = VI. For constant power, E = Pt. If power changes, calculate the energy of each time interval and add them. On a power-versus-time graph, energy is the area beneath the curve.

USE IT CORRECTLY

A 1 kW heater and a 1 kWh battery describe different properties. The heater’s rating is a rate. The battery label describes an amount, subject to the manufacturer’s usable-capacity conditions. “kW per hour” describes a change in power per hour; it is not another way to write kWh.

WORKED EXAMPLE / HYPOTHETICAL

A load draws 2 W for 30 min, then 1 W for 30 min. E = (2 × 0.5) + (1 × 0.5) = 1.5 Wh. Its maximum power was 2 W; its one-hour average was 1.5 W.

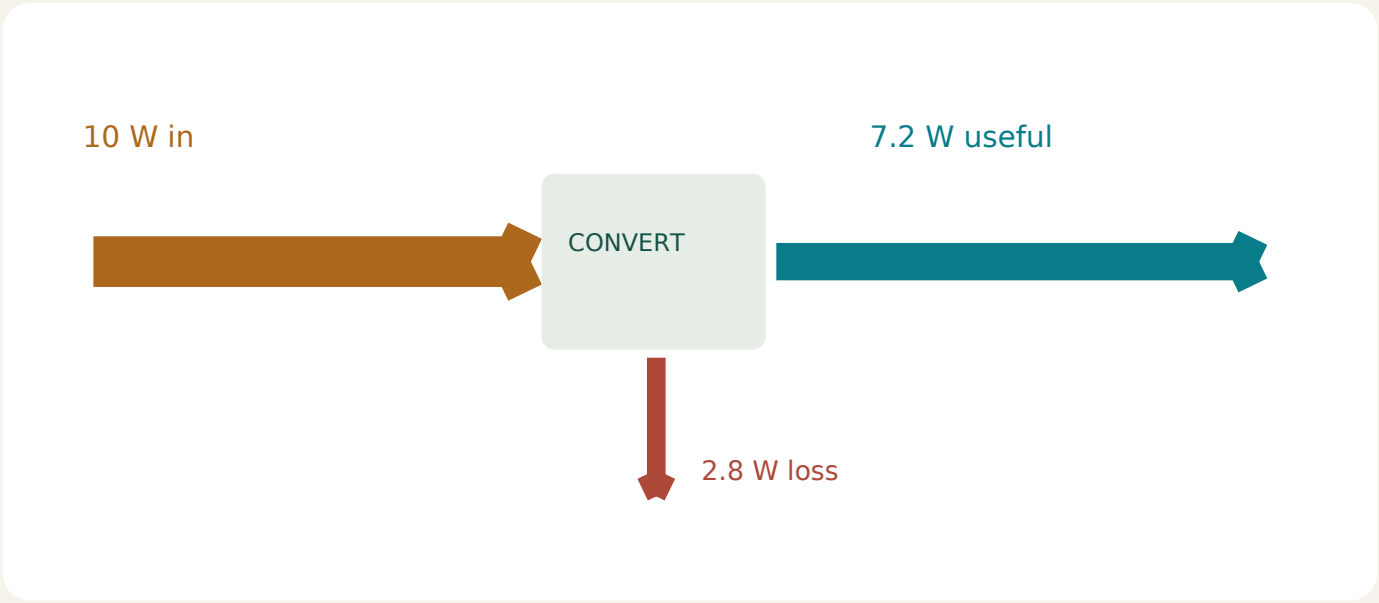
CHECK YOUR UNDERSTANDING

What energy does a steady 3 W load use in 20 minutes?

Answer: 127 | Sources: B1, B2

Efficiency needs a boundary

Efficiency compares useful output with input. Decide what counts as useful before calculating it.



An energy balance must include every output and any change in storage.

HOW IT WORKS

Define efficiency η = useful output energy/input energy over a stated period, or the corresponding power ratio in steady conditions. The difference between input and useful output is not necessarily missing; it may be heat, sound, mechanical friction or energy stored temporarily inside the system.

USE IT CORRECTLY

A chain has several conversion stages. Multiply stage efficiencies when their boundaries line up and the operating conditions match. Do not average them. A converter’s headline peak efficiency may not apply at your low bench load. Auxiliary consumption and standby power can matter strongly at small loads.

WORKED EXAMPLE / HYPOTHETICAL

Two stages are assumed to be 90% and 80% efficient. Combined efficiency = 0.90 × 0.80 = 0.72. A 10 W input then gives 7.2 W useful output and 2.8 W total loss at steady state.

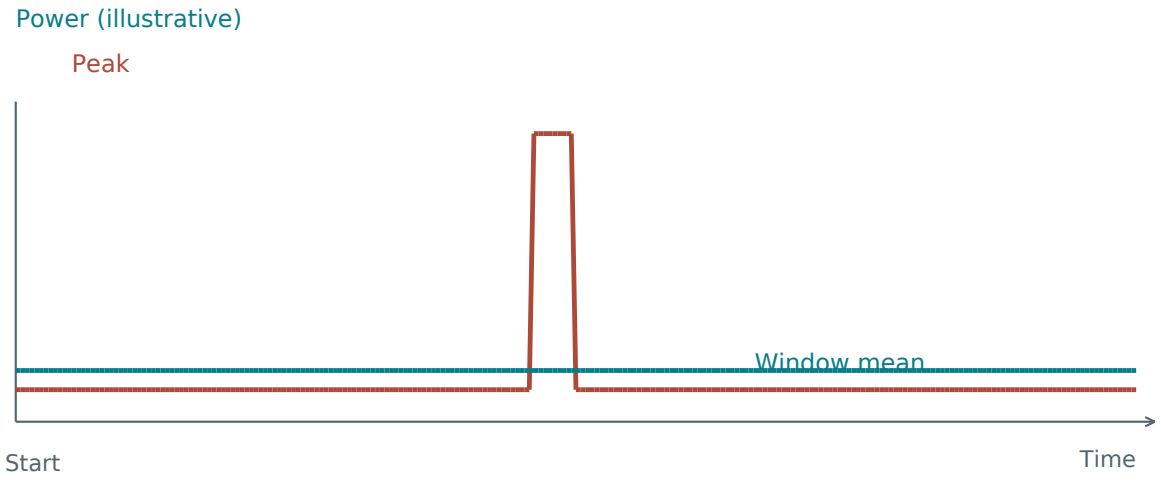
CHECK YOUR UNDERSTANDING

Can a cooling system have a coefficient of performance greater than one without creating energy?

Answer: 127 | Sources: B1, B2

Read a graph before trusting it

Axes, scales and time windows can change what the same data appears to say.



Averaging can preserve energy while concealing a peak.

HOW IT WORKS

First read the title, axis quantities and units. Then check the time interval, sampling interval and whether an axis starts at zero. A line joining samples is an interpolation, not a continuous measurement. A five-minute average can hide a short, large excursion.

USE IT CORRECTLY

Keep instantaneous values, interval averages and cumulative totals distinct. For comparisons, use matching boundaries and periods. Display missing data as gaps or an explicit quality state. Connecting across an outage can make a system look smooth precisely where the evidence is weakest.

WORKED EXAMPLE / HYPOTHETICAL

A 100 W spike lasting 1 s inside a 100 s window contributes only 1 W to that window’s mean if the rest is zero. The average is correct for that window but gives little information about the peak.

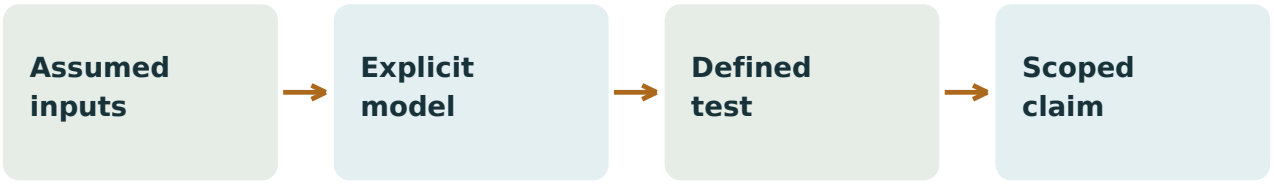
CHECK YOUR UNDERSTANDING

Why can two graphs with the same one-hour mean show very different engineering risks?

Answer: 127 | Sources: B1, B2

Keep claims at the level of evidence

A calculated example, a simulator result and a measurement are useful for different reasons.



The wording of a conclusion must match its evidence.

HOW IT WORKS

An **assumption** is an input you chose. A **model** is a simplified relationship. A **simulation** runs a model through cases. A **measurement** is an observation from an instrument and method. **Verification** checks a stated requirement against evidence; its scope must be named.

USE IT CORRECTLY

These are descriptions, not a ladder that makes a claim certain. A simulation can be verified against its specification while still modelling a physical device poorly. A measured number can be biased. A cryptographic record can preserve a sensor’s wrong answer perfectly. Label inputs and conclusions separately.

WORKED EXAMPLE / HYPOTHETICAL

“The simulated controller recovered from one lost reply in six test cases” is supportable after those tests. “Lupine is safe for battery infrastructure” requires far more evidence about hardware, limits, failure cases and deployment conditions.

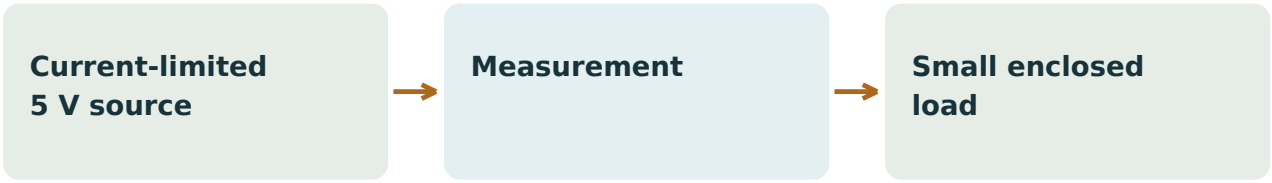
CHECK YOUR UNDERSTANDING

You calculate 78.5 kW using assumed head and flow. Is that a measured plant output?

Answer: 127 | Sources: B1, B2

Set a sensible home-lab boundary

Learn the principles at low energy. Infrastructure equipment comes later, through a reviewed and authorised project.



The starting point is a small reviewed circuit, not a miniature grid installation.

HOW IT WORKS

For this book’s physical projects, stay with supervised, small, current-limited **5 V DC** experiments. Use enclosed loads and suitable connectors. Disconnect power before changing wiring. A low voltage reduces shock risk but does not remove short-circuit, heating or mechanical hazards.

USE IT CORRECTLY

Do not open mains-powered supplies, sealed cooling equipment or battery packs. Do not work on switchboards, inverters or grid wiring. Keep protective covers in place. Any future infrastructure connection needs its own competent design review, device-specific documentation and permissions; this handbook is a learning reference, not an installation approval.

WORKED EXAMPLE / HYPOTHETICAL

A 5 V, 100 Ω resistor draws 50 mA and dissipates 0.25 W. Select a suitably rated resistor with margin, keep it clear of heat-sensitive material and check temperature under supervision. A USB connector alone does not define a safe current limit.

CHECK YOUR UNDERSTANDING

Why is “only 5 V” not enough information to judge a circuit?

Answer: 127 | Sources: B1, B2

Understand circuits

Build a physical picture of current, voltage and the components that shape them.

B From charge to stored energy

EIGHT IDEAS TO UNDERSTAND

09 Charge and current 17

10 Voltage is a difference 18

11 Ohm’s law and a real resistor 19

12 Series and parallel paths 20

13 Kirchhoff keeps the accounts balanced 21

14 Electrical power becomes something else 22

15 Capacitors store electric-field energy 23

16 Inductors resist changes in current 24

Before you move on

Explain one mechanism without looking, solve one example with units and identify one limit of the model. Use the answer section to correct your reasoning, then keep your corrected sketch.

Charge and current

Current describes how quickly charge crosses a point. It is not a substance that disappears inside a load.



A closed loop supports steady current. Arrows show conventional current.

HOW IT WORKS

Electric charge is measured in coulombs (C). Current $I = \Delta Q/\Delta t$, in amperes, tells you the charge crossing a point per second. In a metal, electrons carry charge; conventional current is defined in the direction positive charge would move, opposite electron drift.

USE IT CORRECTLY

A steady series circuit has the same current through each element because charge is conserved. The load converts energy; it does not consume all the charge. An open circuit breaks the steady path. At switch-on, capacitances and electromagnetic effects can create transients before a steady state exists.

WORKED EXAMPLE / HYPOTHETICAL

If 0.6 C crosses a point in 3 s, average current is 0.2 A. In a steady loop with one resistor, 0.2 A enters and leaves that resistor; its electrical energy transfer depends on the voltage across it.

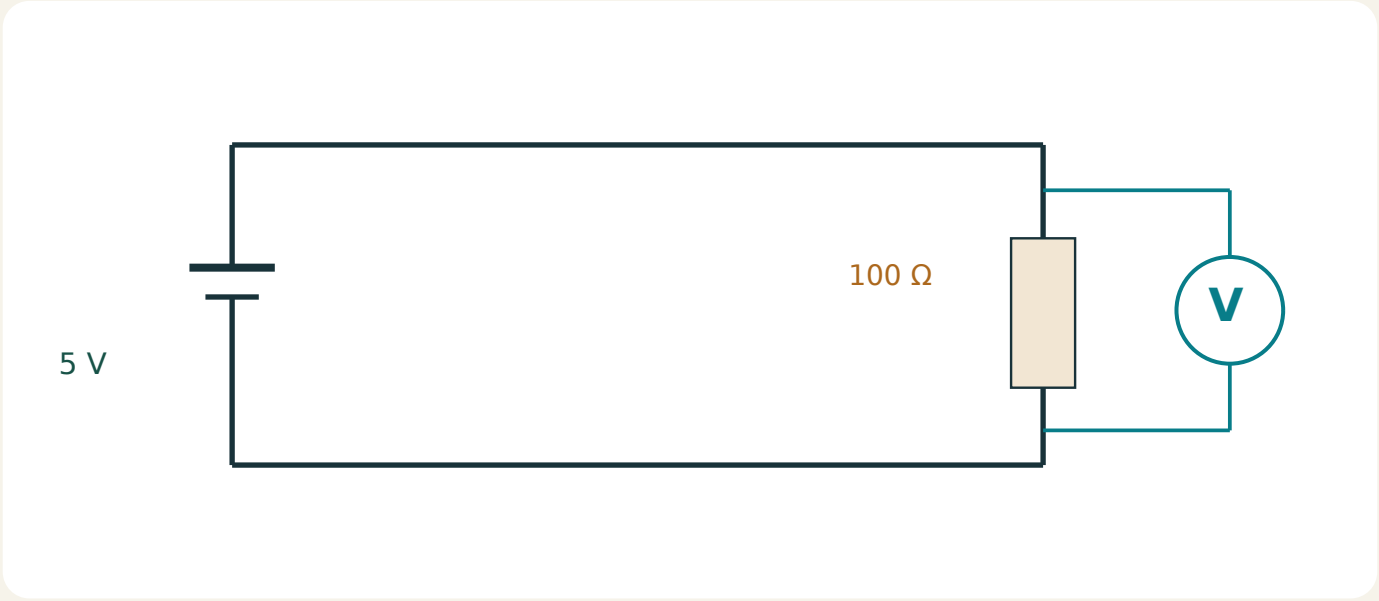
CHECK YOUR UNDERSTANDING

Does current become smaller after passing through one series resistor?

Answer: 128 | Sources: B1, B2

Voltage is a difference

A voltage measurement always compares two points. Naming only one point leaves the reference unstated.



A meter compares two nodes; polarity determines the sign.

HOW IT WORKS

Voltage is electric potential difference: energy transferred per unit charge, in joules per coulomb. Saying a point is “5 V” usually means 5 V relative to a named reference node. The reference is a bookkeeping choice in a circuit; it is not automatically the protective earth of a building.

USE IT CORRECTLY

A voltmeter connects across the two points being compared. Its finite input resistance can affect a delicate circuit. Reversing the leads reverses the sign. A supply label describes a nominal operating value, not a guarantee that the voltage remains unchanged under every load or fault.

WORKED EXAMPLE / HYPOTHETICAL

If 2 C transfers 10 J between two points, V = 10/2 = 5 V. If node A is 5 V and node B is 2 V relative to the same reference, A relative to B is 3 V.

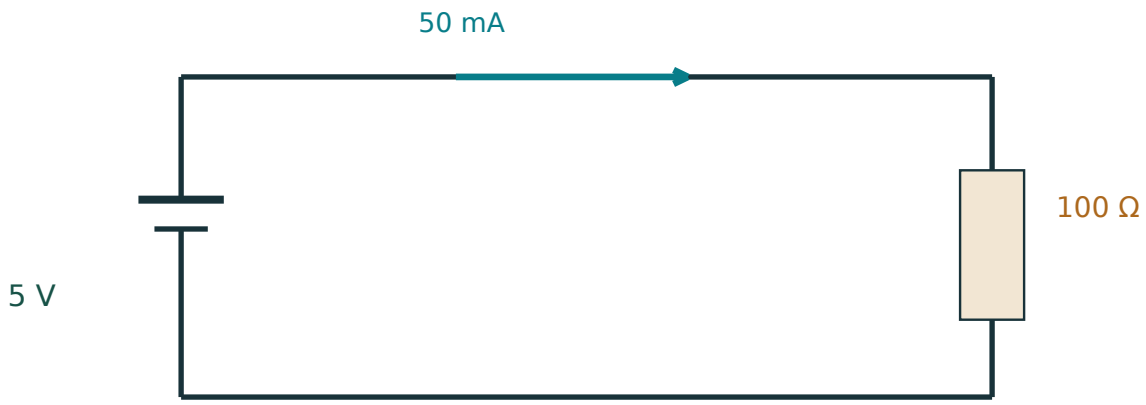
CHECK YOUR UNDERSTANDING

Two readings use different reference nodes. Can you subtract them immediately?

Answer: 128 | Sources: B1, B2

Ohm’s law and a real resistor

For an ohmic component over the conditions being modelled, voltage and current are proportional.



Use a resistor model for a resistor, with stated operating limits.

HOW IT WORKS

Ohm’s law is V = IR. Rearrange it to I = V/R or R = V/I. Real resistors have tolerance, temperature dependence and a maximum power rating. Their printed resistance is not a promise of exact behaviour at every temperature.

USE IT CORRECTLY

Do not apply a constant-resistance model blindly to a motor, LED, battery or regulated power supply. A motor’s current depends on motion and its electronics. An LED has a nonlinear current-voltage relationship. First identify whether the component model fits your question.

WORKED EXAMPLE / HYPOTHETICAL

At 5 V across 100 Ω: I = 5/100 = 0.050 A. Dissipation P = VI = 0.25 W. An ideal 200 Ω resistor at the same voltage draws 0.025 A and dissipates 0.125 W.

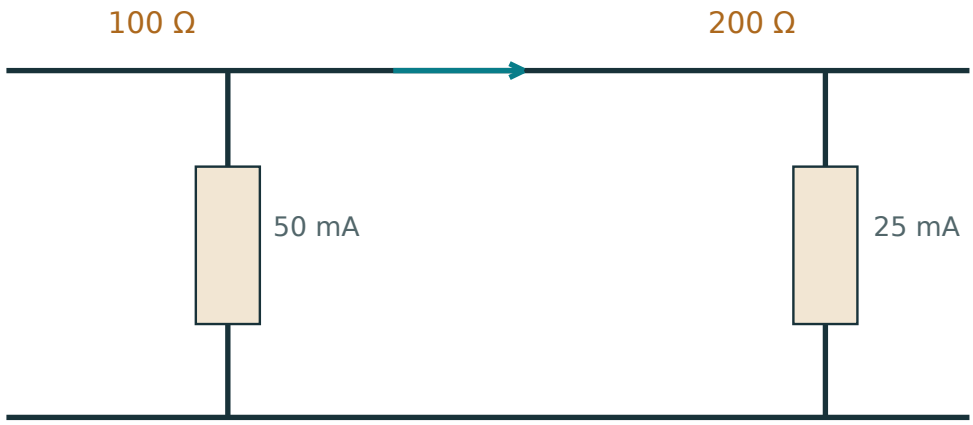
CHECK YOUR UNDERSTANDING

What happens to ideal resistor power if voltage doubles while resistance stays fixed?

Answer: 128 | Sources: B1, B2

Series and parallel paths

Series elements share current. Parallel branches share voltage. Current divides across all conducting paths.



Both branches share 5 V.

Both branches conduct; they share the source voltage.

HOW IT WORKS

In series, resistances add: $R = R_1 + R_2$. In parallel, conductances add: $1/R = 1/R_1 + 1/R_2$. Two equal resistors in parallel have half the resistance of one. The equivalent circuit describes behaviour at its terminals, not every internal temperature.

USE IT CORRECTLY

Current does not flow only through the path of least resistance. It divides according to branch behaviour. In a parallel circuit, a lower-resistance branch carries more current, while other finite-resistance branches can still carry current. A short circuit is therefore dangerous even if another intended path exists.

WORKED EXAMPLE / HYPOTHETICAL

Two 100 Ω resistors across 5 V each draw 50 mA. Together they draw 100 mA and appear as 50 Ω. In series, the pair is 200 Ω, drawing 25 mA; each drops 2.5 V.

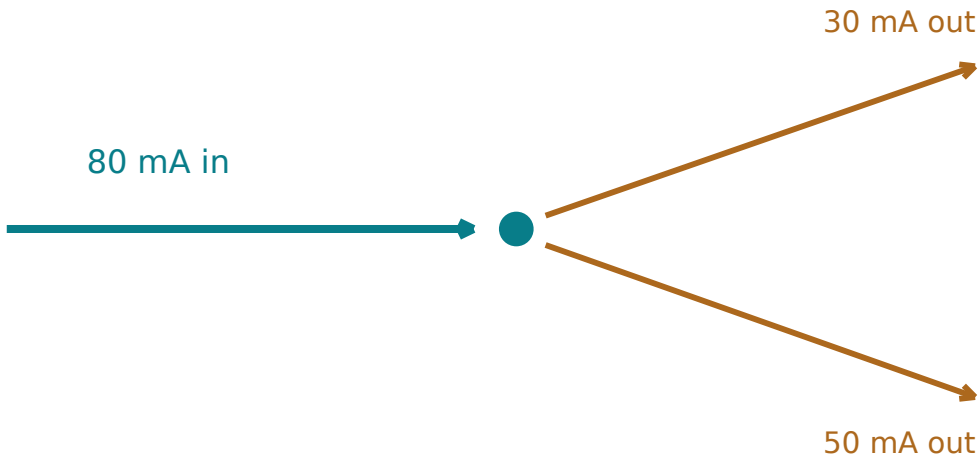
CHECK YOUR UNDERSTANDING

What current flows through a 200 Ω branch placed in parallel with 100 Ω across 5 V?

Answer: 128 | Sources: B1, B2

Kirchhoff keeps the accounts balanced

Conservation laws let you check a circuit without guessing where charge or energy went.



Name the arrows first; then sum signed currents.

HOW IT WORKS

Kirchhoff’s current law says the signed currents at a node sum to zero: total current entering equals total leaving. Kirchhoff’s voltage law says signed voltage changes around a loop sum to zero in the lumped-circuit model, with induced effects represented appropriately.

USE IT CORRECTLY

Choose arrow directions and voltage polarities before writing equations. A negative result means the actual direction is opposite your chosen positive direction. For rapid changes, large loops or high-frequency circuits, a simple wire-only model may miss inductance, capacitance and propagation effects.

WORKED EXAMPLE / HYPOTHETICAL

At a junction, 80 mA enters and 30 mA leaves through one branch. The other branch must carry 50 mA out in the steady model. In a loop with a 5 V source and two drops, 2 V + 3 V balances the source.

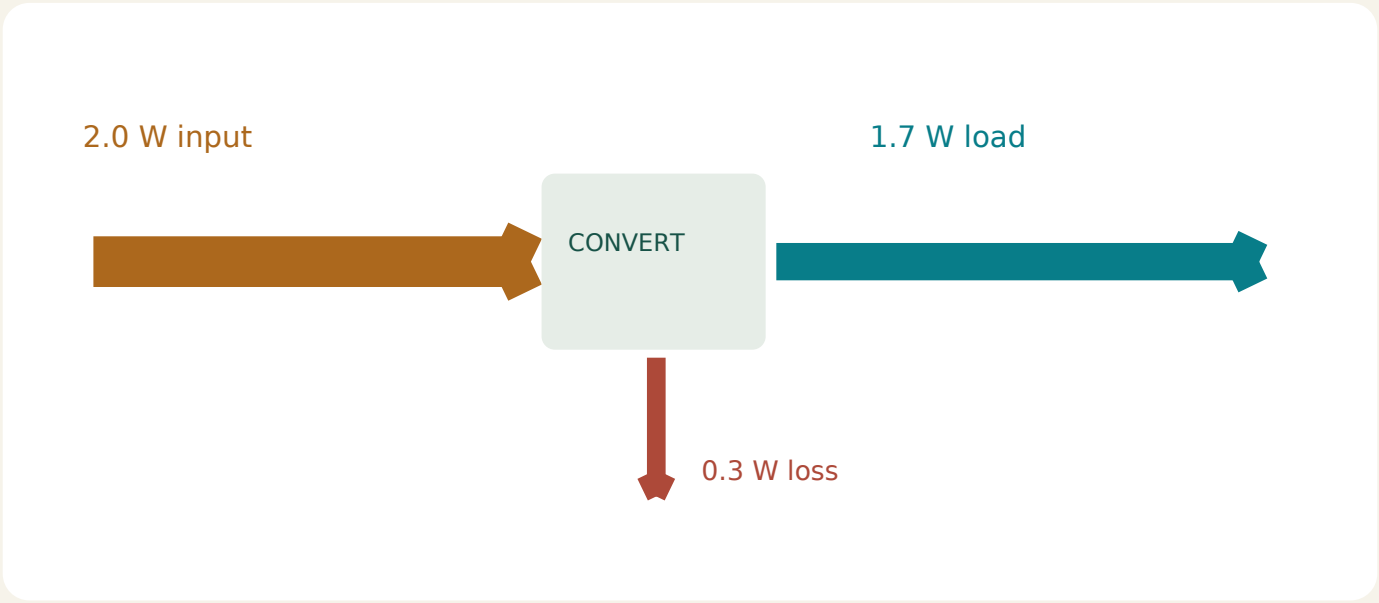
CHECK YOUR UNDERSTANDING

A model shows 100 mA entering a steady node and 70 mA leaving. What should you investigate?

Answer: 128 | Sources: B1, B2

Electrical power becomes something else

A load is an energy converter. The useful output depends on what the device is meant to do.



Follow both useful output and losses across a named boundary.

HOW IT WORKS

For steady DC, P = VI. For a resistor, combine this with Ohm's law to obtain $P = I^2R$ or $P = V^2/R$. For a fan, electrical input becomes mechanical work, heat and sound. The electrical input alone does not identify how much air is moved.

USE IT CORRECTLY

State the measurement point. Power measured before a converter includes the energy later lost in that converter. Power measured at the load terminals excludes upstream losses. When you compare devices, ensure the same useful task, operating condition and boundary are being compared.

WORKED EXAMPLE / HYPOTHETICAL

A hypothetical converter takes 2.0 W and delivers 1.7 W to a load. Loss is 0.3 W; efficiency is 1.7/2.0 = 85%. A load-side power meter cannot by itself establish the 2.0 W input.

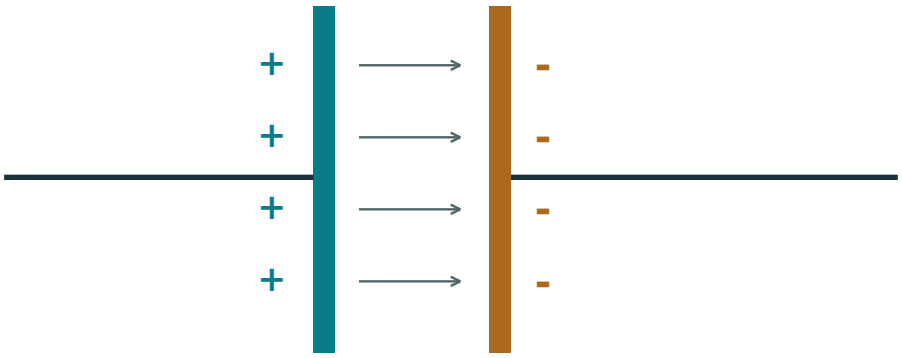
CHECK YOUR UNDERSTANDING

Does an input power reading prove that a fan is rotating?

Answer: 128 | Sources: B1, B2

Capacitors store electric-field energy

A capacitor can absorb or release energy while its voltage changes. It is not simply a tiny battery.



Separated conductors + dielectric

Opposite charges sit on separated conductors; energy resides in the electric field.

HOW IT WORKS

Capacitance C relates charge separation to voltage: Q = CV. Stored energy is E = ½CV². Current through an ideal capacitor depends on the rate of voltage change: i = C dv/dt. The dielectric separates conductors; steady DC does not pass through an ideal dielectric.

USE IT CORRECTLY

Capacitors can smooth supply fluctuations, couple changing signals and help manage transients. Their voltage rating, polarity where applicable, equivalent series resistance and ripple-current limits matter. A capacitor can remain charged after a supply is disconnected. Large or high-voltage capacitors are outside the home projects here.

WORKED EXAMPLE / HYPOTHETICAL

An ideal 1,000 µF capacitor at 5 V stores 0.5 × 0.001 × 25 = 0.0125 J. A 10 mA load discharging it by 1 V takes approximately CΔV/I = 0.1 s if the current stays constant.

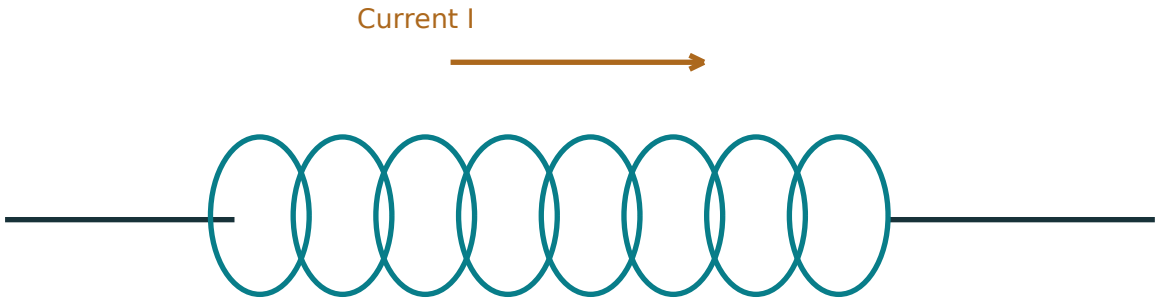
CHECK YOUR UNDERSTANDING

Why does doubling capacitor voltage quadruple stored energy?

Answer: 128 | Sources: B1, B2

Inductors resist changes in current

A changing current produces a changing magnetic field. That field can store and return energy.



Changing current ↔ induced voltage

Current creates magnetic-field storage; switching must manage that stored energy.

HOW IT WORKS

For an ideal inductor, $v = L\,di/dt$. Inductance L is measured in henries (H); energy is $E = \frac{1}{2}LI^2$. A sudden interruption of current can therefore produce a large voltage unless a suitable path absorbs or redirects the stored energy.

USE IT CORRECTLY

Coils appear in motors, transformers and switching converters. Real coils also have winding resistance, magnetic losses and current limits. Flyback protection is chosen for the actual load and switching design. A two-wire brushless fan contains electronics and must be treated according to its datasheet, not automatically as a bare coil.

WORKED EXAMPLE / HYPOTHETICAL

An ideal 10 mH inductor carrying 0.20 A stores $0.5 \times 0.010 \times 0.20^2 = 0.00020$ J. Its small energy can still create a sharp voltage spike if released very quickly.

CHECK YOUR UNDERSTANDING

Why is “open the switch” not a complete inductive-load protection design?

Answer: 128 | Sources: B1, B2

Meet the electronics

Understand the bridge between code and physical equipment before connecting a board.

C Small signals controlling real loads

EIGHT IDEAS TO UNDERSTAND

17 Diodes and LEDs are directional 26

18 A transistor lets a signal control power 27

19 A microcontroller is a small computer 28

20 Digital inputs are not exact voltmeters 29

21 PWM changes time, not the supply label 30

22 ADCs turn analogue signals into codes 31

23 Sensors convert one quantity to another 32

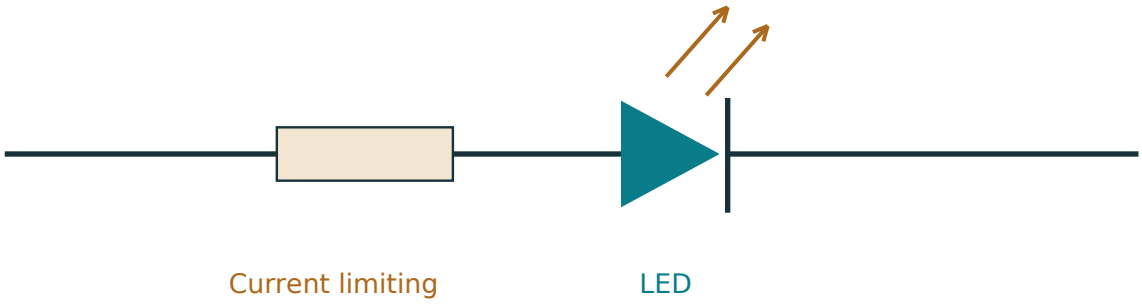
24 Read a datasheet as an engineer 33

Before you move on

Explain one mechanism without looking, solve one example with units and identify one limit of the model. Use the answer section to correct your reasoning, then keep your corrected sketch.

Diodes and LEDs are directional

A diode conducts differently in its two directions. An LED also emits light when operated appropriately.



An LED branch needs a defined current-limiting method.

HOW IT WORKS

A semiconductor diode has a nonlinear current-voltage relationship. Its forward voltage varies with current, temperature and device type. Reverse blocking also has limits. The simple “one-way valve” picture is useful, but does not capture leakage, switching behaviour or breakdown.

USE IT CORRECTLY

An LED usually needs a resistor or regulated current driver. Connecting it directly across a stiff supply can allow damaging current. Calculate a resistor using the intended current and a realistic LED voltage, then check both component ratings. Check the actual LED datasheet rather than relying on colour alone.

WORKED EXAMPLE / HYPOTHETICAL

With assumed 5 V supply, 2 V LED forward drop and 5 mA target: R = (5 - 2)/0.005 = 600 Ω. A 680 Ω resistor gives about 4.4 mA in that simplified model. This is an illustration, not a universal LED specification.

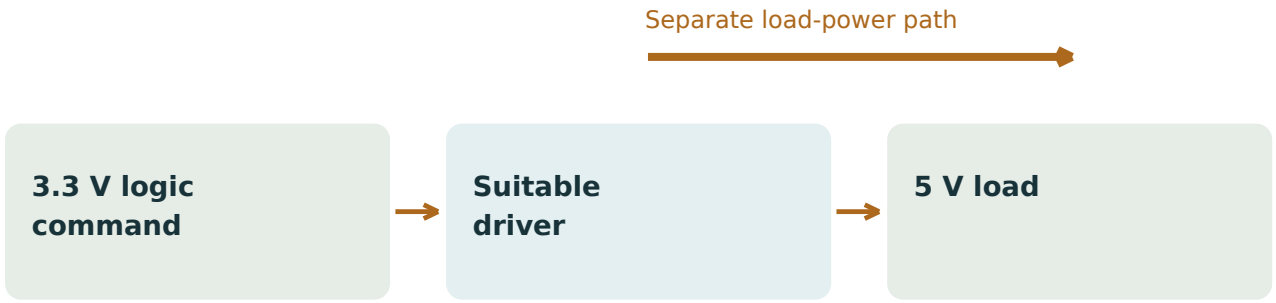
CHECK YOUR UNDERSTANDING

Why is an LED not adequately described by one fixed resistance?

Answer: 129 | Sources: B1, B2

A transistor lets a signal control power

A microcontroller output is a command signal. A driver supplies the load current through a separate path.



GPIO carries a signal. It does not supply the motor current.

Signal and load-power paths have different jobs.

HOW IT WORKS

A MOSFET can act as an electronically controlled switch. Its gate voltage is defined relative to its source. For low-loss switching, the datasheet must specify suitable on-resistance at your actual gate-drive voltage. Threshold voltage only indicates the beginning of conduction under specified test conditions.

USE IT CORRECTLY

Check load current, voltage, heat dissipation and switching behaviour. A default-off gate arrangement and a suitable driver can prevent unintended start-up. Do not power a fan from a GPIO pin. Use a reviewed module or circuit compatible with the actual load, logic level and protection requirements.

WORKED EXAMPLE / HYPOTHETICAL

If a fully enhanced switch has an assumed 0.10 Ω on-resistance and carries 0.20 A, conduction loss is I^2R = 0.004 W. This estimate excludes switching losses and assumes the stated on-resistance really applies at the gate voltage and temperature.

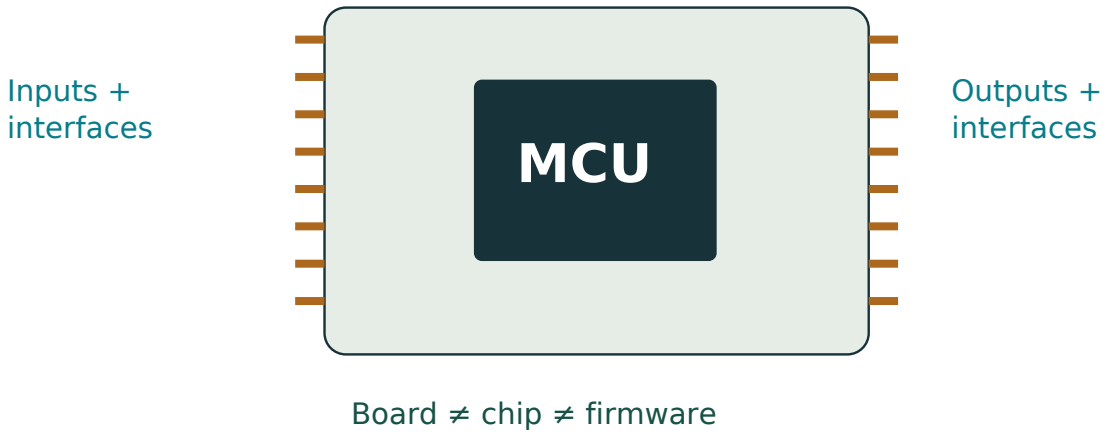
CHECK YOUR UNDERSTANDING

Does a 1 V gate-threshold rating prove a MOSFET is suitable for a 3.3 V control output?

Answer: 129 | Sources: B1, B2; device-specific datasheets govern implementation

A microcontroller is a small computer

It reads inputs, runs instructions and changes outputs. Its timing and electrical limits are part of the system.



A development board includes more than the processor.

HOW IT WORKS

A microcontroller combines a processor, memory and peripherals such as timers, digital inputs and analogue converters. Firmware is the program stored on the device. A development board adds supporting parts such as power regulation and a USB interface; the board and the chip are not interchangeable descriptions.

USE IT CORRECTLY

An ESP32-family board is one possible learning platform, not a single universal pin map. Board variants differ. Record the exact module, board revision and firmware. A USB-powered board can still use 3.3 V logic; a 5 V supply connector does not make its GPIO pins tolerant of 5 V.

WORKED EXAMPLE / HYPOTHETICAL

Before using any pin, make a table with its board label, chip GPIO number, input/output restrictions, boot behaviour and permitted electrical range. Leave a pin unused until you can explain every column.

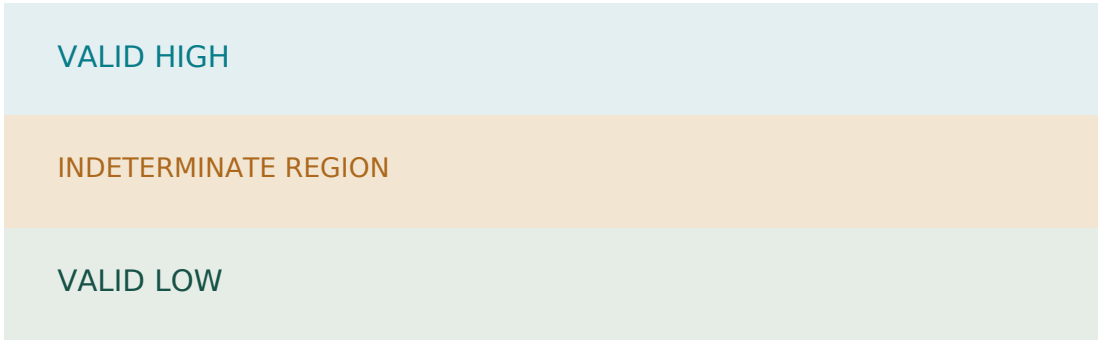
CHECK YOUR UNDERSTANDING

Why can two boards both called ESP32 need different wiring instructions?

Answer: 129 | Sources: S8, S9

Digital inputs are not exact voltmeters

A digital input classifies a voltage using thresholds. It does not report the precise voltage.



Exact thresholds come from the actual input specification.

An electrical threshold decision and a physical-state claim are not the same.

HOW IT WORKS

A digital input recognises a valid low or high region defined by the device specifications. A voltage between those regions may be indeterminate. A floating input has no reliable defined level. Pull-up or pull-down components establish a default where the circuit design permits them.

USE IT CORRECTLY

A physical switch can bounce, producing several transitions during one press. Debouncing rejects brief repeated transitions according to a chosen timing rule. That rule adds delay. Input protection, voltage compatibility and shared references must be considered before a signal is connected to a board.

WORKED EXAMPLE / HYPOTHETICAL

A simulated button produces transitions at 0, 2 and 5 ms before remaining pressed. A 20 ms stable-state debounce would recognise one press only after the input has stayed consistent for that interval. The delay is a deliberate trade-off.

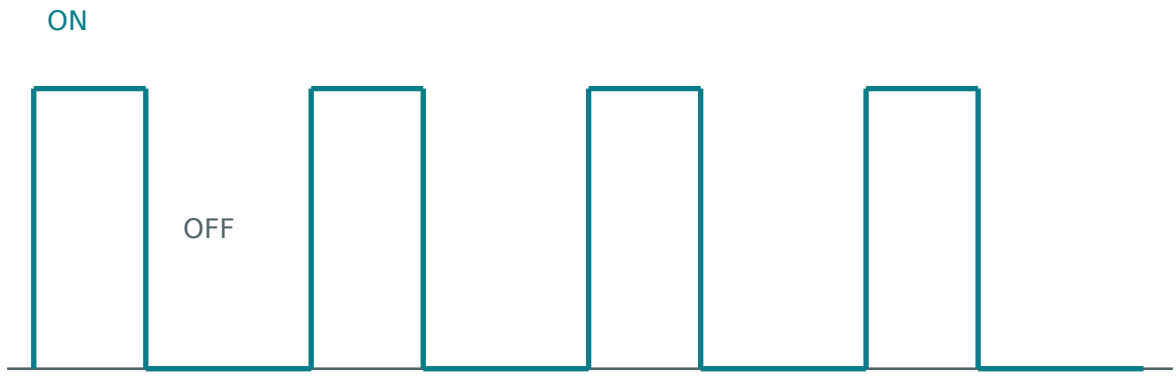
CHECK YOUR UNDERSTANDING

Is a software “high” proof that a relay contact physically closed?

Answer: 129 | Sources: B1; S9

PWM changes time, not the supply label

Pulse-width modulation switches a signal on and off. Duty cycle describes the fraction of each period spent on.



40% on-time in each repeated period

Duty cycle is a time fraction. Device response needs its own model.

HOW IT WORKS

Duty cycle D = on-time/period. An ideal 0-to-5 V waveform with 40% duty has a 2 V time-average voltage. Its heating effect on a resistor depends on the average of voltage squared, not the square of average voltage. A meter may report something else depending on its measurement method.

USE IT CORRECTLY

A motor’s response depends on electrical and mechanical dynamics. A brushless fan may require constant supply and a separate specified PWM input, or may restrict supply switching. Frequency, start-up behaviour and minimum useful duty are device-specific. Do not assume 50% duty means 50% speed or half the power.

WORKED EXAMPLE / HYPOTHETICAL

For an ideal 100 Ω resistor switched between 0 and 5 V at 40% duty: average power = D × V^2/R = 0.4 × 25/100 = 0.10 W. Using the 2 V average in V^2/R would incorrectly predict 0.04 W.

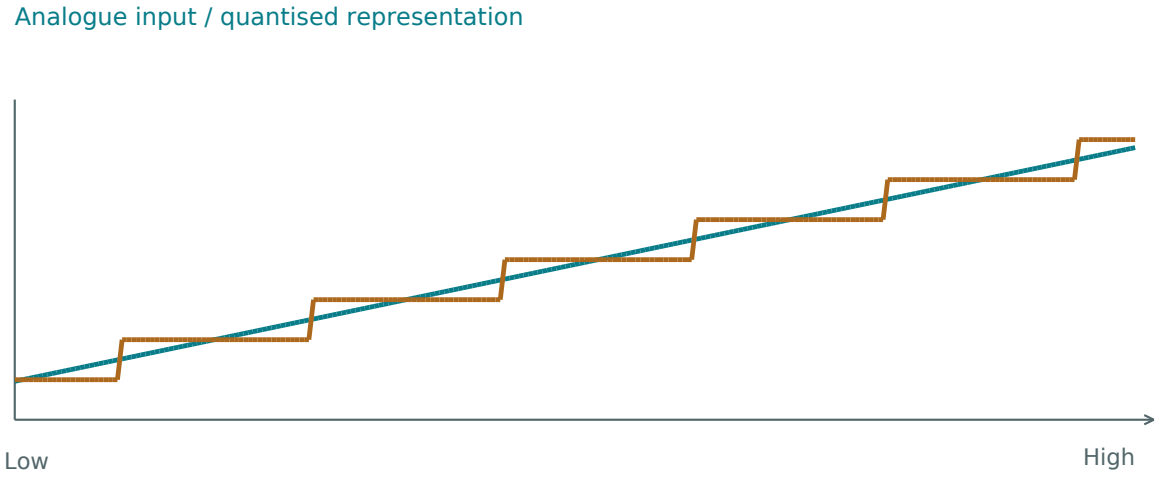
CHECK YOUR UNDERSTANDING

At 1 kHz and 25% duty, what are the period and on-time?

Answer: 129 | Sources: B1, B2

ADCs turn analogue signals into codes

An analogue-to-digital converter produces discrete codes. More bits do not automatically mean better accuracy.



Quantisation divides an input range into codes; calibration addresses other errors.

HOW IT WORKS

An ideal N-bit ADC provides 2^N possible codes. For a nominal full-scale range V_{FS}, a useful approximate code width is $V_{FS}/2^N$. Actual converters also have offset, gain error, nonlinearity, noise and input constraints. Calibration maps codes to a better voltage estimate.

USE IT CORRECTLY

The ideal arithmetic is not a board specification. In ESP32 devices, permitted input ranges and calibration procedures depend on the exact chip and configuration. Check input attenuation, reference variation and pin restrictions. Never connect a 5 V signal merely because the board itself receives USB power.

WORKED EXAMPLE / HYPOTHETICAL

For a hypothetical ideal 12-bit, 0-to-3.3 V converter, code width is approximately 3.3/4,096 = 0.000806 V, or 0.806 mV. This is resolution, not a ±0.806 mV accuracy claim and not an ESP32 range specification.

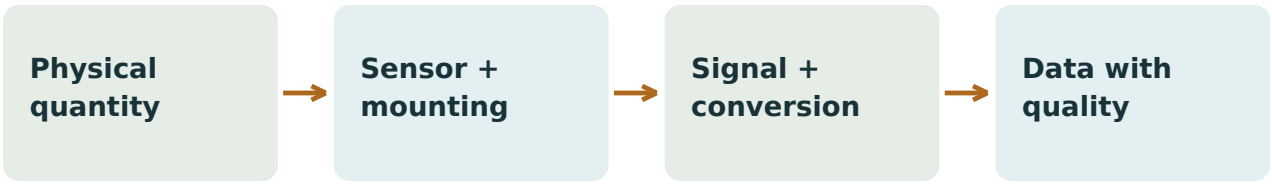
CHECK YOUR UNDERSTANDING

Can averaging noisy ADC readings remove every measurement error?

Answer: 129 | Sources: S10, S11

Sensors convert one quantity to another

A temperature probe, current sensor and speed sensor answer different physical questions.



Position, bandwidth and interface affect the complete measurement.

Measure the variable named in the requirement, at the relevant location.

HOW IT WORKS

A sensor responds to a physical quantity and produces a signal. A measurement chain includes the sensor, mounting, conditioning, conversion, timing and software. Errors can enter at every stage. Position matters: a room temperature probe does not automatically represent a chip junction temperature.

USE IT CORRECTLY

Select a sensor by range, response time, required uncertainty, environmental suitability and interface compatibility. A shunt measures current using a small voltage drop; it changes the circuit slightly and dissipates heat. A Hall-effect current sensor uses magnetic response and has its own offset and bandwidth limits.

WORKED EXAMPLE / HYPOTHETICAL

An assumed 0.1 Ω shunt at 0.20 A produces 20 mV and dissipates 4 mW. To infer current, the measurement circuit must resolve that small differential voltage while meeting its allowed input conditions.

CHECK YOUR UNDERSTANDING

Which sensor best supports a claim that a fan reached a target rotational speed?

Answer: 129 | Sources: B1, B2

Read a datasheet as an engineer

Separate normal operating conditions, guaranteed characteristics and absolute limits.



Read the test conditions, not only the headline number.

HOW IT WORKS

Start with the exact part number and revision. Read the functional description, recommended operating conditions, electrical tables and application notes. A typical value is not a guarantee for every unit. Absolute maximum ratings describe stress limits, not a normal operating target.

USE IT CORRECTLY

Make a compatibility table before buying or wiring: supply range, logic levels, input range, maximum current, heat, interface, start-up state and fault behaviour. Note the conditions attached to every figure. A product listing or tutorial may omit restrictions that the manufacturer explicitly states.

WORKED EXAMPLE / HYPOTHETICAL

A board is powered from 5 V USB, but its sensor interface is specified for 3.3 V signals. A 5 V-output sensor needs an appropriate interface design; the shared product label “5 V compatible” is not enough to establish compatibility.

CHECK YOUR UNDERSTANDING

Why is “typical current: 100 mA” a weak basis for selecting a current limit?

Answer: 129 | Sources: S8-S11; actual component datasheets

Make measurements

Treat every reading as a claim with a location, a time, an instrument and uncertainty.

D Numbers you can reason from

EIGHT IDEAS TO UNDERSTAND

25 Use a multimeter for the right task 35

26 Accuracy, precision and resolution differ 36

27 Calibration connects indication to reference 37

28 Build an uncertainty budget 38

29 Sampling can conceal a signal 39

30 Use statistics without hiding failure 40

31 Timestamps describe different events 41

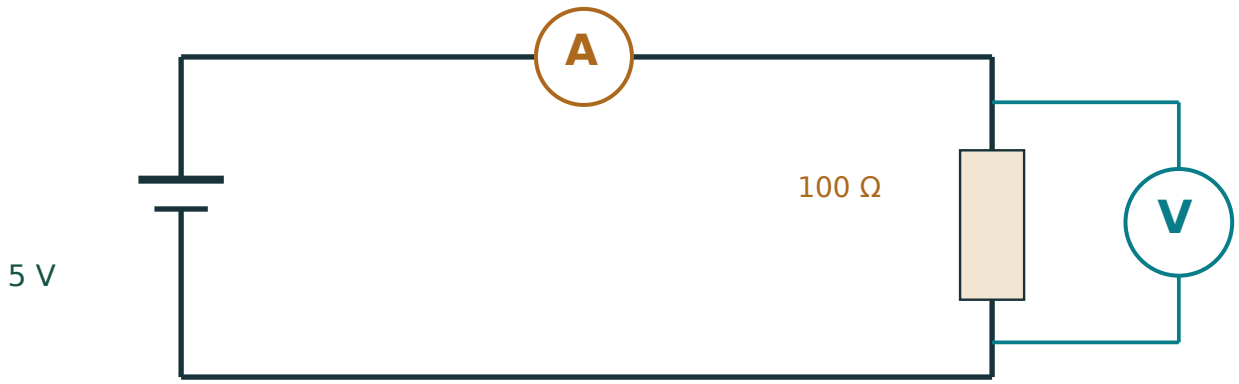
32 Design a test that could disprove you 42

Before you move on

Explain one mechanism without looking, solve one example with units and identify one limit of the model. Use the answer section to correct your reasoning, then keep your corrected sketch.

Use a multimeter for the right task

Voltage, current and resistance measurements use different connections and operating conditions.



Conceptual measurement topology only; disconnect power before changing leads.

HOW IT WORKS

A voltmeter is placed across two points and ideally draws little current. An ammeter is inserted into a current path and ideally adds little resistance. Its internal resistance and fuse affect real measurements. A resistance measurement generally requires an unpowered, suitably isolated component.

USE IT CORRECTLY

For the supervised 5 V projects here, plan the connection with power disconnected. Confirm the lead sockets, mode and range. A meter left in current mode across a supply can create a short. Never use these bench instructions for mains measurements or assume a cheap meter is suitable for infrastructure work.

WORKED EXAMPLE / HYPOTHETICAL

A 100 Ω resistor across an assumed 5.00 V draws about 50 mA. Measure the voltage across the resistor and current in its branch, then compare VI with V^2/R using the resistor’s actual tolerance and meter limitations.

CHECK YOUR UNDERSTANDING

Why does changing from volts to amps require more than turning the selector?

Answer: 130 | Sources: B1, B2

Accuracy, precision and resolution differ

A stable display can still be wrong. A detailed display can still be uncertain.



Precise, biased

Centred, variable

Tightly grouped observations can share a common bias.

HOW IT WORKS

Resolution is the smallest distinguishable indicated change. **Precision** concerns agreement among repeated observations under stated conditions. **Accuracy** concerns closeness to the true quantity and is not established by repeatability alone. Measurement uncertainty expresses the dispersion of values reasonably attributed to the quantity.

USE IT CORRECTLY

Manufacturer specifications often combine a percentage of reading with a number of display counts. Interpret the actual formula and range used. Do not round a computed result to six decimals if its inputs support only two. Keep sufficient digits internally, then report a sensible result with its uncertainty context.

WORKED EXAMPLE / HYPOTHETICAL

Assume a 5.00 V reading on a range with 0.01 V resolution and a bound ±(1% of reading + 2 counts). The stated bound is ±(0.05 + 0.02) = ±0.07 V under the specified conditions.

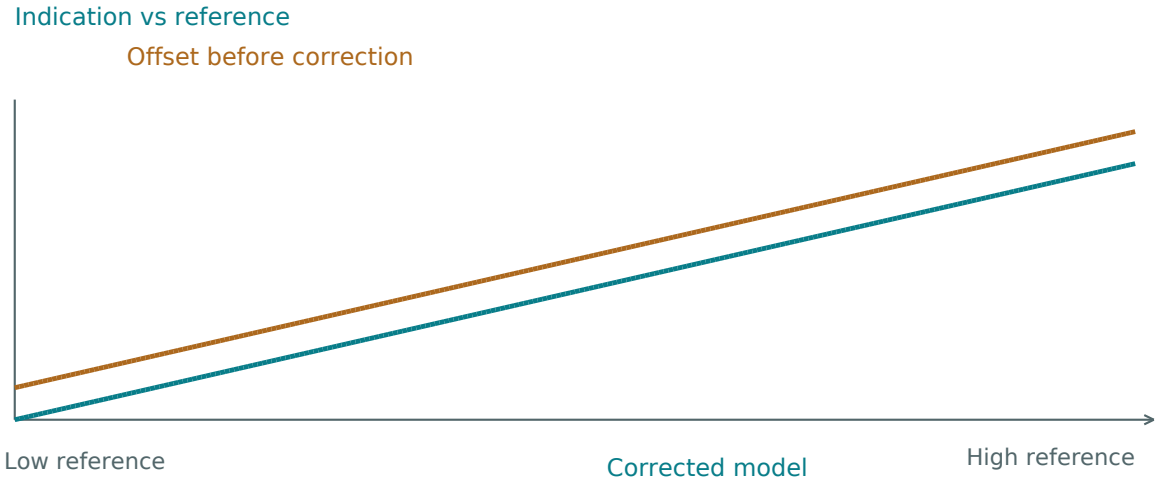
CHECK YOUR UNDERSTANDING

Ten identical readings of 5.00 V prove what?

Answer: 130 | Sources: B2; S6, S7

Calibration connects indication to reference

Calibration characterises a measuring system. A correction is useful only within its justified conditions.



Check the residuals after a correction; do not only admire the fitted line.

HOW IT WORKS

Compare the instrument against a suitable reference at several points spanning the intended range. Record reference uncertainty, environment, method and instrument identity. A single-point adjustment cannot establish a full-range response. A home cross-check is valuable but is not automatically traceable calibration.

USE IT CORRECTLY

A simple linear model is corrected value = a × indicated value + b. The offset b shifts the response; gain a changes its slope. Check residual errors after fitting. Do not apply a correction beyond the tested range without justification. A more complex fit can hide poor data rather than fix it.

WORKED EXAMPLE / HYPOTHETICAL

A sensor indicates 1.1 units at a reference of 1.0 and 3.1 at 3.0. An offset correction of -0.1 matches those two points. More points and repeated observations are needed to test linearity, stability and uncertainty.

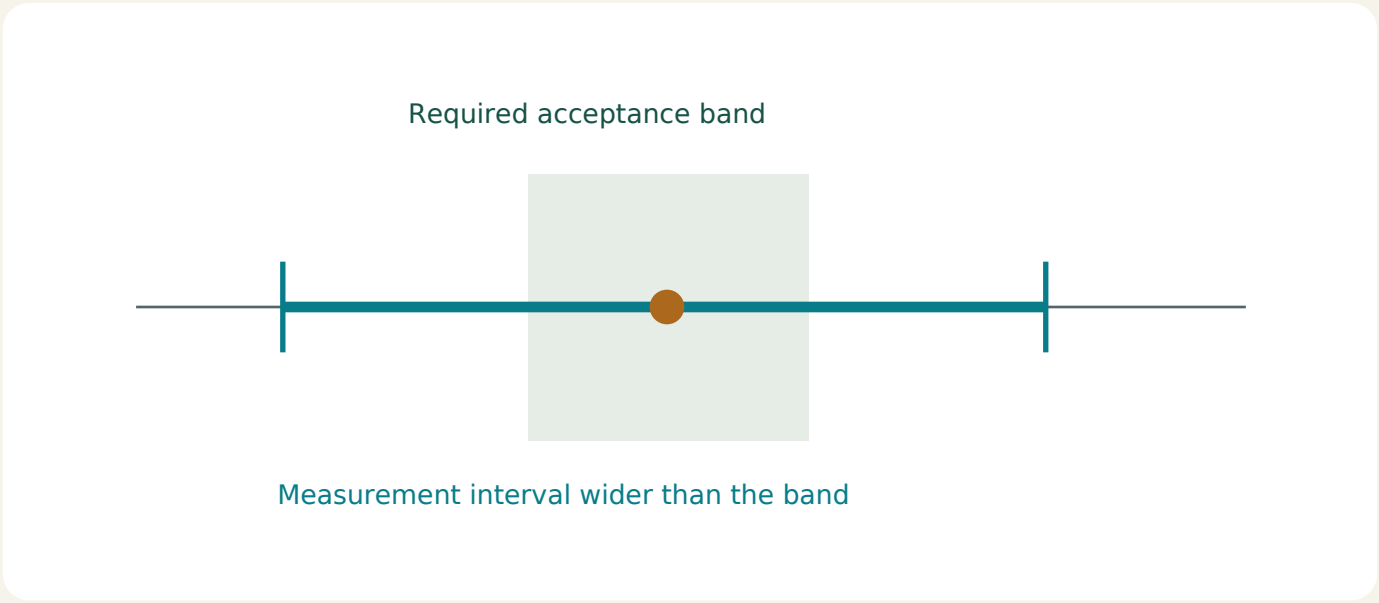
CHECK YOUR UNDERSTANDING

Does agreement with another uncalibrated sensor prove traceability?

Answer: 130 | Sources: S6, S7

Build an uncertainty budget

Choose a decision rule that accounts for what you do not know about the measurement.



An acceptance band and a measurement-error interval are different objects.

HOW IT WORKS

List contributors such as reference uncertainty, instrument specification, mounting, noise, resolution and timing. Statistical evaluation from repeated data and evaluation from other information are different methods. Do not equate them automatically with “random” and “systematic” error categories.

USE IT CORRECTLY

For a simple product P = VI, a conservative small-error bound can add relative voltage and current bounds. Combining independent standard uncertainties by root-sum-square is a different calculation requiring justified assumptions. Keep a manufacturer’s worst-case bound distinct from a standard uncertainty and from a confidence interval.

WORKED EXAMPLE / HYPOTHETICAL

At 5.0 V ±0.1 V and 0.20 A ±0.01 A, nominal power is 1.0 W. Relative bounds are 2% and 5%; first-order worst-case addition gives about 7%, or ±0.07 W. This is a bound estimate, not a 95% confidence statement.

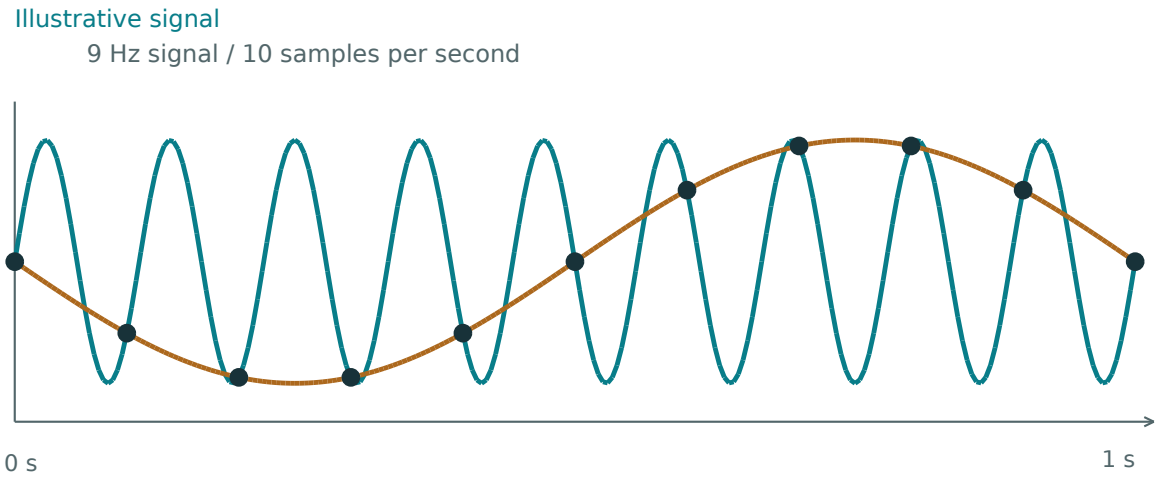
CHECK YOUR UNDERSTANDING

A target band is ±0.02 W but your bound is ±0.07 W. Can an in-band display settle the decision?

Answer: 130 | Sources: S6, S7

Sampling can conceal a signal

A sequence of readings represents only the times and bandwidth actually observed.



Different physical signals can produce similar sparse samples.

HOW IT WORKS

Sampling frequency f_s is readings per second. For ideal reconstruction of a band-limited signal, sampling must exceed twice its highest frequency. Real systems need margin and suitable anti-alias filtering before sampling. Taking more digital points after aliasing does not reconstruct lost information.

USE IT CORRECTLY

A sensor’s bandwidth and response time can be slower than the logger. Temperature sampled ten times per second may still reflect a probe that takes seconds to respond. A one-second logger cannot characterise millisecond power spikes. Match the instrument and acquisition method to the question.

WORKED EXAMPLE / HYPOTHETICAL

A 9 Hz sine sampled at 10 samples/s can appear as a 1 Hz oscillation. The low-frequency appearance is an alias, not evidence of a real 1 Hz source. Increasing sample rate without checking analogue bandwidth is also incomplete.

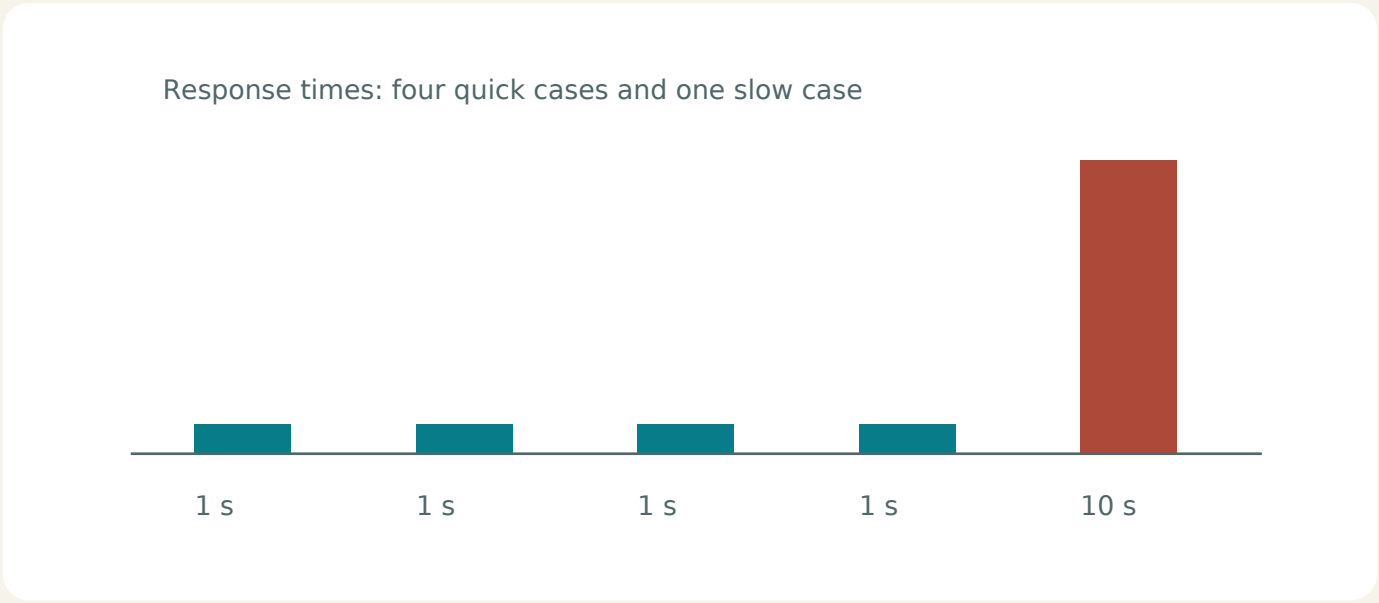
CHECK YOUR UNDERSTANDING

Can five-minute grid data prove that a controller suppressed a 10 Hz oscillation?

Answer: 130 | Sources: B1, B2

Use statistics without hiding failure

Summaries help compare behaviour, but the raw sequence and test conditions still matter.



The same central tendency can coexist with a troublesome tail.

HOW IT WORKS

The mean is the sum divided by sample count. The median is the middle ordered value. Standard deviation describes dispersion under a stated calculation. A percentile identifies a threshold in an ordered distribution; its estimate is uncertain with small samples. None of these alone explains why a failure happened.

USE IT CORRECTLY

Separate repeat tests under comparable conditions from time samples taken during one run. Consecutive readings are often correlated, so 1,000 samples need not mean 1,000 independent experiments. Report missingness and outliers before choosing whether to exclude them. Keep the original data.

WORKED EXAMPLE / HYPOTHETICAL

Response times of 1, 1, 1, 1 and 10 s have mean 2.8 s and median 1 s. Reporting only the median hides a slow case that could matter to a timeout requirement. Investigate the 10 s result rather than deleting it.

CHECK YOUR UNDERSTANDING

Which summary would you report for a small safety-relevant timing test?

Answer: 130 | Sources: B1, B2

Timestamps describe different events

Distinguish when a physical quantity was sampled from when its report was received or stored.



The timestamp attached to a value must identify the event it describes.

HOW IT WORKS

Record capture time, receive time and relevant command times separately. A wall clock gives a calendar timestamp; a monotonic clock measures local elapsed time without wall-clock adjustments. Clock synchronisation has uncertainty. A future-dated sample should not become “fresh” because its computed age is negative.

USE IT CORRECTLY

Freshness is a policy tied to a physical requirement. Check capture time, clock quality, sequence number and expected cadence. A repeated sample with a new receive timestamp remains old evidence. When clocks cannot be compared reliably, preserve that limitation rather than inventing simultaneity.

WORKED EXAMPLE / HYPOTHETICAL

A sample is captured at 12:00:00 and received at 12:00:10. At receive time it is 10 s old, assuming aligned clocks. With a 5 s freshness limit, it is stale even though the network message just arrived.

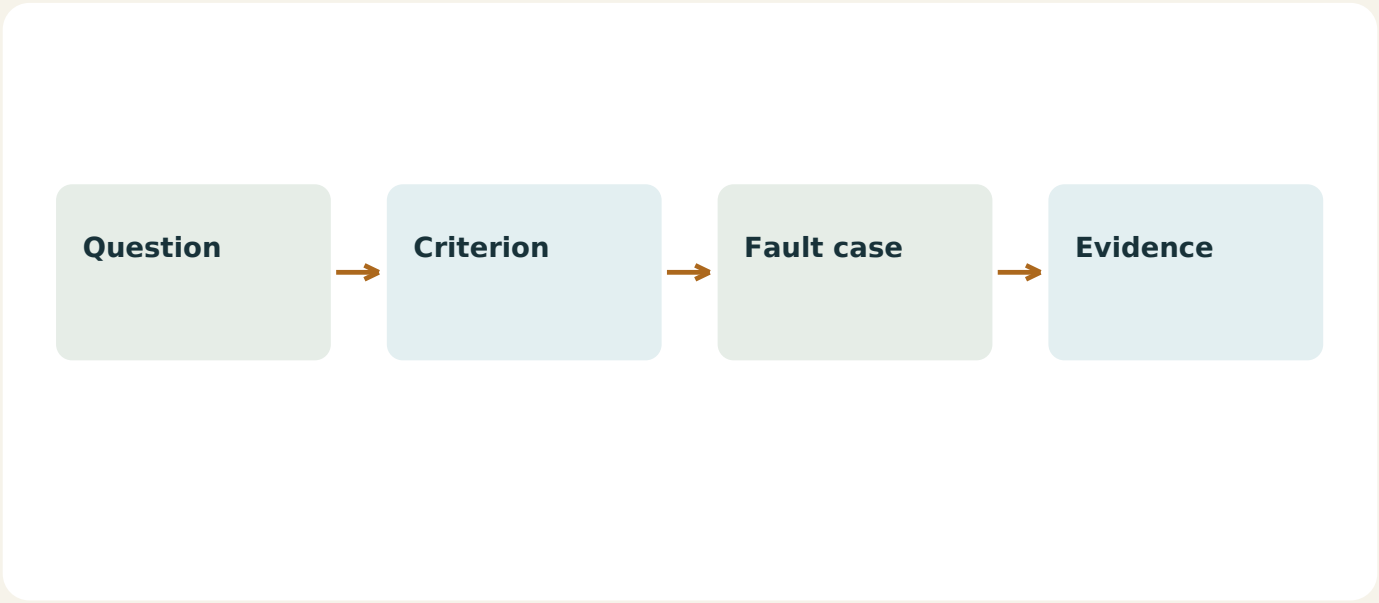
CHECK YOUR UNDERSTANDING

Why use a monotonic clock for a local timeout?

Answer: 130 | Sources: B1, B2

Design a test that could disprove you

A useful experiment has a question, a method, an acceptance criterion and a way to reveal a bad result.



An experiment must allow the expected conclusion to be wrong.

HOW IT WORKS

Choose one claim and define the independent variable you change and the dependent variable you observe. Hold other conditions as stable as practical. Record preconditions and a baseline. Include a negative control: a case that should fail. Decide the acceptance rule before seeing the outcome.

USE IT CORRECTLY

Repeat across relevant conditions and report contradictions. A measurement at the target does not prove your command caused it if it was already there. For causal claims, consider baseline behaviour, competing changes, order effects and an appropriate comparison. A small bench can test a mechanism without establishing field reliability.

WORKED EXAMPLE / HYPOTHETICAL

To test stale-data handling, first supply a valid new sample, then resend the identical capture timestamp repeatedly. The required result is eventual stale/uncertain status, even while messages keep arriving. Save the inputs and verdicts.

CHECK YOUR UNDERSTANDING

Why include a deliberately bad sensor reading in a test suite?

Answer: 130 | Sources: B1, B2

Understand power systems

Follow electrical energy through the equipment that generates, converts and delivers it.

E AC, machines, networks and protection

EIGHT IDEAS TO UNDERSTAND

33 AC repeats; DC keeps its polarity 44

34 RMS and phase explain AC behaviour 45

35 Real, reactive and apparent power 46

36 Three phases share the work 47

37 Generators and motors exchange energy 48

38 Transformers trade voltage for current 49

39 Networks have thermal and voltage limits 50

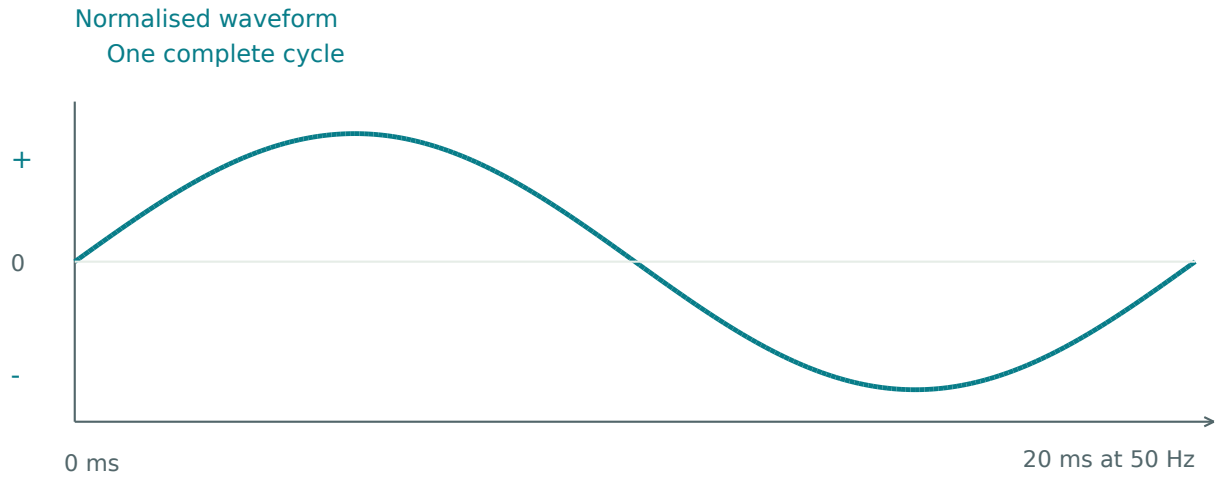
40 Protection must act without your software 51

Before you move on

Explain one mechanism without looking, solve one example with units and identify one limit of the model. Use the answer section to correct your reasoning, then keep your corrected sketch.

AC repeats; DC keeps its polarity

An alternating waveform reverses polarity periodically. Direct current can vary in magnitude while retaining its direction.



One sinusoidal cycle contains a positive and a negative half-cycle.

HOW IT WORKS

For a sine wave, $v(t) = V_{pk} \sin(2\pi ft)$. Peak voltage V_{pk} is its largest magnitude; frequency f is cycles per second, in hertz. Period $T = 1/f$ is the duration of one cycle. These describe different properties of the same signal.

USE IT CORRECTLY

Tasmania’s nominal grid frequency is 50 Hz. A USB supply provides DC to a load, although switching ripple and transients may remain. Draw the waveform you mean rather than assuming every DC line is perfectly flat. Grid waveforms and power-electronic signals require instruments suited to their voltage and frequency.

WORKED EXAMPLE / HYPOTHETICAL

At 50 Hz, $T = 1/50 = 0.020$ s = 20 ms. A 1 kHz PWM signal repeats every 1 ms, but it is a switched waveform rather than a 1 kHz sine wave.

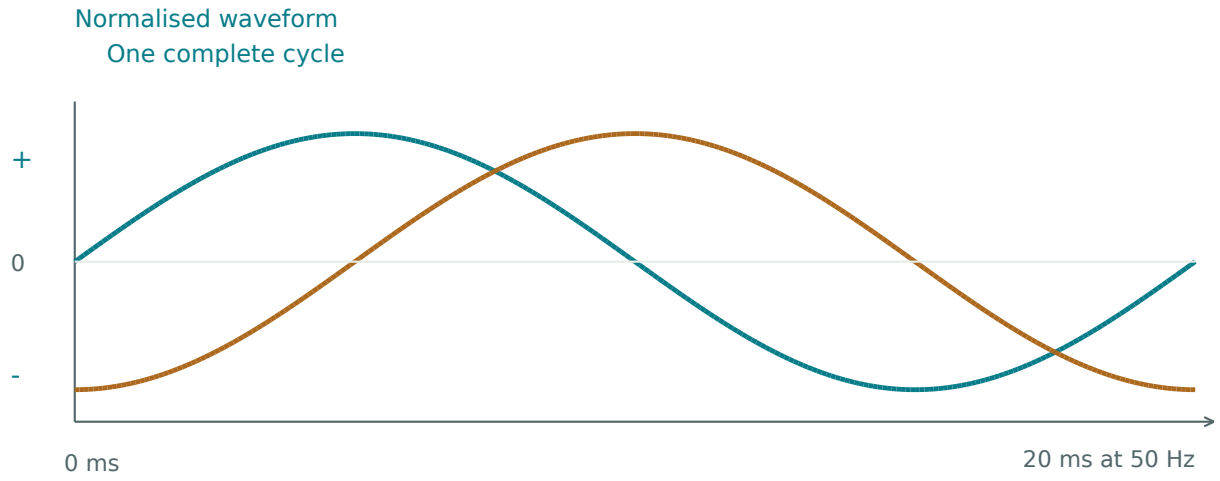
CHECK YOUR UNDERSTANDING

If frequency rises from 50 to 100 Hz, what happens to the period?

Answer: 131 | Sources: B1, B2; S1

RMS and phase explain AC behaviour

Root-mean-square values describe effective magnitude. Phase describes relative timing within a cycle.



Phase compares timing. RMS compares effective magnitude.

HOW IT WORKS

For a sine wave, $V_{rms} = V_{pk}/\sqrt{2}$. The RMS value gives the same heating in an ideal resistor as that DC voltage. The mean of a complete symmetrical sine is zero, so its ordinary average does not represent its heating effect.

USE IT CORRECTLY

Two sine waves at the same frequency can peak at different times. Their phase difference is often expressed in degrees: one complete cycle is 360°. Motors and converters can shift or distort current relative to voltage. The simple sinusoidal phase model is useful but incomplete for distorted waveforms.

WORKED EXAMPLE / HYPOTHETICAL

For a hypothetical 10 V peak sine, RMS is 10/1.414 ≈ 7.07 V. At 50 Hz, a 90° phase displacement is one quarter of a 20 ms cycle, or 5 ms.

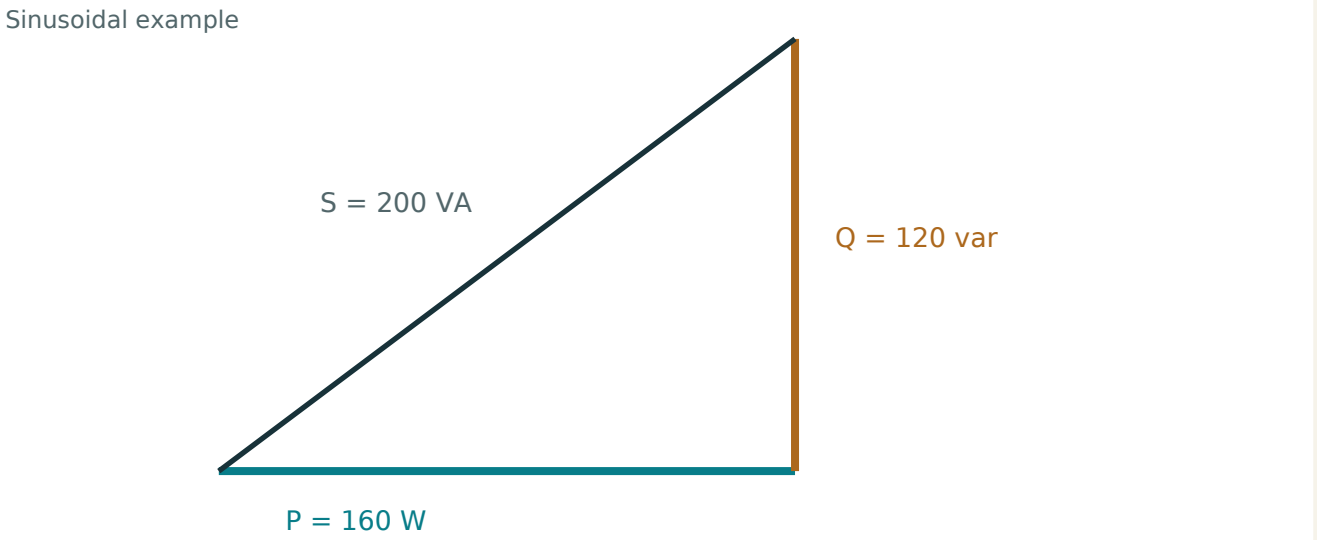
CHECK YOUR UNDERSTANDING

Does a 10 V peak sine have the same resistor heating as 10 V DC?

Answer: 131 | Sources: B1, B2

Real, reactive and apparent power

In AC, volts multiplied by amps does not always equal the useful average electrical power.



The triangle applies to the stated sinusoidal model; distortion needs fuller treatment.

HOW IT WORKS

Apparent power $S = V_{rms}I_{rms}$, measured in volt-amperes (VA). Real power P is the average of instantaneous v(t)i(t), in watts. Power factor PF = P/S. For sinusoidal voltage and current, PF = cos φ and reactive power Q = VI sin φ, in var.

USE IT CORRECTLY

Reactive exchange is associated with electric and magnetic energy storage. It is not simply another quantity of useful energy consumed. Distortion also affects power factor, so cos φ alone can be insufficient. Lower power factor can require more current for the same real power, affecting equipment loading and losses.

WORKED EXAMPLE / HYPOTHETICAL

Assume a single-phase load at 100 V RMS, 2 A RMS and PF = 0.8. S = 200 VA; P = 160 W. Under the sinusoidal model, $|Q| = \sqrt{(200^2 - 160^2)} = 120$ var. The sign depends on the chosen convention.

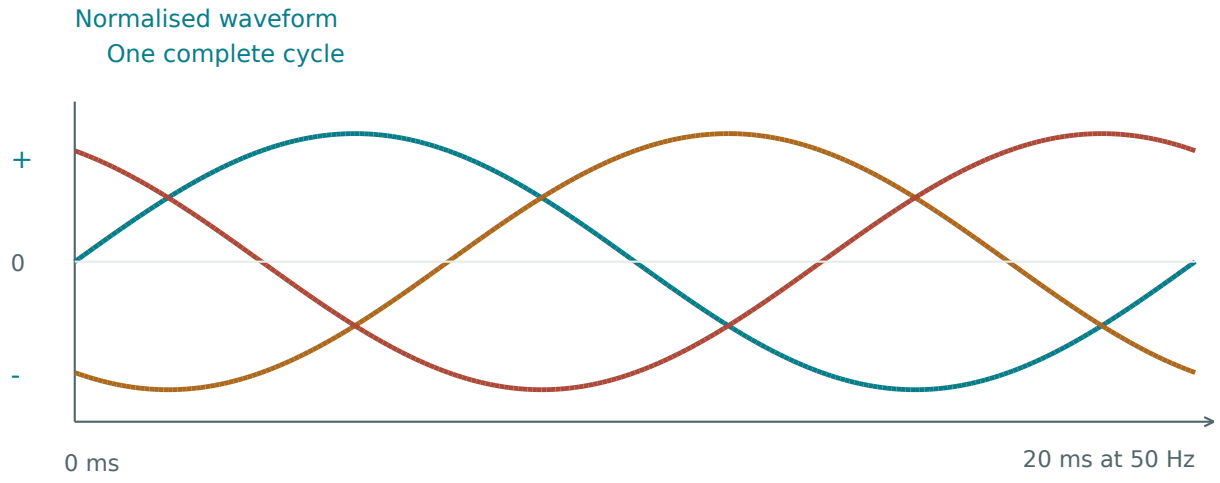
CHECK YOUR UNDERSTANDING

A meter reports 500 VA and PF 0.9. What real power does that imply?

Answer: 131 | Sources: B1, B2

Three phases share the work

A balanced three-phase system has three similar waveforms displaced by 120 electrical degrees.



Three equal-frequency waveforms, separated by one third of a cycle.

HOW IT WORKS

Three-phase supply is common for large machines and facilities. In a balanced sinusoidal system, total real power is $P = \sqrt{3} V_{LL} I_L PF$, where V_{LL} is line-to-line RMS voltage and I_L is line current. Do not insert line-to-neutral voltage into that formula.

USE IT CORRECTLY

In a balanced wye connection, line-to-line voltage is √3 times line-to-neutral voltage. In practice, imbalance, harmonics and neutral current complicate the picture. Equipment ratings and protection depend on the actual connection. This is a conceptual calculation, not guidance for connecting a three-phase supply.

WORKED EXAMPLE / HYPOTHETICAL

Assume 400 V line-to-line, 10 A line current and PF 0.90. P = 1.732 × 400 × 10 × 0.90 ≈ 6,235 W = 6.24 kW. Balanced sinusoidal conditions are part of this estimate.

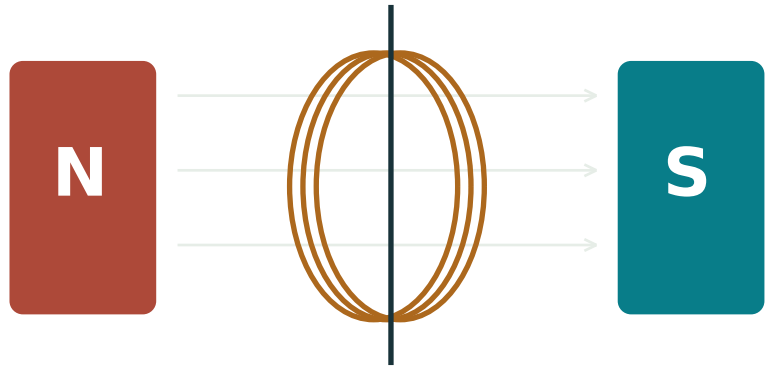
CHECK YOUR UNDERSTANDING

What are the three phase displacements in a balanced set?

Answer: 131 | Sources: B1, B2

Generators and motors exchange energy

Electromagnetism links electrical power with rotating mechanical equipment.



Rotating conductor / field

A turbine supplies mechanical power; the generator converts it.

HOW IT WORKS

A generator converts mechanical input into electrical output. A changing magnetic flux through a circuit induces voltage. Loading the generator produces an opposing mechanical effect, so its prime mover must supply torque. A motor uses electrical input to produce torque and motion, with losses.

USE IT CORRECTLY

Torque is turning effect; mechanical power $P = \tau\omega$, with torque τ in newton-metres and angular speed ω in radians per second. Rotating speed and electrical frequency are linked by pole count in a synchronous machine. Different machine types and converter interfaces need different models.

WORKED EXAMPLE / HYPOTHETICAL

A hypothetical shaft carries 10 N·m at 100 rad/s: mechanical power is 1,000 W. If electrical output is 900 W at steady conditions, conversion efficiency is 90% and the remaining 100 W is loss.

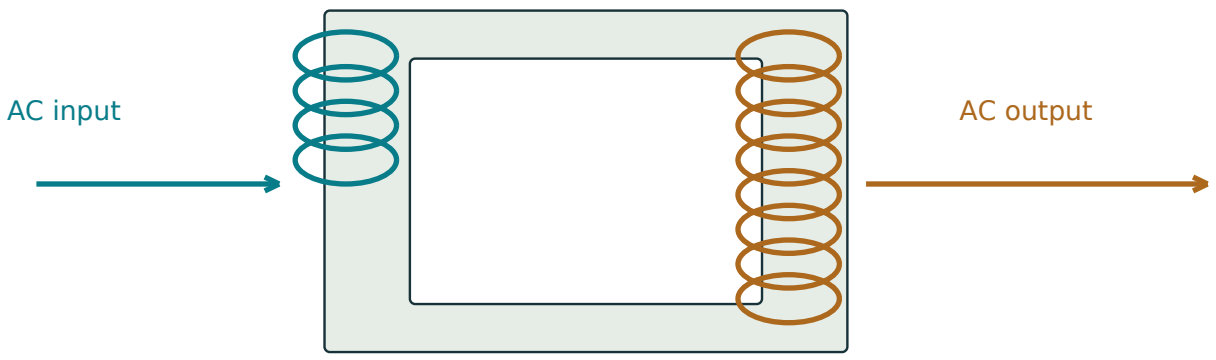
CHECK YOUR UNDERSTANDING

Why is a permanent magnet not an endless source of generator energy?

Answer: 131 | Sources: B1, B2

Transformers trade voltage for current

A transformer couples changing magnetic flux between windings. It cannot create additional power.



Magnetic core couples the windings.

Windings share changing flux through a magnetic core.

HOW IT WORKS

For an ideal transformer, $V_2/V_1 = N_2/N_1$, the ratio of winding turns. With negligible losses, input and output power are equal; increasing voltage reduces the available current for the same power. A normal power transformer requires changing flux and is not operated from steady DC like an AC supply.

USE IT CORRECTLY

Real transformers have winding losses, core losses, leakage effects and thermal limits. Ratings depend on operating conditions and cooling. A transformer also does not automatically make an entire system isolated: the specific winding arrangement and other connections determine that.

WORKED EXAMPLE / HYPOTHETICAL

An ideal turns ratio of 10:1 steps 100 V down to 10 V. If output is 2 A, output power is 20 W and ideal input current is 0.20 A. Actual input power would be higher because of losses.

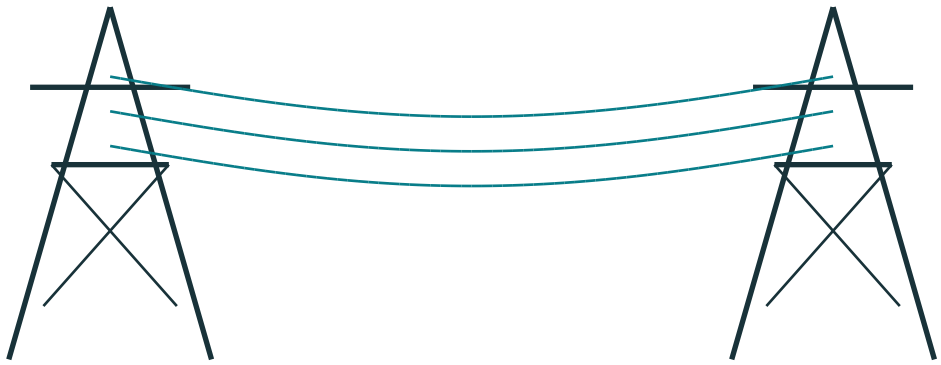
CHECK YOUR UNDERSTANDING

Does a 10:1 voltage reduction mean energy is reduced by a factor of ten?

Answer: 131 | Sources: B1, B2

Networks have thermal and voltage limits

A network is more than a wire with a maximum power label.



Thermal limits + voltage + stability + contingencies

Generation adequacy and local network capability answer different questions.

HOW IT WORKS

Conductor resistance causes loss $P_{loss} = I^2R$. Current also creates temperature rise; the allowable loading depends on conductor, ambient conditions and heat removal. Voltage drop and AC reactance matter too. Power flows through connected paths according to network behaviour, not a selected contractual route.

USE IT CORRECTLY

At equal transferred power, higher voltage generally allows lower current and lower resistive loss for the same conductor resistance. This motivates transmission at high voltage. Real network analysis also includes reactive power, stability, fault levels, topology and contingency conditions. A spare MW at one site does not prove another connection is feasible.

WORKED EXAMPLE / HYPOTHETICAL

In an illustrative equal-power comparison, 1,000 W at 100 V needs 10 A; at 1,000 V it needs 1 A. For a 1 Ω path, I^2R is 100 W versus 1 W. This isolates resistive loss, not a complete AC design.

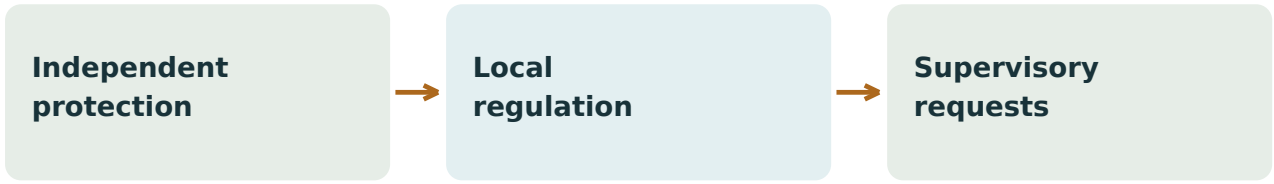
CHECK YOUR UNDERSTANDING

Why can a network constraint remain even when the state has enough energy overall?

Answer: 131 | Sources: B1, B2

Protection must act without your software

A fault can change current and voltage faster than a supervisory application can safely manage.



Supervisory requests sit above independent equipment protection.

HOW IT WORKS

Protection detects specified abnormal conditions and initiates isolation or another protective action. Fuses, circuit breakers and protective relays have different roles and ratings. Coordination aims to isolate the appropriate part of a system while preserving the rest where possible. Fault current is not the same as normal load current.

USE IT CORRECTLY

Protective earthing, insulation and residual-current protection address different hazards. A residual-current device detects imbalance; it is not a substitute for every overcurrent protection function. Exact design, installation and testing are specialist work. Lupine must not become the only protection against excessive current, temperature or unsafe equipment state.

WORKED EXAMPLE / HYPOTHETICAL

If a supervisory controller freezes while a load overheats, local protection should still enforce its defined limit. A Wi-Fi alarm may help inform a person, but it cannot substitute for a protective function designed for that hazard.

CHECK YOUR UNDERSTANDING

What should happen to essential protective limits if the Lupine computer loses power?

Answer: 131 | Sources: B1, B2

Ground it in Tasmania

Connect the physics to local infrastructure without depending on a speculative project announcement.

F Hydro, renewable generation and the grid

EIGHT IDEAS TO UNDERSTAND

41 Tasmania is your engineering setting 53

42 Hydro power comes from head and flow 54

43 Water and turbines have dynamics 55

44 Pumped hydro shifts energy in time 56

45 Wind and solar need interfaces 57

46 Frequency is a system balance signal 58

47 System strength is not inertia 59

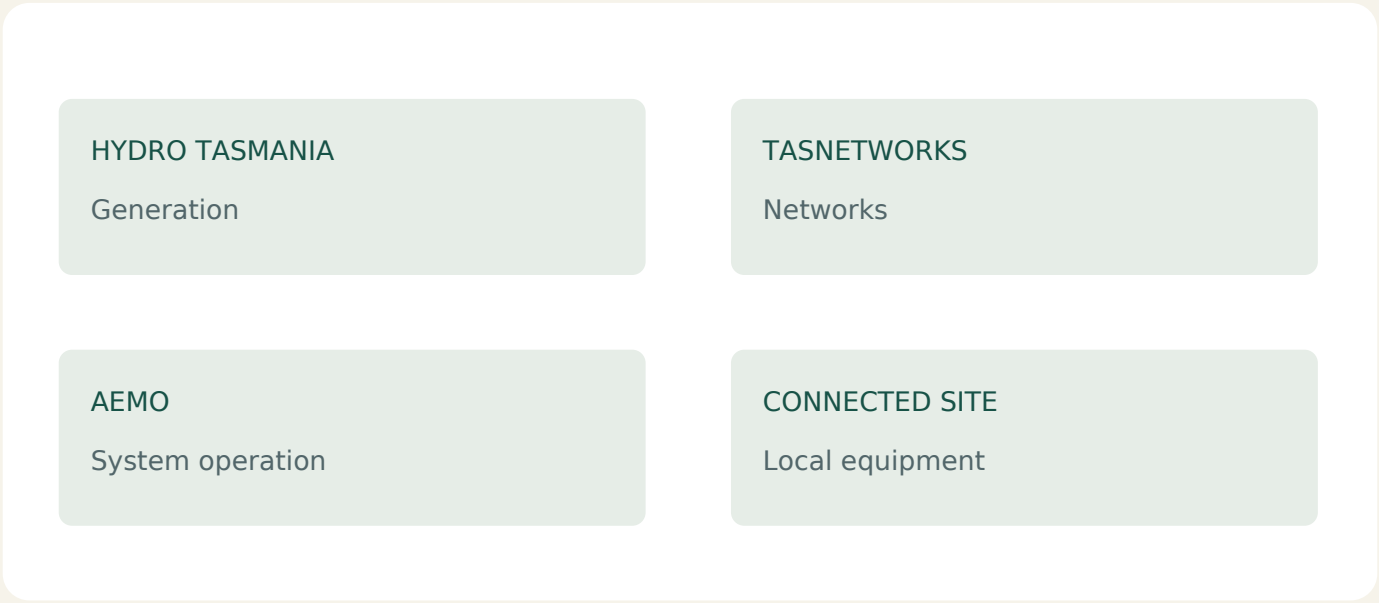
48 A connection is a technical interface 60

Before you move on

Explain one mechanism without looking, solve one example with units and identify one limit of the model. Use the answer section to correct your reasoning, then keep your corrected sketch.

Tasmania is your engineering setting

Understand the roles around the equipment before deciding which problem Lupine should solve.



Functional roles, not a complete ownership map.

HOW IT WORKS

Hydro Tasmania is a generation business with hydro schemes. TasNetworks owns, operates and maintains Tasmania's transmission and distribution networks. AEMO performs NEM system-operation and planning functions. A connected industrial site has its own equipment, controls and responsibilities. These roles meet at defined physical and information interfaces.

USE IT CORRECTLY

For a local technical question, identify the relevant asset and team rather than contacting an organisation with a broad startup pitch. A generator issue, a network-connection issue and a server-cooling issue need different evidence and specialists. The current status of specific projects can change; this book does not depend on a particular announcement.

WORKED EXAMPLE / HYPOTHETICAL

Question: "When a remote setpoint times out on this controller, what evidence distinguishes an unexecuted command from a lost reply?" This names a technical failure mode without claiming that anyone's present system is defective.

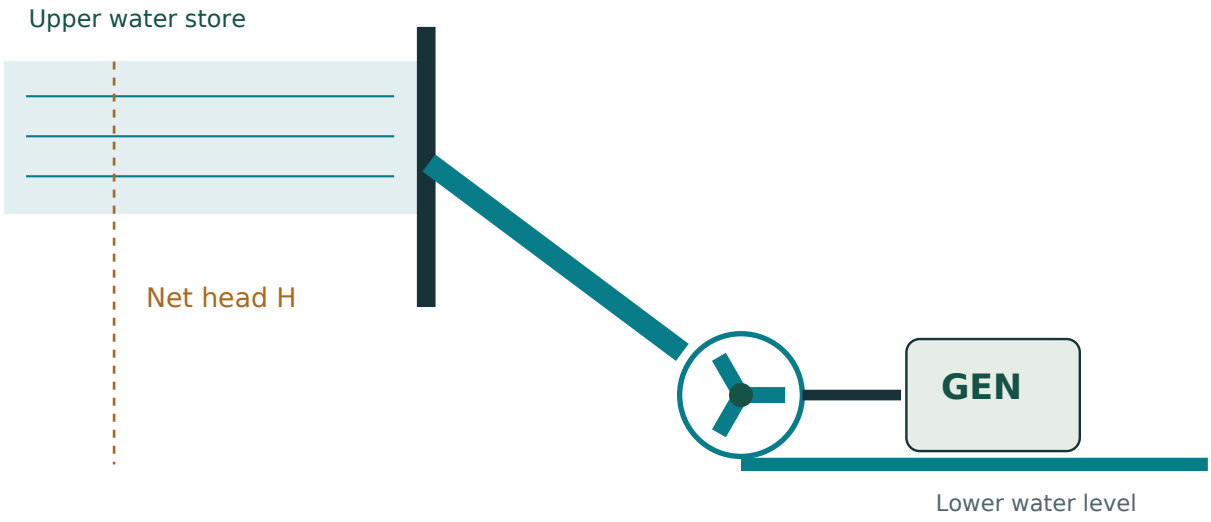
CHECK YOUR UNDERSTANDING

Who would usually own the server's internal fan-control design?

Answer: 132 | Sources: S2-S5

Hydro power comes from head and flow

Water loses gravitational potential as it moves through a usable height difference.



Head and flow supply hydraulic power. The diagram is conceptual.

HOW IT WORKS

A useful estimate is P = ρgQHη. Water density ρ is about 1,000 kg/m³; g is about 9.81 m/s²; Q is volume flow in m³/s; H is net head in metres; η is the combined efficiency. Net head accounts for the usable height after relevant hydraulic losses.

USE IT CORRECTLY

The turbine converts hydraulic power into shaft power; the generator converts shaft power into electrical power. Actual output depends on operating point and equipment limits. In a cascade, the same water can generate at successive stations because it loses additional height, not because one energy drop is counted repeatedly.

WORKED EXAMPLE / HYPOTHETICAL

Assume Q = 1 m³/s, H = 10 m and η = 0.80. P = 1,000 × 9.81 × 1 × 10 × 0.80 = 78,480 W, about 78.5 kW. One hour at that output gives 78.5 kWh.

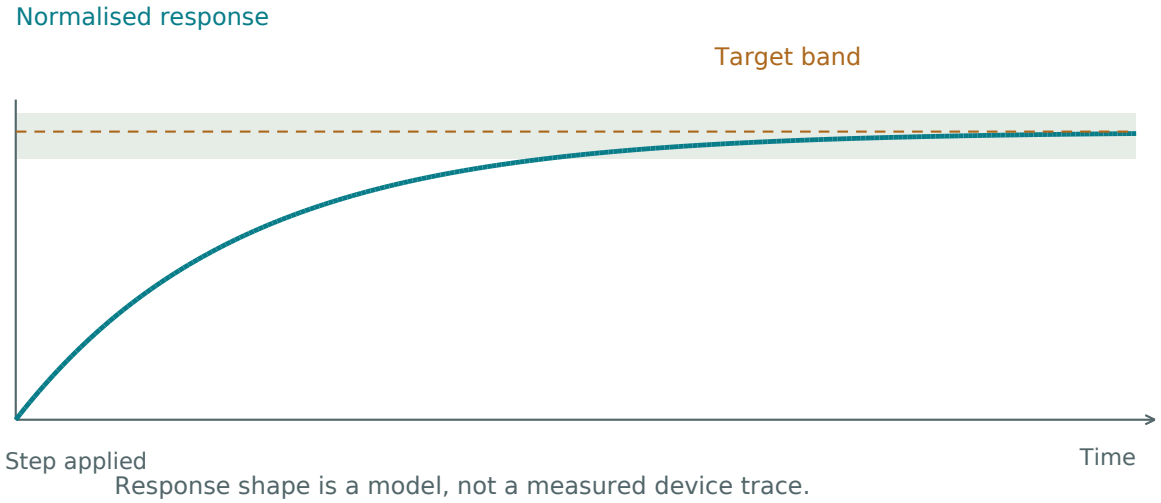
CHECK YOUR UNDERSTANDING

If flow doubles while head and efficiency remain fixed, what does the model predict?

Answer: 132 | Sources: B2; S4, S12

Water and turbines have dynamics

A hydro unit cannot be treated as an instantaneous power source.



A physical response takes time; its shape depends on the actual equipment.

HOW IT WORKS

A governor adjusts turbine input to help control speed or power under its operating mode. Excitation controls the generator's magnetic field and interacts with voltage/reactive-power behaviour. Gates, water passages, rotating inertia and electrical conditions all affect response. Different turbine types suit different head and flow ranges.

USE IT CORRECTLY

Rapid flow changes can create pressure transients, including water hammer. Machines also have mechanical limits and possible resonances. A small simulator can teach cause and effect, but plant-specific control or torsional-fatigue claims need actual machine parameters, validated models and specialist review. Do not infer them from regional five-minute data.

WORKED EXAMPLE / HYPOTHETICAL

A model that instantly changes hydro output from 0 to 100 kW ignores water and machine response. Add a stated ramp or dynamic model to study supervisory timing, while keeping clear that this is a teaching approximation.

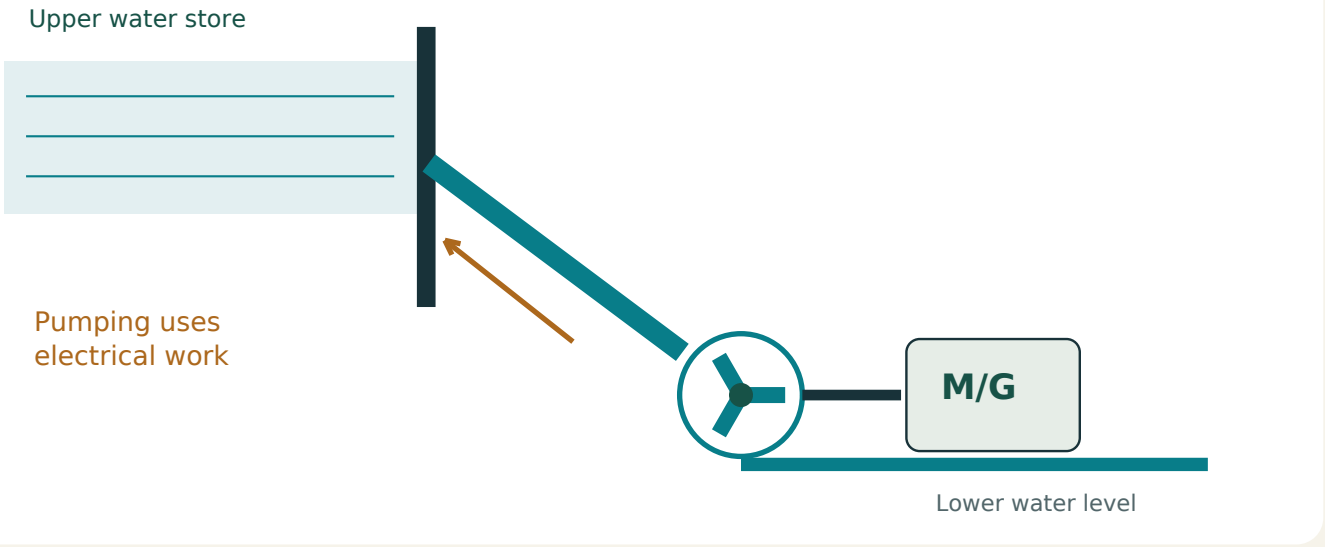
CHECK YOUR UNDERSTANDING

Why should a load-control project care about the time scale of a generator response?

Answer: 132 | Sources: B1, B2

Pumped hydro shifts energy in time

Storage can move energy between periods. It does not generate a net energy surplus over its full cycle.



Charging lifts water; discharging releases part of the stored energy.

HOW IT WORKS

Pumping uses electrical energy to lift water to a higher store. Generating releases some of that potential energy later. Pumping, hydraulic and generating losses make round-trip output smaller than input. Storage value depends on timing and capability, even though the energy balance has a loss.

USE IT CORRECTLY

Power rating defines how quickly a system can pump or generate. Energy capacity describes how long it can sustain a stated output within reservoir and equipment limits. Water availability, head changes, environmental constraints and operating modes all affect practical capability. Separate an existing plant’s verified features from a proposed project.

WORKED EXAMPLE / HYPOTHETICAL

Assume 100 kWh enters a full storage cycle with 75% round-trip efficiency. About 75 kWh is returned within that assumed boundary. A 25 kW discharge would last 3 h if the output stayed constant and other constraints allowed it.

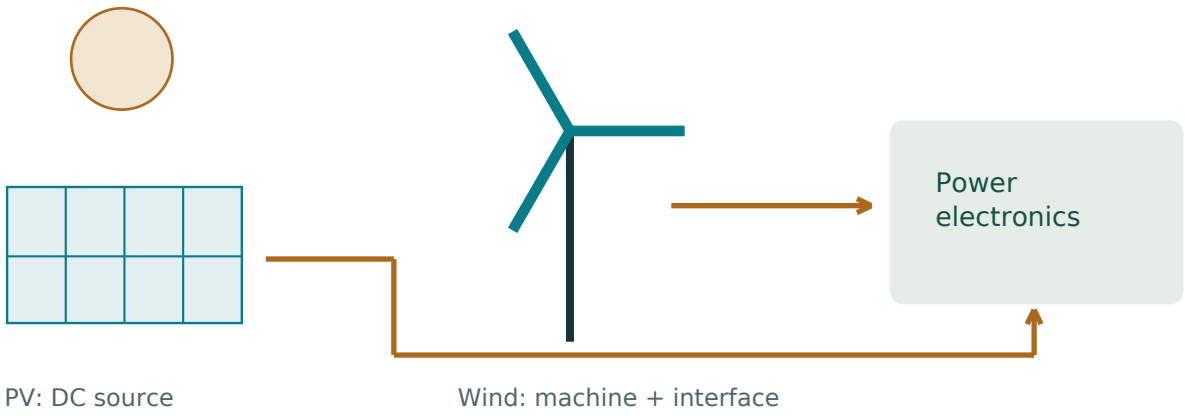
CHECK YOUR UNDERSTANDING

Does a 100 MW pumped-storage power rating tell you its duration?

Answer: 132 | Sources: B1, B2

Wind and solar need interfaces

The renewable energy source and its electrical interface are different parts of the system.



Source availability, conversion capability and grid permission all constrain output.

HOW IT WORKS

Solar photovoltaic cells produce DC whose output depends on conditions and operating point. Power electronics can manage the operating point and convert to AC. A wind turbine converts air-flow energy to mechanical and then electrical energy; many modern designs use converter interfaces, while machine arrangements vary.

USE IT CORRECTLY

Weather availability, equipment rating and grid constraints limit output. A nominal MW rating is not a promise of constant generation. Curtailment is a deliberate restriction below available output. Differentiate a resource shortage, a device fault and a commanded reduction; they can look similar in an aggregate graph.

WORKED EXAMPLE / HYPOTHETICAL

A hypothetical 10 kW PV system produces 6 kW in one interval. That could reflect sunlight, temperature, an inverter limit or a command. Without irradiance, device status and control evidence, the power trace alone does not identify the cause.

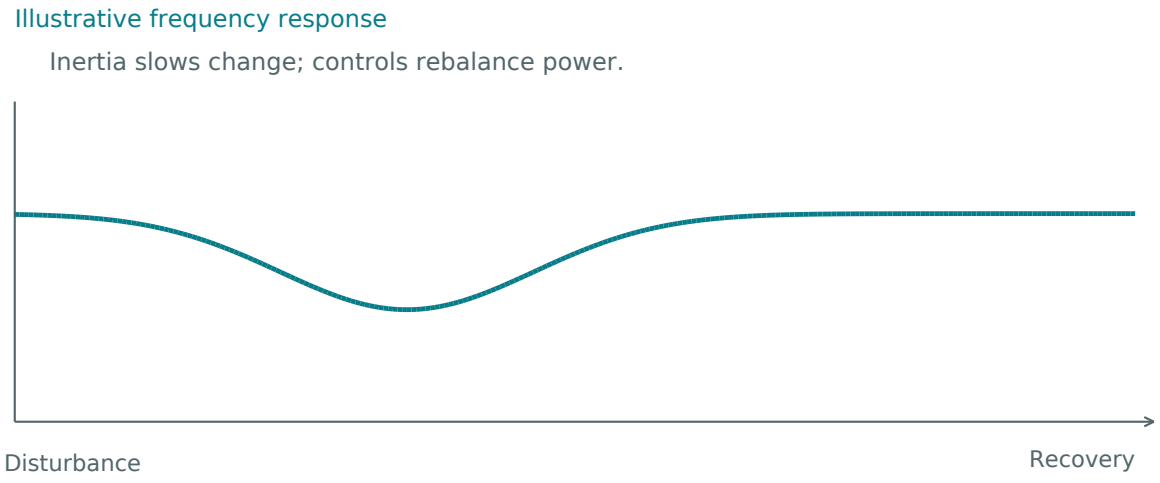
CHECK YOUR UNDERSTANDING

Why can a battery be useful beside variable renewable generation?

Answer: 132 | Sources: B1, B2; S13

Frequency is a system balance signal

Real-power imbalance changes the motion and control response of the connected electrical system.



Power balance, stored rotational energy and controls all shape frequency response.

HOW IT WORKS

A synchronous machine stores rotational energy $E = \frac{1}{2}J\omega^2$. When mechanical input and electrical output differ, rotor speed tends to change. Across a power system, inertia slows rapid frequency change; governors and other frequency controls adjust power. Storage and responsive loads can contribute under suitable controls and conditions.

USE IT CORRECTLY

A frequency-control action needs power capability, energy availability and timing. Five-minute dispatch describes a different time scale from a fast frequency response. Basslink is HVDC: Tasmania is not simply synchronously locked to mainland AC frequency through that link. Converter controls may exchange power to support frequency within their capability.

WORKED EXAMPLE / HYPOTHETICAL

An illustrative 1 MW generation deficit lasting 2 s corresponds to 2 MJ, or about 0.556 kWh. The small energy amount does not make the event trivial: delivering 1 MW promptly is a separate power and control requirement.

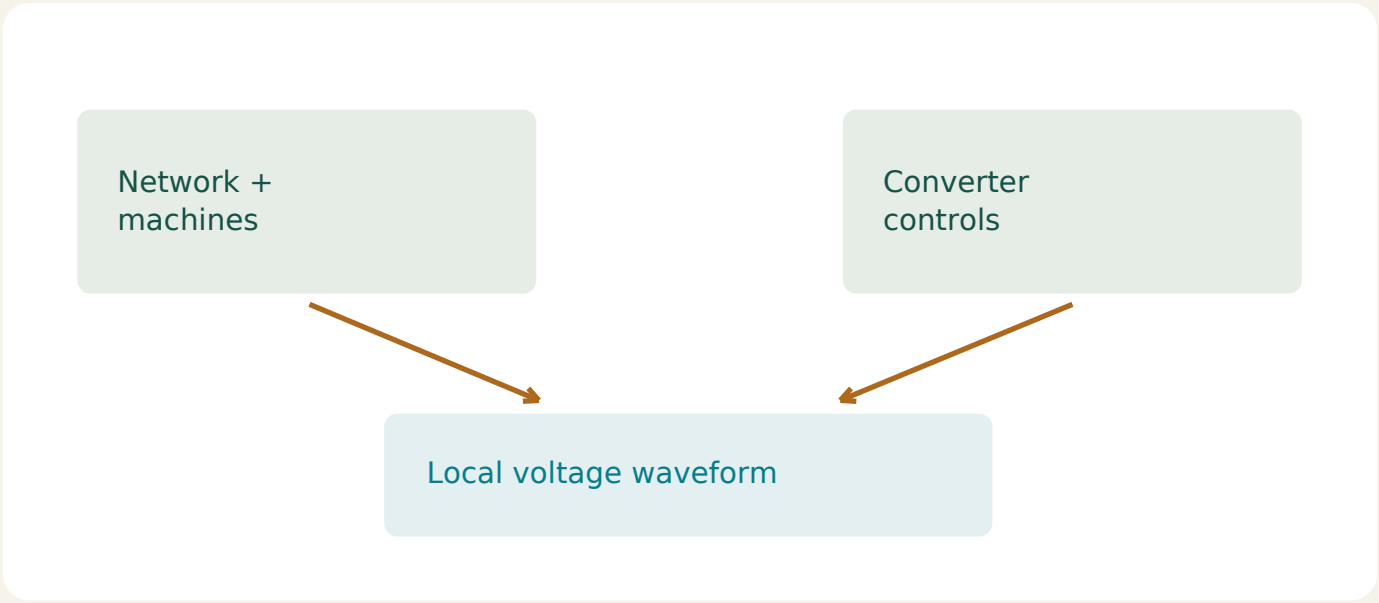
CHECK YOUR UNDERSTANDING

Why is a large battery energy label insufficient to establish fast-frequency capability?

Answer: 132 | Sources: B2; S1, S14

System strength is not inertia

Voltage-waveform behaviour and frequency response are related engineering issues, but they are not the same property.



Voltage stability, waveform quality and frequency cannot be collapsed into one score.

HOW IT WORKS

System strength concerns the ability to maintain and control the voltage waveform at a location, including after disturbances. Inertia concerns energy response associated with changing frequency. A strong-looking average voltage does not establish acceptable dynamic behaviour or inverter interaction under faults.

USE IT CORRECTLY

Synchronous machines, network impedance and power-electronic controls influence local behaviour. Grid-following inverters depend on a voltage reference; grid-forming approaches contribute a controlled voltage reference, subject to limits. Short-circuit strength is a useful indicator in some contexts, but no single number replaces a suitable stability study.

WORKED EXAMPLE / HYPOTHETICAL

Two sites can have the same normal voltage and different fault levels or dynamic interactions. A dashboard showing only average voltage cannot demonstrate that either site meets all connection-performance requirements.

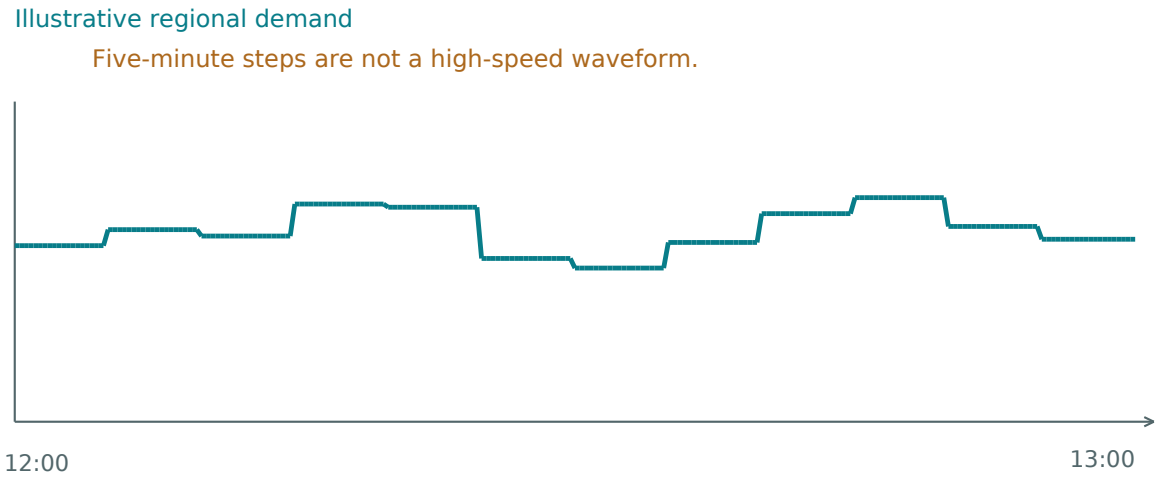
CHECK YOUR UNDERSTANDING

What kind of evidence is needed for a claim about inverter interaction at a specific connection?

Answer: 132 | Sources: S14

A connection is a technical interface

Available energy, network transfer capability and a site’s permission to connect are separate questions.



Regional data is context, not a substitute for device telemetry.

HOW IT WORKS

An engineering connection assessment can involve load/generation characteristics, fault levels, protection, voltage performance, reactive capability, harmonics and dynamic response. The applicable requirements depend on the actual connection and equipment. A successful small prototype does not establish infrastructure compliance.

USE IT CORRECTLY

Use public AEMO data to learn regional behaviour: demand, generation and interconnector flow, with timestamps and field definitions. Public five-minute data can reveal broad patterns. It cannot show every site’s internal state or prove a sub-second event. Record region identifiers, units, interval definitions and any revised records.

WORKED EXAMPLE / HYPOTHETICAL

For a one-hour regional plot, compare successive five-minute demand values. Label them according to the data definition; do not call them continuous high-speed measurements or attribute a change to one data centre without supporting evidence.

CHECK YOUR UNDERSTANDING

Can a public TAS1 demand graph reveal whether one battery obeyed a command?

Answer: 132 | Sources: S3, S5

Storage and continuity

Separate energy capacity from power capability, and supervisory software from local protection.

G Batteries, converters and resilient supply

EIGHT IDEAS TO UNDERSTAND

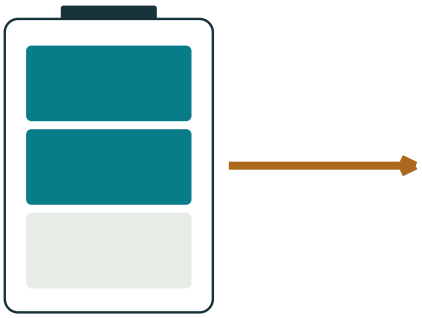
49	A battery has both W and Wh limits	62
50	State of charge is an estimate	63
51	A BMS enforces battery boundaries	64
52	Converters change electrical form	65
53	Grid-following and grid-forming differ	66
54	A microgrid needs a defined boundary	67
55	A UPS is a designed continuity path	68
56	Redundancy can share a common failure	69

Before you move on

Explain one mechanism without looking, solve one example with units and identify one limit of the model. Use the answer section to correct your reasoning, then keep your corrected sketch.

A battery has both W and Wh limits

Power tells you how quickly energy can leave. Energy capacity helps determine how long it can continue.



ENERGY: kWh
POWER: kW
DURATION: h

Duration ≈ usable energy / output power

Energy capacity and instantaneous power capability are separate constraints.

HOW IT WORKS

A battery system stores electrochemical energy and supplies DC through its terminals and controls. Rated capacity depends on specified conditions. Usable energy can be lower because of permitted state-of-charge range, conversion losses, temperature, age and discharge rate. Internal resistance also causes heating and voltage drop.

USE IT CORRECTLY

Do not confuse ampere-hours with watt-hours. An energy estimate from Ah needs an appropriate voltage model; battery voltage changes during use. For early calculations, use a clearly stated average voltage or, preferably, a manufacturer’s energy specification. No loose-cell or opened-pack experiments are included in this book.

WORKED EXAMPLE / HYPOTHETICAL

An assumed 1 kWh system with a 0.5 kW output limit has an ideal full-capacity duration of 2 h at that limit. If usable fraction is 0.8 and discharge-path efficiency 0.9, duration becomes $1 \times 0.8 \times 0.9/0.5 = 1.44$ h.

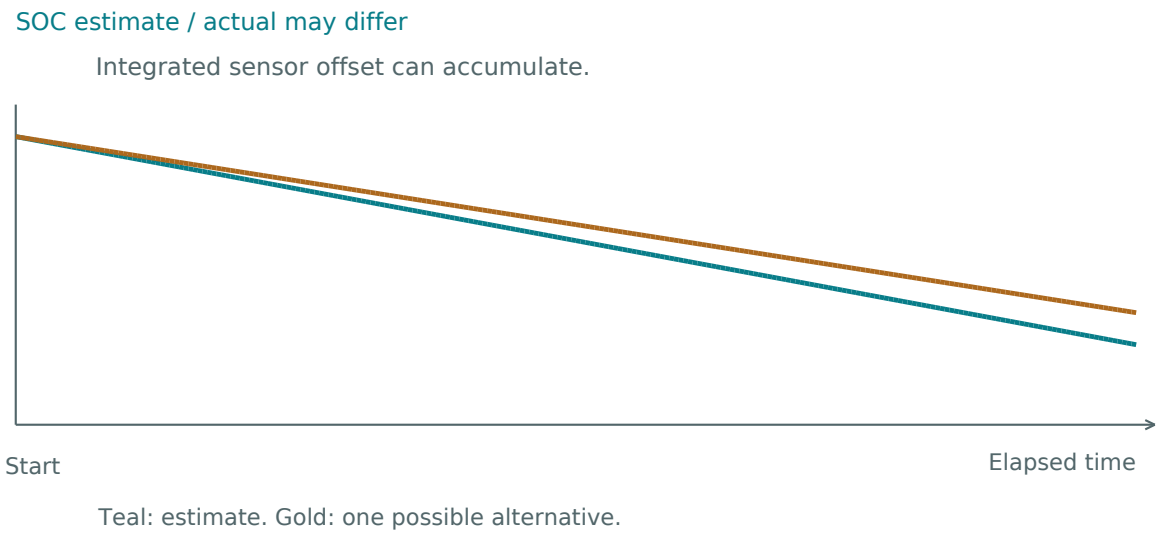
CHECK YOUR UNDERSTANDING

Can a 1 kWh battery necessarily deliver 10 kW for six minutes?

Answer: 133 | Sources: B1, B2

State of charge is an estimate

A displayed percentage is produced by a model and measurements. It is not a direct view inside the chemistry.



An estimator carries assumptions, history and uncertainty.

HOW IT WORKS

State of charge, SOC, estimates how much charge remains relative to a defined capacity. Coulomb counting integrates current over time; errors can accumulate. Voltage-based estimates depend on chemistry, temperature, load and relaxation. Real systems may combine methods and update capacity estimates as the battery ages.

USE IT CORRECTLY

State of health describes changes in capability, not one universal number. Reduced capacity and increased internal resistance can matter differently. Record whether an SOC value is measured, estimated or supplied by a device algorithm. Lupine should respect the battery controller’s operating limits rather than overriding them using its own simple estimate.

WORKED EXAMPLE / HYPOTHETICAL

If a model starts at 100% on an assumed 10 Ah capacity and records a net 1 Ah discharge, its simple estimate is 90%. Current-sensor bias, initial SOC error and changing capacity can make the actual state different.

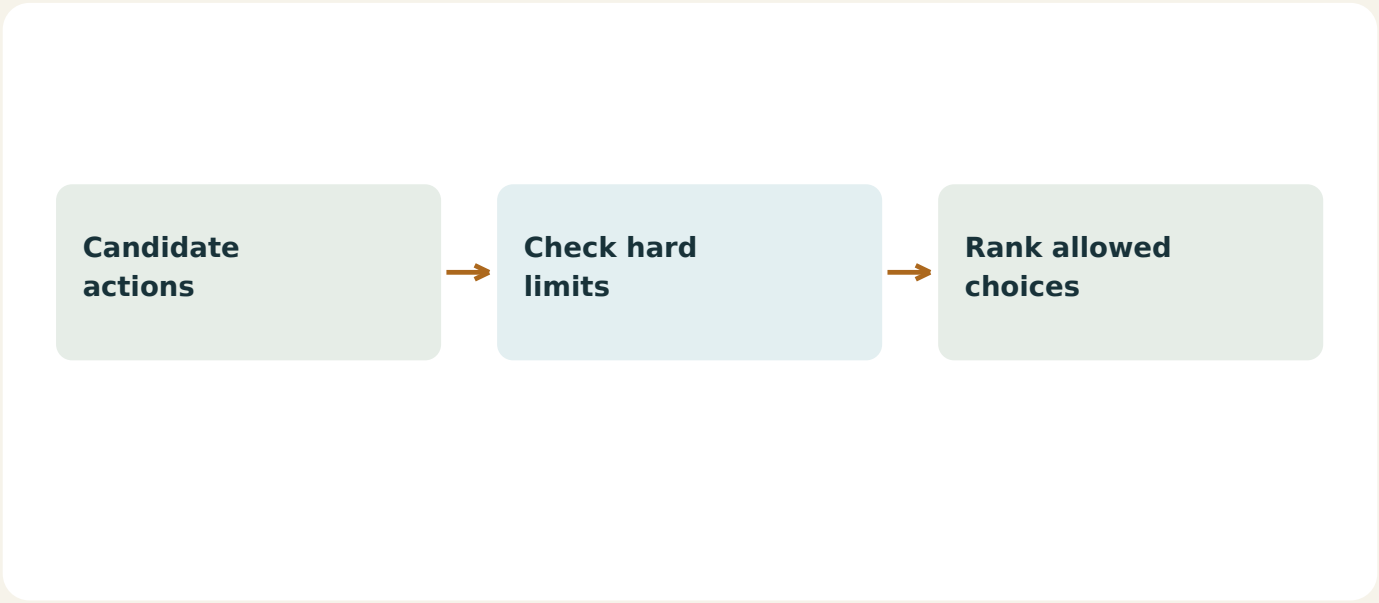
CHECK YOUR UNDERSTANDING

Why can a small current-sensor offset matter over many hours?

Answer: 133 | Sources: B1, B2

A BMS enforces battery boundaries

A battery-management system monitors the pack and manages operation within its design limits.



Hard constraints are evaluated before choosing an action.

HOW IT WORKS

A BMS can monitor cell voltages, pack current and temperatures, estimate battery state and control protective switching. Its exact functions depend on the product. Cell balancing addresses differences among cells; it does not make every damaged or unsuitable pack safe.

USE IT CORRECTLY

A supervisory energy controller requests allowed actions at a higher level. It must handle a BMS refusing or limiting an action, including changed available power. A recorded request is not permission to bypass protection. Pack design, failure management and certification are specialist work outside these home projects.

WORKED EXAMPLE / HYPOTHETICAL

Lupine requests 500 W discharge, but the battery reports only 200 W presently available. Treat 200 W as a binding capability limit, verify the units and status, and preserve the reason. Do not score the limit against a preferred outcome.

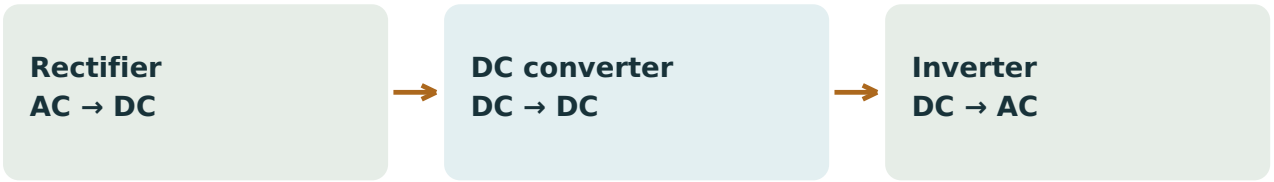
CHECK YOUR UNDERSTANDING

Should a supervisor repeatedly force a rejected command until the BMS accepts it?

Answer: 133 | Sources: B1, B2

Converters change electrical form

Rectifiers, DC-DC converters and inverters are different links in an energy path.



Conversion changes form and operating conditions, with losses.

HOW IT WORKS

A rectifier converts AC to DC. A DC-DC converter changes a DC level or regulates a DC output. An inverter produces AC from DC. Switching devices and energy-storage components shape the conversion. Control loops manage the output, while real losses become heat.

USE IT CORRECTLY

Ratings include voltage ranges, power/current capability and environmental limits. Efficiency varies with load. Isolation is a specific design feature, not a synonym for conversion. Bidirectional capability must be explicit; a converter that supplies a load is not automatically able to absorb power from it.

WORKED EXAMPLE / HYPOTHETICAL

An assumed converter delivers 20 W at 90% efficiency. Input is 20/0.9 = 22.22 W; loss is 2.22 W. At 5 V input, the implied input current is 4.44 A, which exceeds this book’s small bench scope.

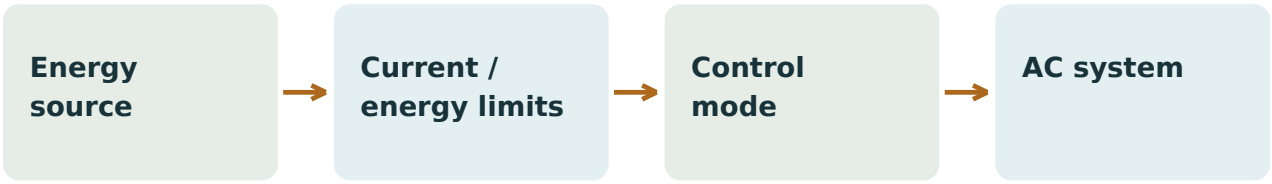
CHECK YOUR UNDERSTANDING

Why might a source be unable to absorb regenerative power from a motor?

Answer: 133 | Sources: B1, B2

Grid-following and grid-forming differ

An inverter’s control behaviour matters as much as its power rating.



Control mode and physical capability both matter.

HOW IT WORKS

A grid-following inverter generally synchronises to an existing voltage waveform and controls injected current. Grid-forming control establishes a controlled voltage reference and responds through its defined dynamics. These labels describe control approaches; they do not remove current, energy or hardware limits.

USE IT CORRECTLY

Grid-forming equipment is not automatically able to black-start any network, supply arbitrary fault current or operate stably with every other controller. System behaviour depends on tuning, protection, source capability and connection conditions. A valid claim needs the specific implementation and tests, not only a product category.

WORKED EXAMPLE / HYPOTHETICAL

A battery inverter can be rated 1 MW while its current limit constrains response during a voltage disturbance. A model that treats it as an unlimited ideal voltage source will miss an important physical boundary.

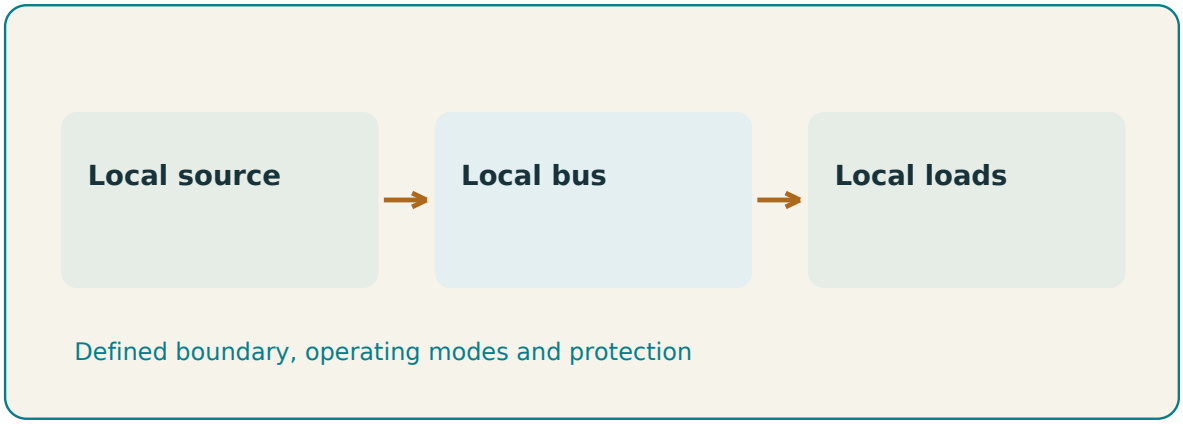
CHECK YOUR UNDERSTANDING

Does “grid-forming” mean unlimited support during a disturbance?

Answer: 133 | Sources: B2; S13, S14

A microgrid needs a defined boundary

A group of local sources and loads is not automatically an islandable microgrid.



The electrical boundary and permitted operating modes must be explicit.

HOW IT WORKS

A microgrid coordinates resources and loads within a defined electrical boundary. Some designs can disconnect from the wider network and maintain an islanded supply; others cannot. Islanding needs suitable voltage/frequency control, power balance, protection and a deliberate transition and reconnection design.

USE IT CORRECTLY

A supervisory controller may schedule resources, but local controls and protection must handle faster physical behaviour. Loads can have different priorities and interruption tolerances. The supervisory layer must know which actions are permitted in each operating mode. A home fan demonstration reproduces only a small control pattern, not a working islanded power system.

WORKED EXAMPLE / HYPOTHETICAL

If an island has 5 kW of available supply and 7 kW of demand, something must change: demand, generation, storage output or operating state. Software cannot resolve the deficit by assigning a higher confidence score.

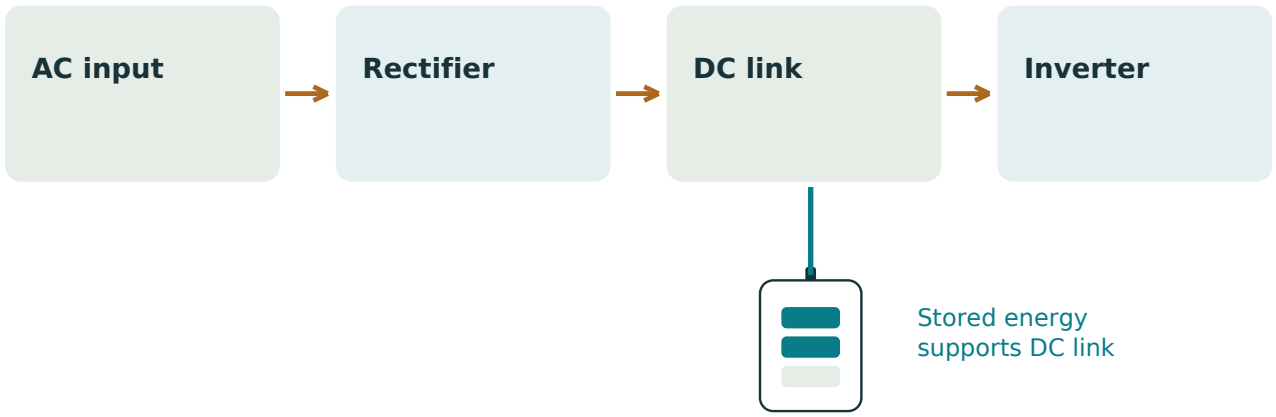
CHECK YOUR UNDERSTANDING

What extra questions arise before reconnecting an island to an AC grid?

Answer: 133 | Sources: B1, B2

A UPS is a designed continuity path

An uninterruptible power supply bridges disturbances within its specific topology and capability.



A conceptual double-conversion path; real bypass and protection details vary.

HOW IT WORKS

UPS designs include standby, line-interactive and double-conversion arrangements. In double conversion, power normally passes through AC-DC and DC-AC stages; stored energy supports the DC link as designed. Other topologies have different normal paths and transfer behaviour. Do not assume every UPS provides the same disturbance response.

USE IT CORRECTLY

Runtime depends on load, storage condition, efficiency and cutoff limits. A VA rating and a W rating address different constraints. Bypass and maintenance paths can change protection and power quality. The equipment’s actual topology and operating mode must be included in a site diagram.

WORKED EXAMPLE / HYPOTHETICAL

A hypothetical UPS has 1 kWh usable stored energy and an assumed 90% discharge-path efficiency. At 500 W output, idealised runtime is 1 × 0.9/0.5 = 1.8 h. Actual manufacturer runtime curves may differ substantially.

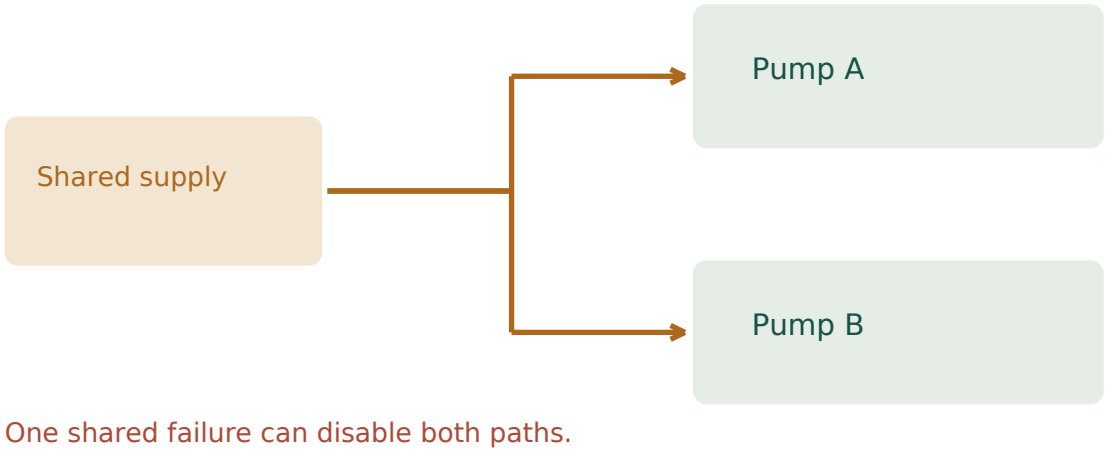
CHECK YOUR UNDERSTANDING

Does “battery present” guarantee uninterrupted supply to every load?

Answer: 133 | Sources: B1, B2

Redundancy can share a common failure

Two components do not provide two independent paths if one fault can disable both.



Count independent end-to-end paths, not only duplicate boxes.

HOW IT WORKS

Redundancy supplies extra capacity or alternative paths. N+1 means one extra unit beyond the required number under a defined design condition. It does not automatically cover shared cooling, control, wiring, fuel, protection or maintenance failures. Trace the complete dependency chain.

USE IT CORRECTLY

Reliability concerns successful function over a specified period and conditions. Availability concerns readiness for use over time. In a simple repairable steady-state model, availability ≈ MTBF/(MTBF + MTTR). That model has assumptions and does not establish the probability of a particular dangerous event.

WORKED EXAMPLE / HYPOTHETICAL

Two pumps each have a separate motor but share one controller and one power feed. Controller or feed failure may stop both. Drawing the common dependencies can be more useful than counting pumps.

CHECK YOUR UNDERSTANDING

What is a common-mode failure in a dual-sensor setup?

Answer: 133 | Sources: B1, B2

Follow the heat

Energy does not vanish inside a server. Learn where heat goes and how to measure its journey.

H Thermal systems and data-centre cooling

EIGHT IDEAS TO UNDERSTAND

57 Heat and temperature are different 71

58 Thermal resistance links heat to rise 72

59 Flow can carry thermal power 73

60 Pumps pay for pressure and flow 74

61 Air cooling needs a managed path 75

62 Liquid cooling still needs heat rejection 76

63 PUE measures a specific ratio 77

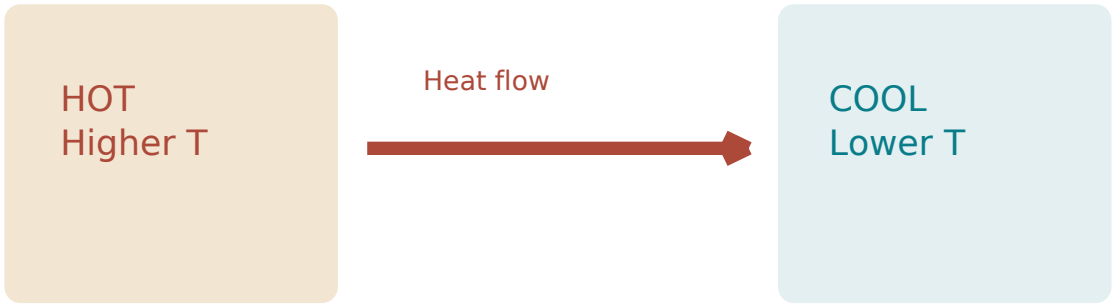
64 Heat reuse needs the right temperature 78

Before you move on

Explain one mechanism without looking, solve one example with units and identify one limit of the model. Use the answer section to correct your reasoning, then keep your corrected sketch.

Heat and temperature are different

Temperature describes thermal state. Heat is energy transferred because of a temperature difference.



Temperature difference drives passive heat transfer.

A temperature reading is not itself an energy or power reading.

HOW IT WORKS

Energy enters a body, changes its internal energy or leaves by heat transfer and other work. For a simple sensible-heating model, $E = mc\Delta T$. Here m is mass, c is specific heat capacity and ΔT is temperature change. Phase changes and varying properties require a fuller model.

USE IT CORRECTLY

Conduction transfers energy through material; convection involves fluid motion and heat exchange; radiation transfers energy electromagnetically. These mechanisms can act together. A hot chip temperature does not alone reveal its heat-transfer rate, and a cool surface does not prove the absence of a hot internal junction.

WORKED EXAMPLE / HYPOTHETICAL

Assume 0.10 kg of water, c = 4,180 J/(kg·K), and a 5 K rise with no losses. $E = 0.10 \times 4{,}180 \times 5 = 2{,}090$ J. At 10 W net heating, the idealised time is 209 s.

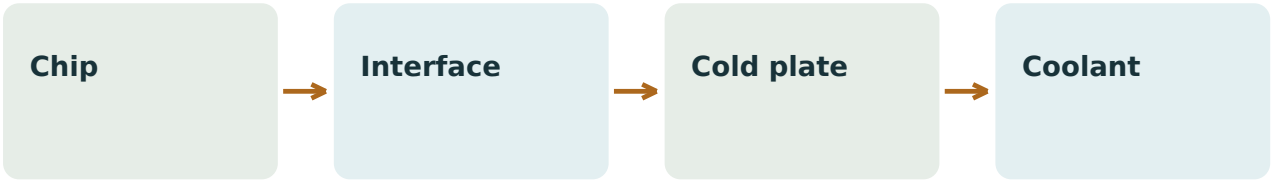
CHECK YOUR UNDERSTANDING

Does a 5 °C temperature difference have the same size as 5 K?

Answer: 134 | Sources: B1, B2

Thermal resistance links heat to rise

A simplified steady thermal path can be analysed using temperature difference divided by heat flow.



Each thermal resistance contributes $\Delta T = P \times R\theta$.

Every interface can add temperature rise; transient storage is a separate effect.

HOW IT WORKS

Thermal resistance $R_\theta = \Delta T/P$, in K/W, describes a stated path and conditions. A chip-to-coolant path includes interfaces, materials and convection. Series thermal resistances add in a simple one-dimensional steady model. Actual heat spreading and changing fluid conditions can be more complex.

USE IT CORRECTLY

Thermal mass stores energy and makes temperature respond slowly. A sensor may lag further because of its mounting and response time. Do not use a steady resistance model to claim a device can tolerate an arbitrary short peak. Junction temperature limits and transient thermal behaviour are device-specific.

WORKED EXAMPLE / HYPOTHETICAL

At an assumed 20 W through a 1.5 K/W path, steady temperature rise is 30 K above the reference boundary. If coolant is 25 °C, the model predicts 55 °C at the upstream point, within the model’s assumptions.

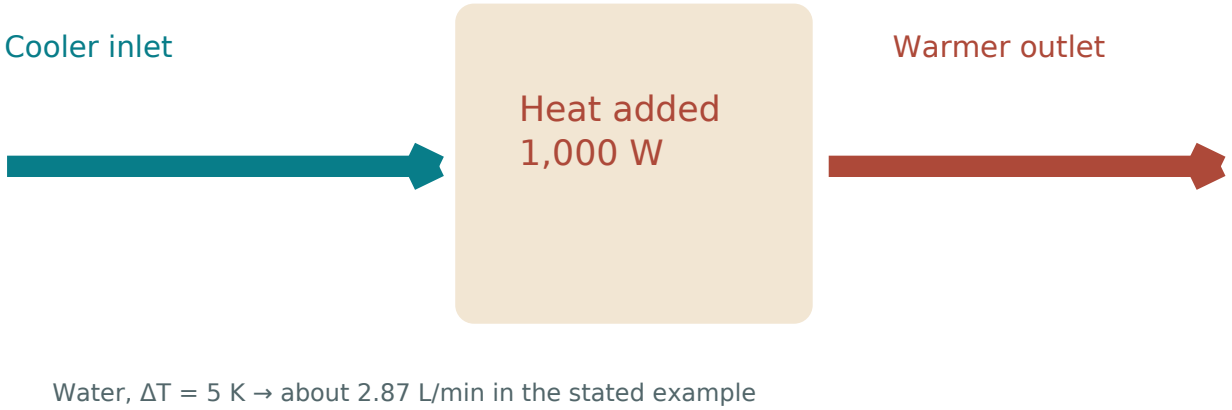
CHECK YOUR UNDERSTANDING

What happens in that model if thermal resistance doubles at the same heat flow?

Answer: 134 | Sources: B1, B2

Flow can carry thermal power

A fluid can remove heat when its temperature rises as it passes the equipment.



Measure both flow and temperature difference; neither alone establishes heat removal.

HOW IT WORKS

For steady sensible heat transport, thermal power $P = \dot{m} c_p \Delta T$. Mass flow $\dot{m}$ is kg/s; specific heat capacity c_p is J/(kg·K). For a liquid, mass flow equals density times volume flow. The formula requires suitable inlet/outlet measurements and a defined boundary.

USE IT CORRECTLY

A small measured temperature difference can have large relative uncertainty. Sensor matching, mounting, mixing and time alignment matter. The result describes heat transferred to that fluid stream, not automatically all electrical power entering a facility. Heat can leave through other paths or accumulate in equipment.

WORKED EXAMPLE / HYPOTHETICAL

For water carrying 1,000 W with a 5 K rise, $\dot{m} = 1{,}000/(4{,}180 \times 5) = 0.0478$ kg/s. At an assumed density of 1 kg/L, this is about 2.87 L/min. This is a calculation, not a prescribed flow for a real cold plate.

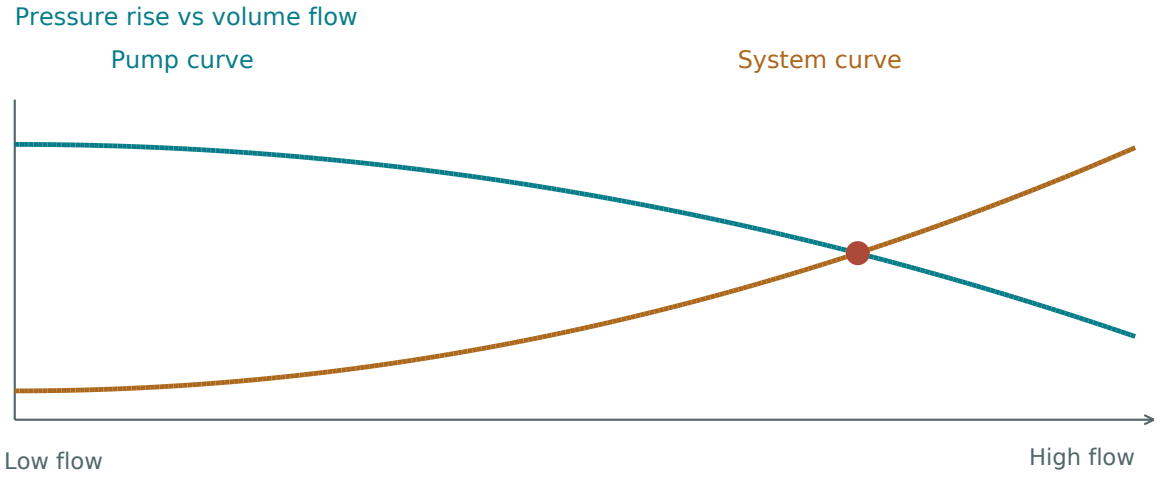
CHECK YOUR UNDERSTANDING

At equal thermal power, what happens to required water flow if permitted ΔT halves?

Answer: 134 | Sources: B1, B2

Pumps pay for pressure and flow

A pump must overcome a system’s pressure losses while supplying the needed flow.



The pump curve and system curve jointly set the operating point.

HOW IT WORKS

Hydraulic power is $P_{hyd} = \Delta pQ$, where pressure rise Δp is in pascals and volume flow Q in m^3/s. Electrical input is higher because of pump and motor losses. The operating point is where the pump characteristic and the system resistance curve meet.

USE IT CORRECTLY

More pump speed does not simply guarantee useful cooling: restrictions, bypasses, trapped gas, cavitation and control behaviour can change the result. Use the actual pump curve and fluid properties. Pressure, flow and temperature are different measurements. Do not build pressurised or mains-powered cooling equipment for the projects here.

WORKED EXAMPLE / HYPOTHETICAL

Assume $\Delta p = 100{,}000$ Pa and $Q = 0.0001$ m^3/s. Hydraulic power is 10 W. If combined pump/motor efficiency is 50%, electrical input is 20 W. The flow is 0.1 L/s, or 6 L/min.

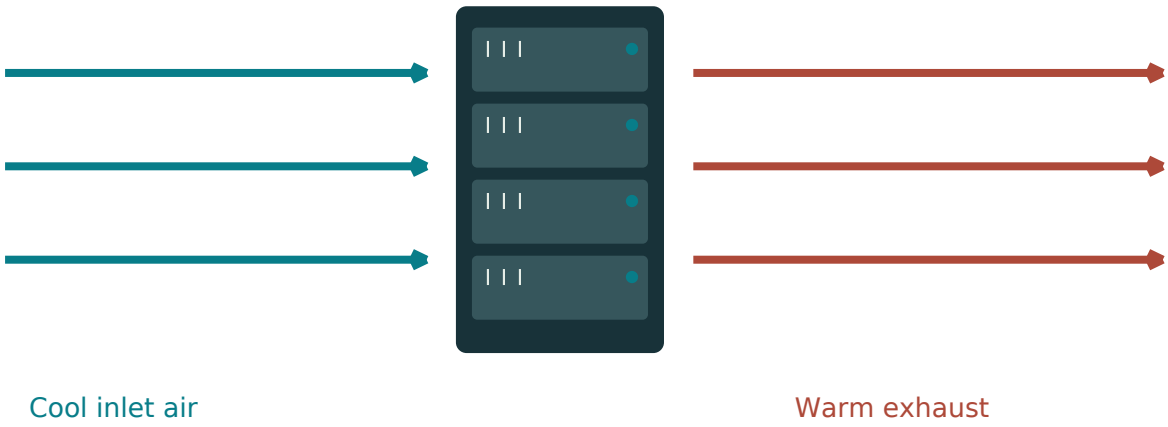
CHECK YOUR UNDERSTANDING

Can pressure alone establish whether sufficient coolant reaches a cold plate?

Answer: 134 | Sources: B1, B2

Air cooling needs a managed path

Moving air is useful only if it carries heat from the intended equipment to the intended exit.



Separate cool inlet and warm exhaust paths; avoid recirculation.

HOW IT WORKS

Server fans move air through heat-producing parts. Facility air handling supplies appropriate inlet conditions and removes exhaust heat. Recirculation can send hot exhaust back to inlets; bypass flow can avoid the useful cooling path. Containment and good airflow management help separate paths.

USE IT CORRECTLY

A fan’s free-air flow rating does not describe its flow through a restrictive server. Flow depends on pressure resistance and the fan curve. Use inlet temperatures and component telemetry appropriately; room averages may hide local hot spots. More fan power can increase airflow while also adding heat and electrical consumption.

WORKED EXAMPLE / HYPOTHETICAL

Using assumed air c_p = 1,000 J/(kg·K), 1 kW carried with a 10 K rise needs 0.10 kg/s of air. At 1.2 kg/m³, volume flow is about 0.083 m³/s. This ignores leakage and other heat paths.

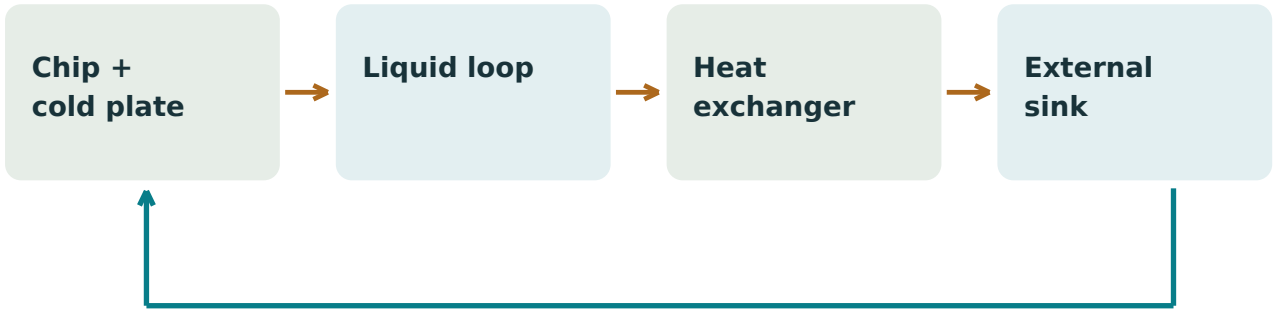
CHECK YOUR UNDERSTANDING

Why can an average room temperature look acceptable while one rack overheats?

Answer: 134 | Sources: B1, B2; S15

Liquid cooling still needs heat rejection

A cold plate moves heat into liquid. The facility must then move that heat to an external sink.



Return path; pumps consume energy too.

Chip → cold plate → fluid loop → heat exchanger → external sink.

HOW IT WORKS

In a direct-to-chip system, thermal interfaces and cold plates transfer heat to a liquid loop. A coolant distribution unit can separate or regulate loops, depending on design. Heat exchangers, pumps and outside heat-rejection equipment complete the thermal path. Some components may still require air cooling.

USE IT CORRECTLY

Fluid choice, material compatibility, cleanliness, leak detection and serviceability matter. Condensation can occur when a surface is below the surrounding air’s dew point. Liquid at the chip is not a guarantee of lower total energy or zero water use; compare complete system boundaries and operating conditions.

WORKED EXAMPLE / HYPOTHETICAL

If a liquid loop carries 80% of an assumed 10 kW rack heat load, it carries 8 kW and leaves 2 kW for other paths in that simplified balance. A facility design must still account for both portions and auxiliary power.

CHECK YOUR UNDERSTANDING

Does liquid cooling eliminate the need to reject heat outside the building?

Answer: 134 | Sources: B1, B2; S15

PUE measures a specific ratio

Power usage effectiveness compares total facility energy with IT equipment energy over a defined boundary and period.



The facility boundary includes IT and the defined non-IT services.

HOW IT WORKS

PUE = facility energy/IT equipment energy. For the same interval and compatible boundary, a value of 1.20 means 0.20 units of non-IT energy per unit delivered to IT. That overhead can include cooling, power conversion, lighting and other included facility services.

USE IT CORRECTLY

Choose compatible measurement boundaries and periods. A short instantaneous power ratio is not automatically comparable to an annual energy PUE. Lower PUE does not prove more useful computation, better reliability, lower carbon impact or better water performance. Each requires its own metric and evidence.

WORKED EXAMPLE / HYPOTHETICAL

Assume IT energy is 10 kWh and facility total is 12 kWh for the same period. PUE = 1.20. If unused IT work increases IT energy without proportional overhead, the ratio could improve while useful-compute efficiency gets worse.

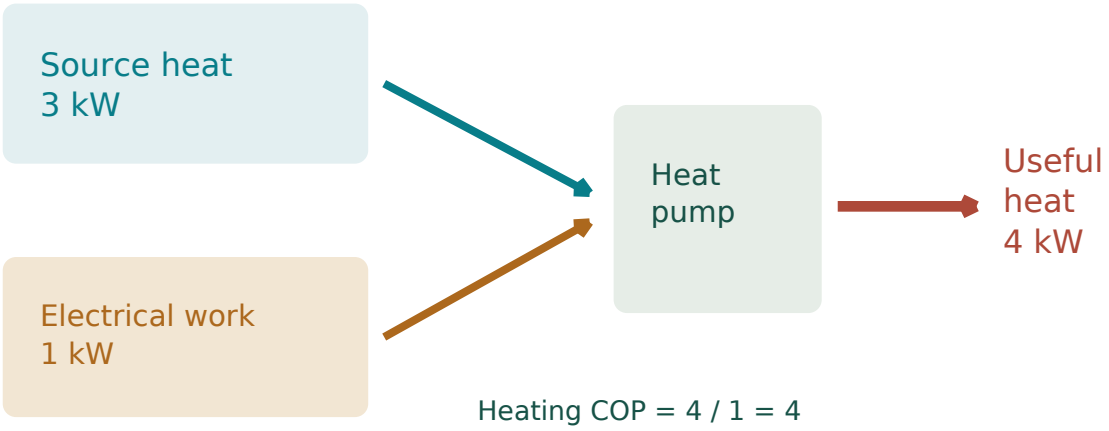
CHECK YOUR UNDERSTANDING

Does a PUE of 1.20 mean the cooling equipment uses exactly 20% of IT energy?

Answer: 134 | Sources: S16

Heat reuse needs the right temperature

Recoverable heat is useful only if a real consumer can accept its temperature, quantity and timing.



Source heat plus electrical work can become higher-temperature useful heat.

HOW IT WORKS

Low-temperature waste heat cannot automatically replace high-temperature heat. Compare the source temperature, consumer requirement, distance, heat losses and coincident demand. A heat pump can raise temperature, but requires additional work and a suitable design. Seasonal mismatch and interruptions can dominate a proposal.

USE IT CORRECTLY

Coefficient of performance, COP, compares heat moved or delivered with electrical work for a defined mode. It can exceed one because the system transfers heat from another source. Keep heating COP and cooling COP distinct. A useful-heat claim needs delivered heat measurements, not only warm outlet water.

WORKED EXAMPLE / HYPOTHETICAL

In an assumed heating case, a heat pump delivers 4 kW while using 1 kW of electricity and taking 3 kW from a source. Heating COP = 4. Energy is conserved: 3 + 1 = 4 kW.

CHECK YOUR UNDERSTANDING

Why does nearby demand matter for a data-centre heat-reuse concept?

Answer: 134 | Sources: B1, B2; S15

Understand the AI factory

Connect useful computation to electricity, networking, cooling and physical constraints.

I Compute, power paths and workloads

EIGHT IDEAS TO UNDERSTAND

65 Compute hardware has a physical job 80

66 Draw the complete rack power path 81

67 Workloads can change power quickly 82

68 The network can be the bottleneck 83

69 A workload is more than a power trace 84

70 Choose metrics that match the claim 85

71 Flexibility has a feasible envelope 86

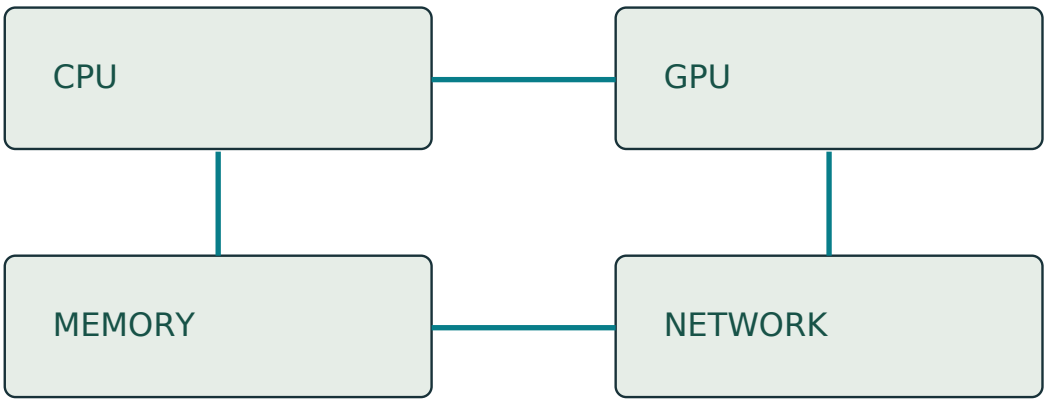
72 Trace failure through the dependencies 87

Before you move on

Explain one mechanism without looking, solve one example with units and identify one limit of the model. Use the answer section to correct your reasoning, then keep your corrected sketch.

Compute hardware has a physical job

A data centre transforms electrical energy into computation and heat. Useful work depends on the task, not just hardware utilisation.



Compute performance is shaped by several interacting resources.

HOW IT WORKS

A CPU handles general-purpose instructions and coordination. A GPU is designed for large amounts of parallel numerical work. Memory holds active data; storage retains datasets and results; networks exchange data between systems. A workload can be limited by compute, memory movement, storage or communication.

USE IT CORRECTLY

Training adjusts a model’s parameters using data. Inference uses a model to produce outputs. Both have quality and performance requirements. High utilisation is not necessarily useful work: repeated failures, redundant computation and low-quality outputs can consume energy without meeting the intended task.

WORKED EXAMPLE / HYPOTHETICAL

Two systems each use 1 kWh. One completes 100 valid tasks and the other 80 under the same quality and latency requirements. Their useful-task efficiencies are 100 and 80 tasks/kWh, despite identical energy use.

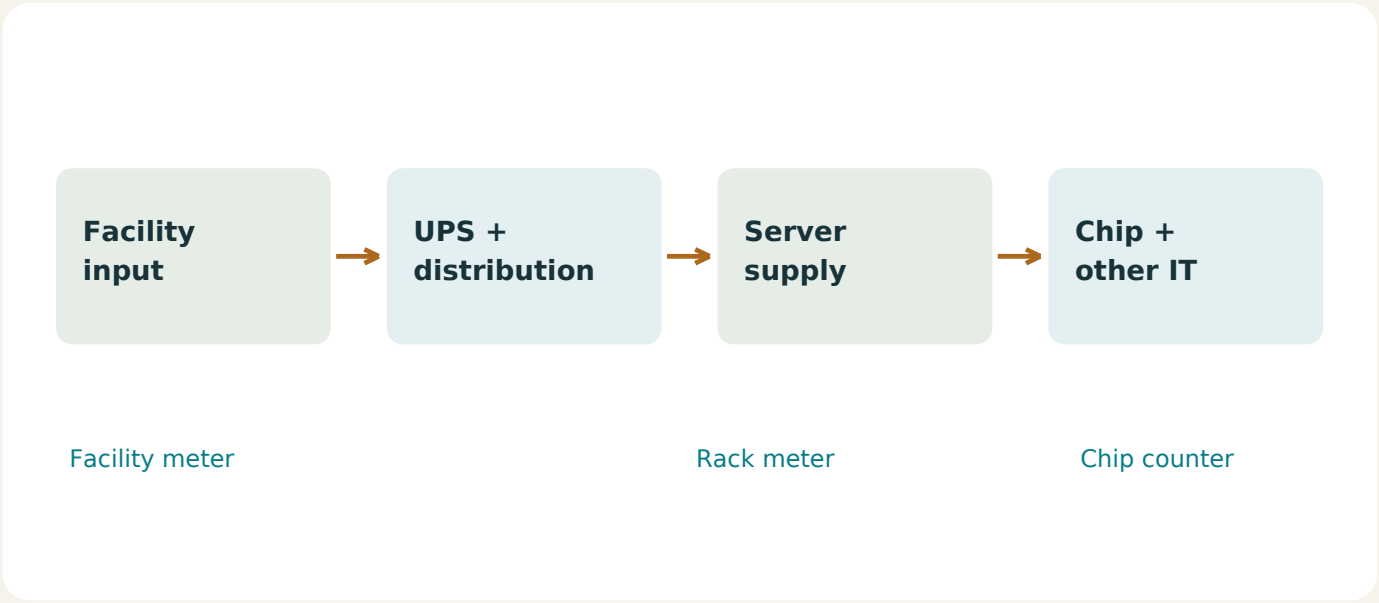
CHECK YOUR UNDERSTANDING

Why is “GPU busy” not a complete efficiency measure?

Answer: 135 | Sources: B1, B2

Draw the complete rack power path

Facility input, rack input and chip power are different measurement boundaries.



A chip counter and a facility meter do not measure the same boundary.

HOW IT WORKS

A typical path can include utility supply, switchgear, UPS, distribution, server power supplies and on-board voltage regulation. Architectures vary, so draw the actual one. Every conversion stage has operating limits and losses. Backup and bypass paths may change where power flows.

USE IT CORRECTLY

A server’s reported package power is not automatically the whole server’s electrical input. Fans, memory, storage, networking and conversion losses may be outside that counter. Record whether telemetry is measured, estimated or vendor-modelled. Compare like boundaries when claiming an improvement.

WORKED EXAMPLE / HYPOTHETICAL

Assume a rack input meter reports 10 kW and summed reported chip power is 7 kW. The difference is 3 kW, but cannot automatically be called “wasted conversion”: it may include useful memory, networking, fans and measurement differences.

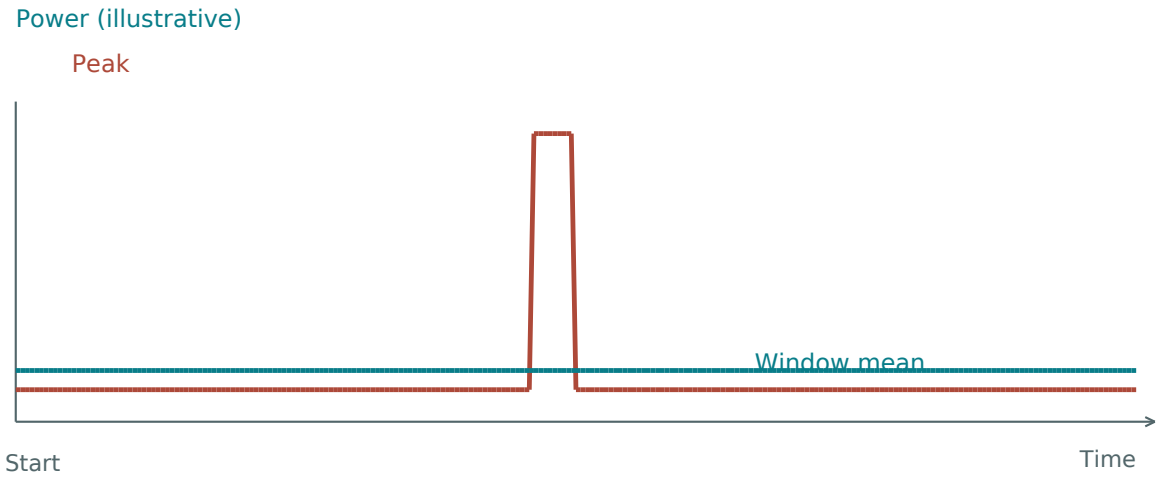
CHECK YOUR UNDERSTANDING

Which boundary would you choose to compare total rack energy for the same workload?

Answer: 135 | Sources: B1, B2; S15

Workloads can change power quickly

Electrical demand follows the behaviour of hardware, software and their control systems.



Peak shape, timing and aggregation affect the upstream demand.

HOW IT WORKS

Computation phases, communication, memory access, synchronisation and checkpointing can produce changing demand. A workload’s average power does not describe its peaks, periodicity or ramp rates. Power caps and local energy storage can alter what reaches the facility supply.

USE IT CORRECTLY

A server trace cannot be scaled blindly to a whole site. Jobs may align or diversify; conversion, storage and control layers can smooth or reshape demand. To study a claimed grid interaction, use appropriate time resolution and representative aggregation. A sinusoidal teaching model is not a measured AI-factory load.

WORKED EXAMPLE / HYPOTHETICAL

Ten identical simulated nodes peaking together create a larger combined peak than ten staggered nodes with the same total energy. The result depends on synchronisation assumptions, so record those assumptions with the plot.

CHECK YOUR UNDERSTANDING

What evidence would you need before attributing a grid oscillation to one workload pattern?

Answer: 135 | Sources: B1, B2

The network can be the bottleneck

A cluster is a connected system. Moving data may limit useful computation even when processors are available.



Bandwidth (rate) ≠ latency (delay)

Communication performance can constrain the useful-compute path.

HOW IT WORKS

Bandwidth is a data-transfer rate, often bits per second. Latency is delay for a particular operation. Throughput is work completed per time under stated conditions. Congestion, retransmission and synchronisation can make a distributed job wait, altering both completion time and electrical demand.

USE IT CORRECTLY

Do not mix bits with bytes: one byte is eight bits. Link speed is not guaranteed application throughput. A job’s communication pattern and data placement matter. The physical network also consumes energy and requires cooling, so it belongs in a complete IT-energy boundary.

WORKED EXAMPLE / HYPOTHETICAL

A hypothetical 8 GB transfer over an ideal 8 Gb/s link takes at least 8 s using decimal units: 8 GB = 64 Gb. Protocol overhead and other traffic make actual time longer. Binary GiB would change the conversion.

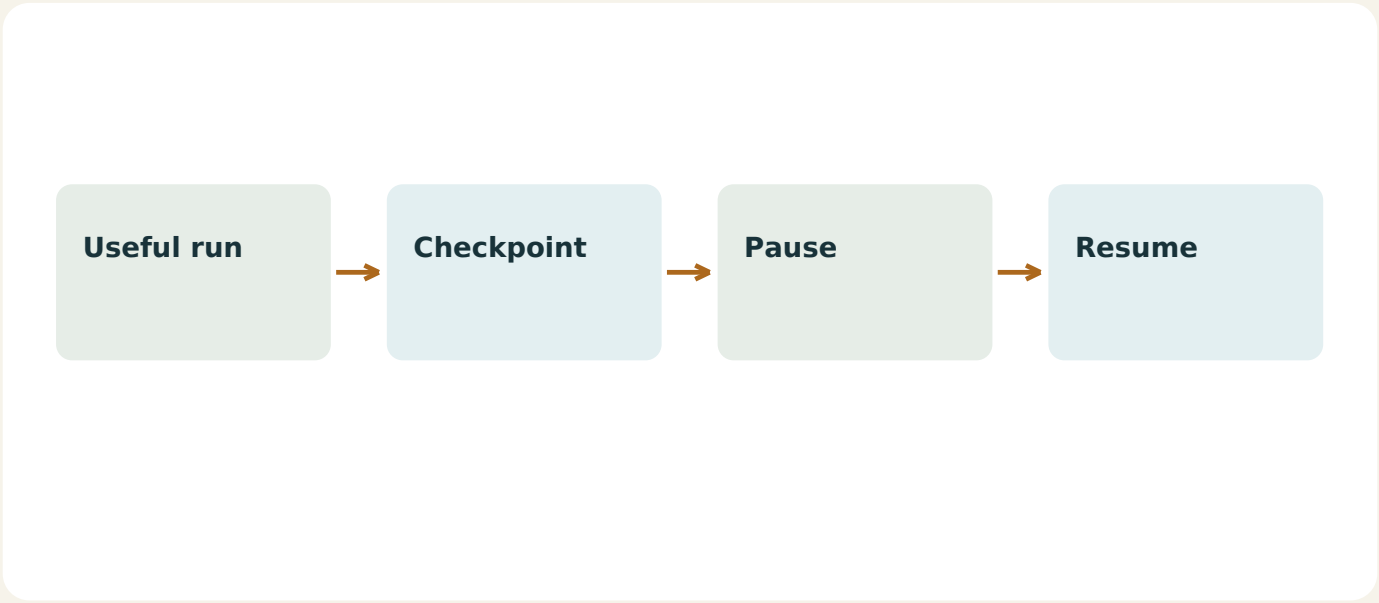
CHECK YOUR UNDERSTANDING

Can a faster GPU alone guarantee a faster distributed training job?

Answer: 135 | Sources: B1, B2

A workload is more than a power trace

To shift or limit computation responsibly, define what the task must still achieve.



A useful-work requirement must survive the energy-control action.

HOW IT WORKS

A batch job may tolerate waiting; an interactive service may have strict latency requirements. Checkpointing saves recoverable progress but consumes time, storage and energy. Stopping and resuming can lose work or create extra computation. The application’s guarantees are part of the physical control problem.

USE IT CORRECTLY

Before proposing flexible compute, define minimum useful output, permitted delay, interruption frequency, recovery conditions and data integrity. A load reduction that breaks the task is not a successful result. Keep the workload owner’s authorised limits separate from a controller’s preferred action.

WORKED EXAMPLE / HYPOTHETICAL

A job paused for 10 min may resume from its latest checkpoint or may restart completely. The same electrical curtailment can therefore cause very different useful-work loss. Record the recovery behaviour instead of assuming every job is freely interruptible.

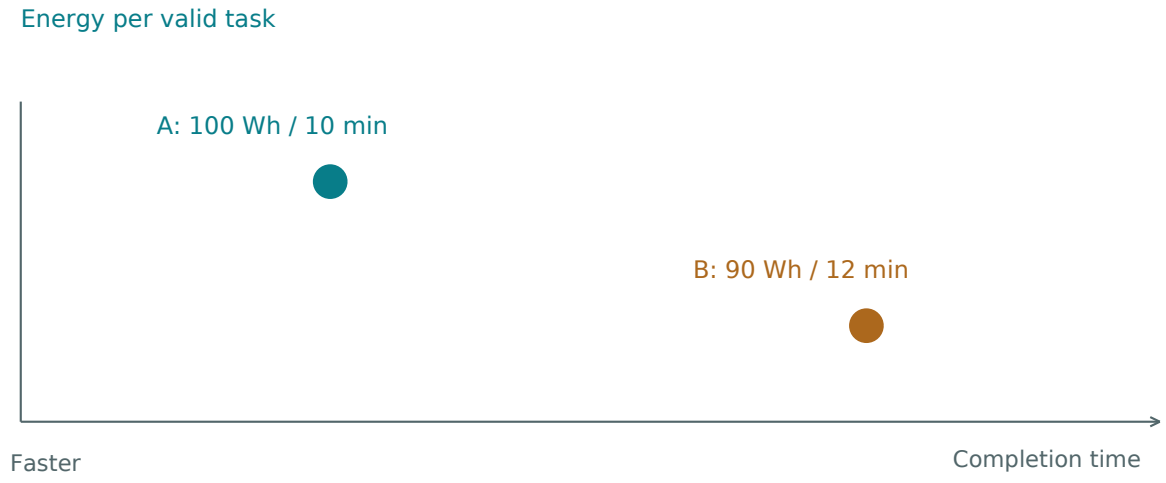
CHECK YOUR UNDERSTANDING

What should be specified before testing a compute power cap?

Answer: 135 | Sources: B1, B2

Choose metrics that match the claim

Energy, performance, temperature and reliability answer different questions.



Compare the full useful-task requirement, not one favourable number.

HOW IT WORKS

For a fixed, valid task, energy per completed task can compare efficiency. Throughput and latency describe performance; temperature describes a thermal condition. Report the task quality and operating environment alongside the metric. An apparent improvement may disappear if the workload or quality changed.

USE IT CORRECTLY

Use matched baseline tests and repeat them. Randomising test order can reduce drift effects. Plot the trade-off rather than forcing everything into one score. A hard temperature or latency requirement must remain a constraint. A few runs can demonstrate a bench effect without establishing general performance.

WORKED EXAMPLE / HYPOTHETICAL

Assume a valid job uses 100 Wh in 10 min in case A and 90 Wh in 12 min in case B. B uses 10% less energy but takes 20% longer. Whether it satisfies the requirement depends on the permitted completion time.

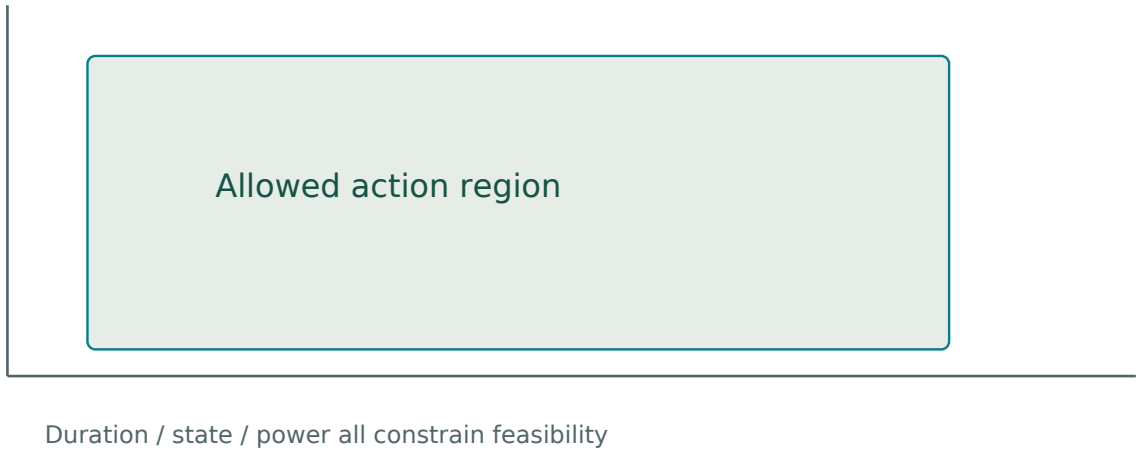
CHECK YOUR UNDERSTANDING

Is lower energy per task always a successful change?

Answer: 135 | Sources: B1, B2

Flexibility has a feasible envelope

A load can provide a controlled change only within physical and application constraints.



Available action depends on state, duration and recovery conditions.

HOW IT WORKS

Define a feasible envelope: minimum and maximum power, ramp limits, permitted duration, recovery behaviour and state-dependent capability. For cooling, stored heat and temperature limits constrain duration. For compute, service requirements and checkpoints matter. Available flexibility changes with current state.

USE IT CORRECTLY

A request to reduce power is not proof of delivered reduction. Device measurements can establish observed behaviour within limits; a claim about reduction relative to what would otherwise have happened needs a credible baseline. Preserve that distinction even if every command and reading is cryptographically recorded.

WORKED EXAMPLE / HYPOTHETICAL

A site drops from 10 kW to 8 kW after a request, but the workload was already scheduled to finish. The observed drop is 2 kW. The portion caused by the request remains unestablished without a suitable comparison.

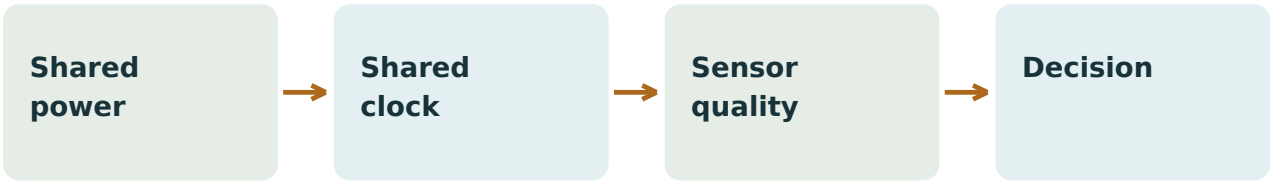
CHECK YOUR UNDERSTANDING

What does a signed instruction plus a lower meter reading prove about causation?

Answer: 135 | Sources: B1, B2

Trace failure through the dependencies

A small component can disable a large function if the system depends on it.



Look for shared dependencies and believable but wrong information.

HOW IT WORKS

A failure-mode analysis asks what can fail, why, how it is detected and what consequence follows. A sensor frozen at a plausible value can be more misleading than a clean disconnect. A shared clock, power supply or configuration can defeat several apparently separate components.

USE IT CORRECTLY

Write one failure at a time, then examine combinations and common causes. Severity, likelihood and detectability are distinct judgements; invented numerical scores should not masquerade as measured probabilities. Use the analysis to choose tests and protective boundaries, not to create a decorative risk table.

WORKED EXAMPLE / HYPOTHETICAL

If a cooling controller trusts a frozen temperature, its command may remain unchanged while the real system heats. Test freshness, plausibility, independent protection and behaviour after loss of measurement. Do not intentionally overheat a physical system to prove the point.

CHECK YOUR UNDERSTANDING

Why might a plausible frozen reading be difficult to detect?

Answer: 135 | Sources: B1, B2

Control the system

Learn why a requested target, a controller output and a physical result are different things.

J Feedback, dynamics and limits

EIGHT IDEAS TO UNDERSTAND

73 Feedback compares target and observation 89

74 Dynamics determine settling time 90

75 Hysteresis prevents constant switching 91

76 PID has three different memories 92

77 Feasibility comes before optimisation 93

78 Separate control time scales 94

79 SCADA observes and commands 95

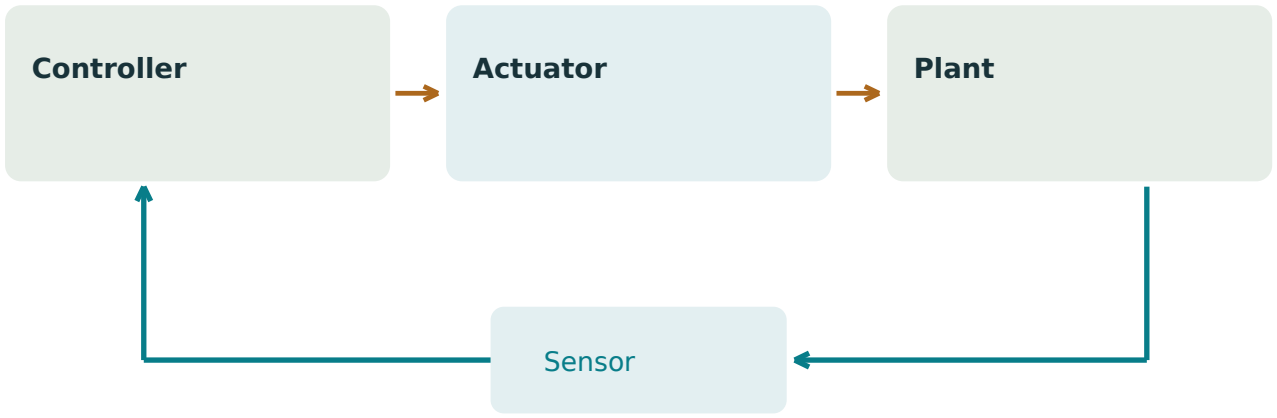
80 A model is not automatically a digital twin 96

Before you move on

Explain one mechanism without looking, solve one example with units and identify one limit of the model. Use the answer section to correct your reasoning, then keep your corrected sketch.

Feedback compares target and observation

A controller changes an input in response to a difference between the requested and observed result.



Target → controller → actuator → plant, with measured feedback.

HOW IT WORKS

The **setpoint** is the target. The **plant** is the physical process. The **actuator** changes an input to that process. The **sensor** observes a chosen output. Error e = target - measured output drives the control rule. State the signs so the controller corrects rather than amplifies error.

USE IT CORRECTLY

The measured variable must match the claim. Fan input power, fan speed and cooling performance are different variables. Open-loop control applies an input without using measured output to correct it; closed-loop control uses feedback. Neither is automatically safe or accurate without an appropriate design.

WORKED EXAMPLE / HYPOTHETICAL

A temperature controller increases fan command when temperature exceeds the target. A speed controller may instead compare measured revolutions per minute. They can use the same fan but solve different control problems.

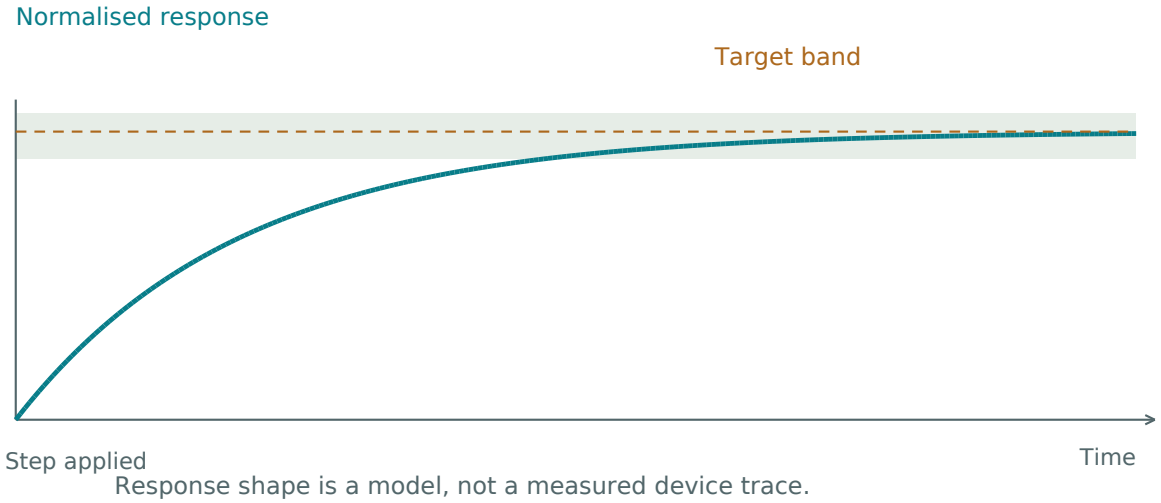
CHECK YOUR UNDERSTANDING

If a controller tracks input current, can it claim to regulate airflow?

Answer: 136 | Sources: B1, B2; S22

Dynamics determine settling time

A physical process takes time to respond. Measurement and communication add their own delays.



The band, entry time and hold requirement define a settling test.

HOW IT WORKS

A simple first-order step response approaches a new steady value exponentially. After one time constant τ, it has completed about 63% of the change; after three, about 95%. This is a model, not a universal device law. Overshoot, dead time and multiple time constants need other descriptions.

USE IT CORRECTLY

Define settling as entry into a stated band that is maintained for a required hold time. A single sample inside the band is not settling. Measure response across relevant conditions and include uncertainty. The sample rate must be sufficient to observe the behaviour; stale data must not satisfy the hold requirement.

WORKED EXAMPLE / HYPOTHETICAL

For a hypothetical first-order response with $\tau = 2$ s and no dead time, about 95% of the change occurs after 6 s. A ±1% settling requirement would take longer and still need suitable measurements.

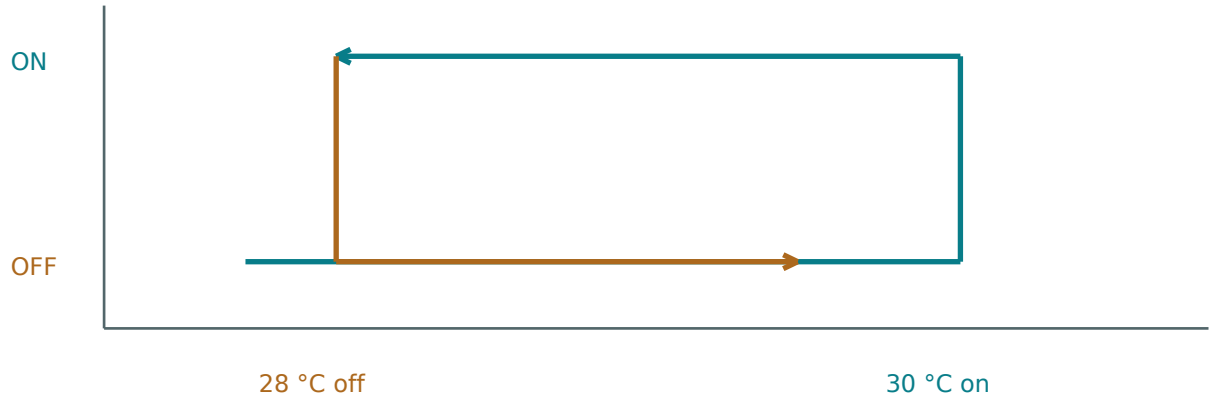
CHECK YOUR UNDERSTANDING

Can you select a real device’s settle time solely from a simulator you invented?

Answer: 136 | Sources: B1, B2

Hysteresis prevents constant switching

Separate on and off thresholds can make a simple controller less sensitive to noise.



Different transition thresholds create a state-dependent band.

HOW IT WORKS

In on/off control, an actuator changes state when a threshold is crossed. Hysteresis gives different thresholds for the two directions. A cooling fan might turn on above a higher temperature and off below a lower one. Within the gap, it retains its state.

USE IT CORRECTLY

Hysteresis reduces chatter, but can allow a wider operating band. Minimum on/off times, sensor delay and thermal inertia also matter. The loss-of-sensor behaviour must be chosen from the real hazard: “off” is not a universally safe fallback, especially in an actual cooling system.

WORKED EXAMPLE / HYPOTHETICAL

In a purely simulated thermostat, turn the fan on above 30 °C and off below 28 °C. At 29 °C, the prior state matters. These numbers teach hysteresis; they are not server temperature limits or a physical lab heating target.

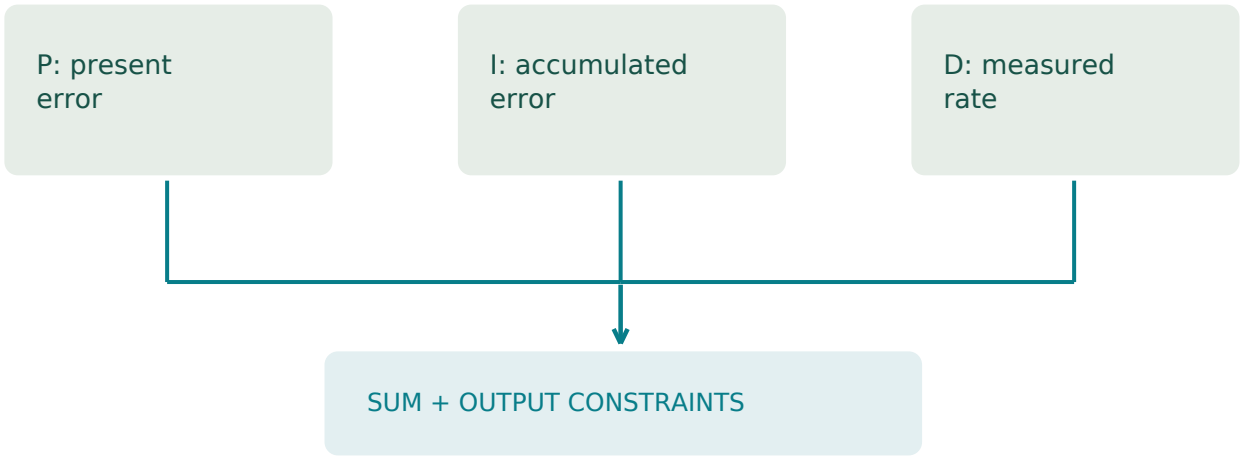
CHECK YOUR UNDERSTANDING

If the simulated fan is already on and the temperature becomes 29 °C, what happens?

Answer: 136 | Sources: B1, B2

PID has three different memories

Proportional, integral and derivative terms respond to error in different ways.



Present error, accumulated error and rate of change need separate treatment.

HOW IT WORKS

A common continuous form is:
$u = K_p e + K_i \int e\,dt + K_d de/dt$.
Proportional action responds to present error; integral action accumulates error; derivative action responds to rate of change. Practical implementations include timing, filtering and output limits.

USE IT CORRECTLY

Too much gain can cause oscillation or instability. Integral windup occurs when accumulation continues while the actuator is saturated; anti-windup addresses that behaviour. Derivative action can amplify measurement noise and is often applied to measured output. Tuning must suit the actual plant, not an attractive animation.

WORKED EXAMPLE / HYPOTHETICAL

If a proportional controller has $K_p = 2$ command-units/K and error is 3 K, its proportional term is 6 command units. The output still needs permitted bounds and a defined mapping to the actuator.

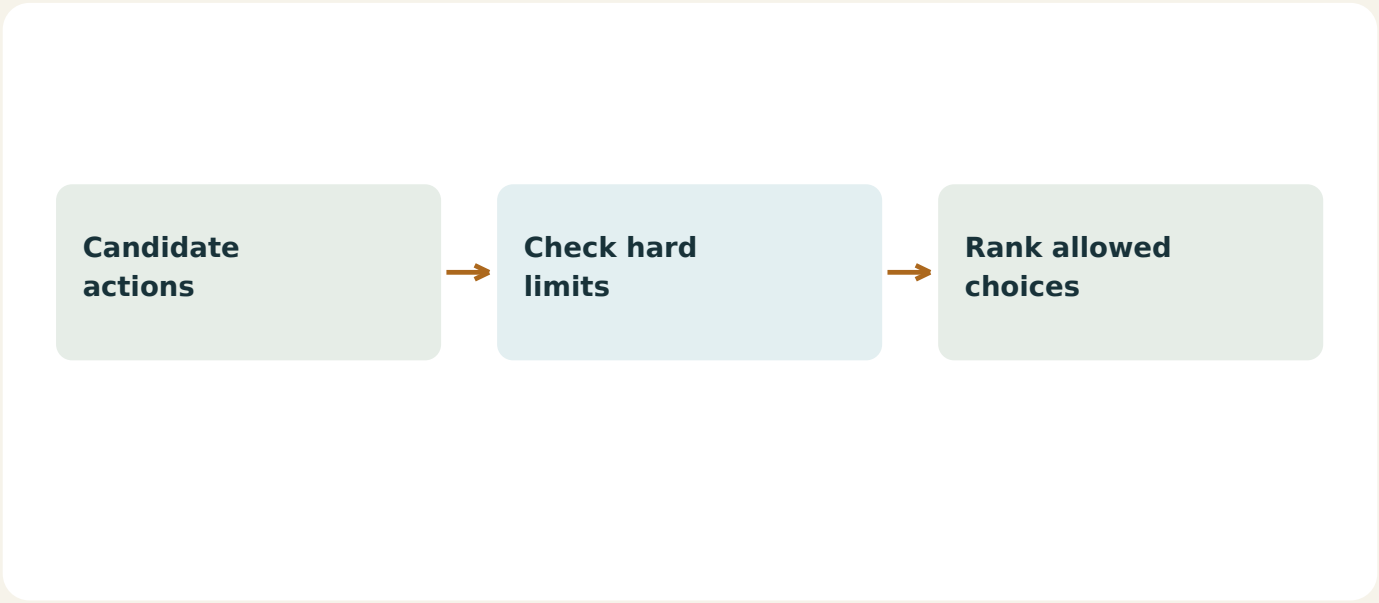
CHECK YOUR UNDERSTANDING

What should happen when a controller asks for an output above a hard actuator limit?

Answer: 136 | Sources: B2; S22

Feasibility comes before optimisation

A weighted preference must never compensate for violating a hard operating constraint.



Filtering for feasibility precedes ranking.

HOW IT WORKS

Define hard limits as pass/fail conditions on an action: allowed power, current, temperature, ramp rate, mode, state and permissions. Reject infeasible actions first. Only then compare remaining actions using preferences. Unknown or stale capability data may make feasibility uncertain rather than permissive.

USE IT CORRECTLY

Limits can change while an action is pending. Check them at appropriate stages and let local equipment enforce its own protective boundaries. Avoid a single number that trades a forbidden condition against an attractive benefit. In Lupine, Aiva’s proposed choice must remain separate from Rail’s execution evidence.

WORKED EXAMPLE / HYPOTHETICAL

Candidate A scores well but requests 300 W when available power is 200 W. Candidate B scores less well and requests 150 W. A is infeasible; its score cannot make it allowed. B still needs all other constraints checked.

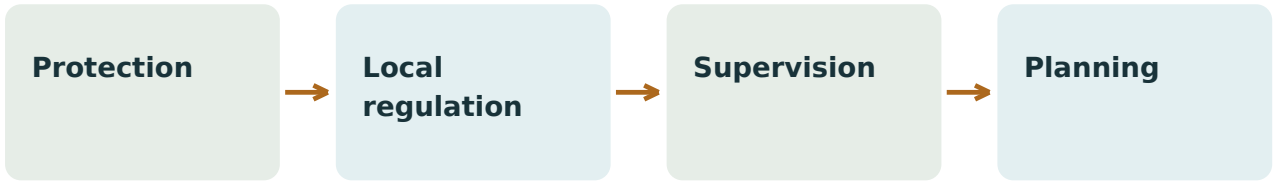
CHECK YOUR UNDERSTANDING

What is wrong with giving a temperature violation a small scoring penalty?

Answer: 136 | Sources: B1, B2

Separate control time scales

Fast local regulation and slower supervisory decisions should not depend on the same timing guarantees.



Different time scales require different responsibilities and failure handling.

HOW IT WORKS

A device can regulate current or switching at a fast local rate. A local controller may manage a thermal loop. A supervisor requests operating targets and checks outcomes on a slower time scale. A planning layer considers longer horizons. These layers exchange limits and state, not unlimited authority.

USE IT CORRECTLY

The permitted delay and failure behaviour should be explicit at each boundary. Wi-Fi and ordinary application software do not provide the same guarantees as dedicated protection. A supervisor should not fight a local regulator or rapidly alternate targets without understanding the underlying dynamics.

WORKED EXAMPLE / HYPOTHETICAL

A battery supervisor requests 100 W. The inverter regulates its electrical output locally and may limit it under a fault. Rail checks reported acceptance and suitable physical evidence; it does not implement the inverter’s switching loop.

CHECK YOUR UNDERSTANDING

Which layer should prevent excessive device current if the supervisor disconnects?

Answer: 136 | Sources: B1, B2

SCADA observes and commands

Supervisory control and data acquisition links remote equipment with a higher-level view.



Keep request, reported setting and measured process value distinct.

HOW IT WORKS

SCADA gathers measurements and status, presents alarms and can issue authorised controls. A PLC or local controller runs device/process logic. Tags are named values in the system; a tag must carry units, quality and timing, not merely a display number. An HMI is the human-machine interface.

USE IT CORRECTLY

A control screen can be out of date while equipment continues operating. Reported setpoint, acknowledged command and measured output should be separate fields. Alarm acknowledgement means someone acknowledged the alarm; it does not necessarily mean the physical cause was fixed.

WORKED EXAMPLE / HYPOTHETICAL

Display three distinct items: requested power = 100 W, device-reported setpoint = 100 W, measured output = 70 W with timestamp and quality. Collapsing them into one “power” field conceals a potentially important mismatch.

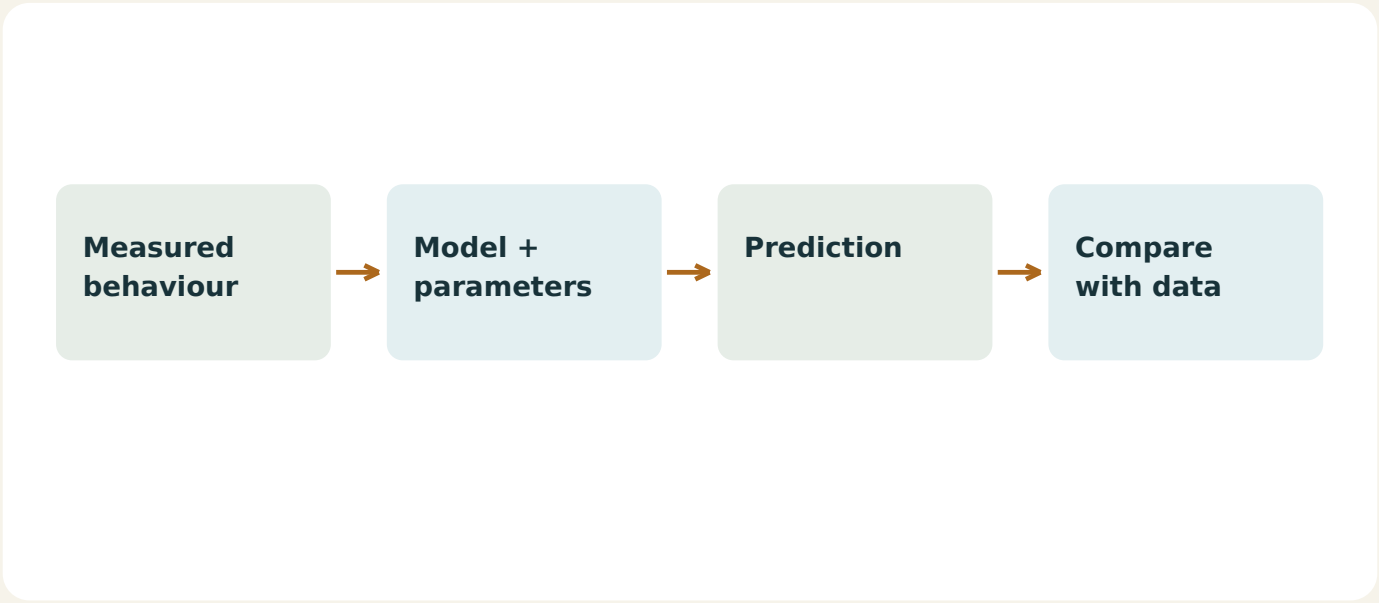
CHECK YOUR UNDERSTANDING

Does a green communications indicator prove the plant is in the desired physical state?

Answer: 136 | Sources: B2; S19, S20

A model is not automatically a digital twin

Models help answer questions when their assumptions, parameter sources and validation limits are visible.



Validation compares independent evidence with the model’s stated purpose.

HOW IT WORKS

A model represents selected behaviour, not the whole device. A simulator executes that representation. A digital-twin claim usually implies an ongoing relationship to a specific physical system and its data; the exact definition should be stated. A hand-tuned animation is not validated by the label.

USE IT CORRECTLY

Identify parameters from appropriate evidence, test against data not used to fit them and record where the model fails. Separate structural error from parameter uncertainty. Use the simplest model that can answer the question, then add complexity only when a specific mismatch justifies it.

WORKED EXAMPLE / HYPOTHETICAL

A first-order fan model reproduces electrical power settling but not rotation stalls. It can test a power-reading timeout rule; it cannot verify a stall-detection requirement unless the missing behaviour is modelled and validated.

CHECK YOUR UNDERSTANDING

What is a useful limitation statement for an unvalidated simulator?

Answer: 136 | Sources: B1, B2

Connect the equipment

Give every value meaning and every device connection an explicit boundary.

K Data, code, protocols and security

EIGHT IDEAS TO UNDERSTAND

81 Code should preserve physical meaning 98

82 Give each record a schema 99

83 TCP delivers bytes, not outcomes 100

84 Modbus exposes a data model 101

85 A register map is part of the interface 102

86 An adapter isolates device differences 103

87 Quality gates precede verdicts 104

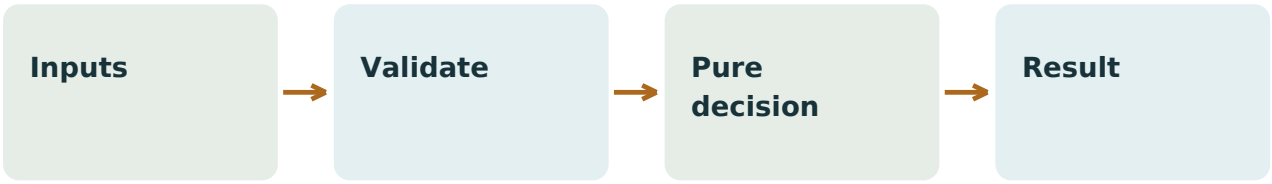
88 Secure the control boundary 105

Before you move on

Explain one mechanism without looking, solve one example with units and identify one limit of the model. Use the answer section to correct your reasoning, then keep your corrected sketch.

Code should preserve physical meaning

A program stores representations of the world. A value’s type and meaning influence every later decision.



Meaningful types and explicit absence are part of the engineering design.

HOW IT WORKS

In Python, a variable names a value. A function accepts inputs and returns a result. A dictionary groups named fields; a list groups ordered items. A Boolean is true or false. None represents an absent value and must remain distinct from numeric zero. Exceptions report conditions the caller needs to handle.

USE IT CORRECTLY

Keep calculation and decision logic separate from device input/output. A pure function can classify given evidence without opening a connection or changing equipment. Test it with explicit inputs, including missing and invalid values. Reject non-finite numbers such as NaN before ordinary range comparisons.

WORKED EXAMPLE / HYPOTHETICAL

A useful function is power_w(voltage_v, current_a). If either reading is absent or invalid, return an explicit invalid result or raise a defined error. Do not multiply a missing value that was silently replaced with zero.

CHECK YOUR UNDERSTANDING

Why is “reading or 0” dangerous when classifying a failed sensor?

Answer: 137 | Sources: S23

Give each record a schema

A schema states what each field means and which combinations are valid.

quantity	input_power
value / unit	1.0 / W
quality	valid
capture time	explicit timestamp

Every important value needs units, identity, time and quality.

HOW IT WORKS

A telemetry record should identify device, quantity, value, unit, capture time, receive time, quality and sequence or sample identity where available. A command record needs a unique ID, target, unit, permitted range, deadline and lifecycle state. Keep observed values separate from requested values.

USE IT CORRECTLY

Validate required fields and reject incompatible units, impossible times and non-finite numbers. Preserve raw payloads separately from normalised values when appropriate. Version the schema so a future parser can interpret old records. JSON is a representation format; it does not provide these semantics automatically.

WORKED EXAMPLE / HYPOTHETICAL

Use quantity = input_power, value = 1.0, unit = W, quality = valid. A missing sample can use value = null with quality = missing. Do not use value = 0 and quality = valid to stand in for a disconnected sensor.

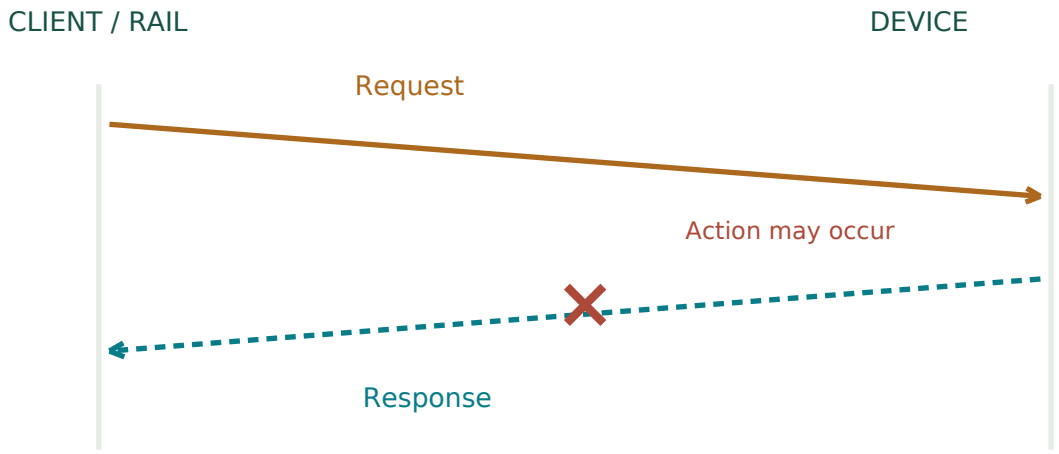
CHECK YOUR UNDERSTANDING

Why should a record include both a device ID and a command ID?

Answer: 137 | Sources: S23

TCP delivers bytes, not outcomes

A reliable network stream is useful, but it does not establish that a physical action completed.



The action and its acknowledgement can have different outcomes.

HOW IT WORKS

An IP address identifies a network endpoint in its context; a port identifies a service. TCP provides an ordered byte stream between endpoints while the connection exists. Applications still need message framing, validation, timeouts and meaning. Receiving bytes is not the same as committing a device action.

USE IT CORRECTLY

If a connection drops after send, the peer may have received, processed or physically executed the request. A reconnect does not erase that history. Wi-Fi adds variable delay and disconnections; a local network is not a timing guarantee. Use defined command IDs and a reconciliation protocol where the device supports them.

WORKED EXAMPLE / HYPOTHETICAL

Client sends “set target to 1 W”; device acts; reply is lost. The client observes a timeout even though the action occurred. Resending a non-idempotent “increase by 1 W” could create an additional change.

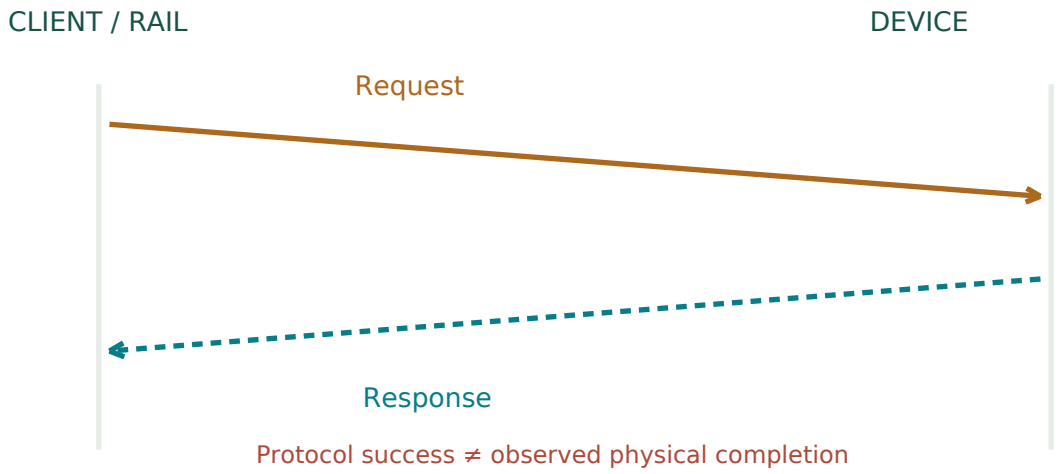
CHECK YOUR UNDERSTANDING

What does a TCP timeout establish about physical execution?

Answer: 137 | Sources: B1, B2

Modbus exposes a data model

Modbus lets a client read or write defined data items. The device documentation supplies the physical meaning.



Address, function and representation are separate from physical outcome.

HOW IT WORKS

The common model has discrete inputs and coils as single bits, and input/holding registers as 16-bit words. Function codes select operations. Protocol addresses are zero-based, while manuals may use one-based reference labels. Check the actual convention before interpreting an address.

USE IT CORRECTLY

A register write response indicates a protocol result, not guaranteed physical completion. Modbus TCP adds a transaction header; its transaction identifier matches requests and responses but is not a durable physical-operation ID. Plain Modbus does not provide modern authentication by itself. Start with a local simulator or authorised read-only connection.

WORKED EXAMPLE / HYPOTHETICAL

A manual might label the first holding register “40001” while a library expects offset 0. Passing 40001 without checking can access the wrong item or fail. An exact documented mapping is required.

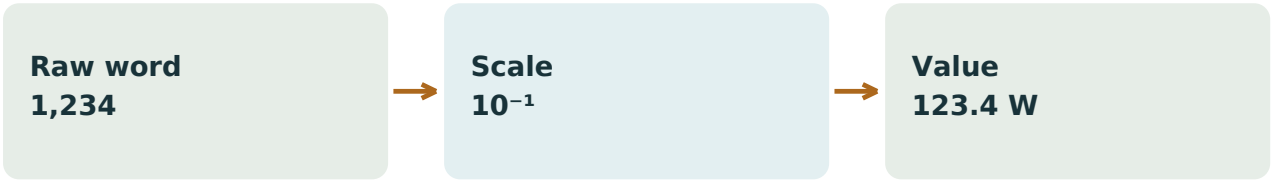
CHECK YOUR UNDERSTANDING

Does a successful Modbus write prove a valve moved or a battery changed power?

Answer: 137 | Sources: S17

A register map is part of the interface

Raw words need signedness, scaling, word order and validity rules before they become engineering values.



Also check sign, multi-word order and unavailable-value codes.

Correct decoding requires the complete documented representation.

HOW IT WORKS

A 16-bit word can represent unsigned or signed data. Multiple registers may encode a larger integer or floating-point value. Modbus defines byte order within its protocol data items, but multi-register ordering and device-specific representation still need documentation. Sentinel codes can mean unavailable, not a huge real value.

USE IT CORRECTLY

A scale factor may define value = raw × 10^{sf}. SunSpec models standardise many data definitions, but support and exact model versions must be checked on the device. Log raw words and the mapping revision during integration. Never guess a writable address by analogy with another product.

WORKED EXAMPLE / HYPOTHETICAL

Hypothetical map: raw = 1,234, scale factor = -1, unit = W. Value is 123.4 W. If 65,535 is documented as invalid for that field, do not interpret it as 6,553.5 W.

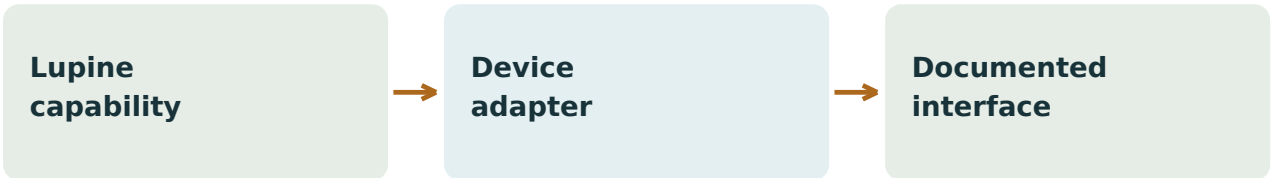
CHECK YOUR UNDERSTANDING

Why can byte-perfect communication still produce a dangerously wrong displayed value?

Answer: 137 | Sources: S17, S18

An adapter isolates device differences

Lupine should express a capability without pretending every controller behaves identically.



Normalise the representation while preserving semantic limits.

HOW IT WORKS

A device adapter translates a common internal request into a documented interface and converts responses into normalised evidence. It should expose supported operations, units, limits and status-query semantics. If a device lacks durable command identity, the adapter must report that limitation rather than invent it.

USE IT CORRECTLY

OpenEMS is an existing energy-management framework with Modbus integration components. Integrating with it can reuse established infrastructure; it does not itself establish a new product need. Keep transport, device semantics and Rail’s decision logic separated so one failure does not become an unsupported outcome claim.

WORKED EXAMPLE / HYPOTHETICAL

Adapter A supports absolute power setpoints and a reported active setting. Adapter B exposes only a toggle. They should not implement the same “retry safe” promise. The common interface must preserve capability differences.

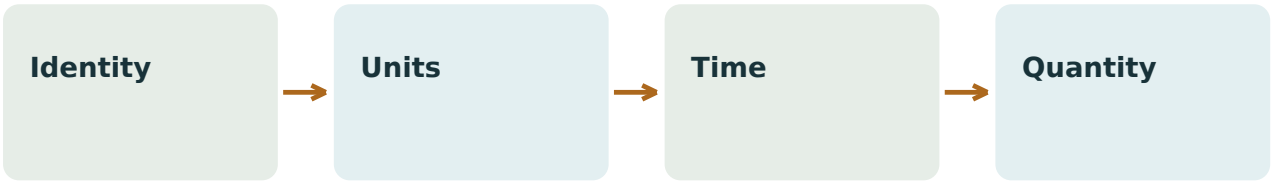
CHECK YOUR UNDERSTANDING

What should an adapter return when the device has no trustworthy execution-status query?

Answer: 137 | Sources: S19; S17

Quality gates precede verdicts

A fresh-looking number should pass validation before it influences an execution decision.



All relevant evidence gates must pass before numerical comparison.

HOW IT WORKS

Check identity, units, allowed representation, capture time, clock quality, sample sequence and relevant device mode. Then assess whether the reading belongs to the post-command observation window and matches the required quantity. A cached pre-command value must not satisfy a new outcome requirement.

USE IT CORRECTLY

Keep missing, stale, invalid and contradictory as distinct quality conditions. A maximum sample age is only one gate; a new value from the wrong sensor is still wrong evidence. If time uncertainty overlaps the command boundary, preserve the ambiguity. Log the failed gate so a reviewer can understand the verdict.

WORKED EXAMPLE / HYPOTHETICAL

A 0 W reading is valid for a request only if it is a real, suitable sample from the correct device and time window. A missing measurement represented as null must never be promoted into the same evidence.

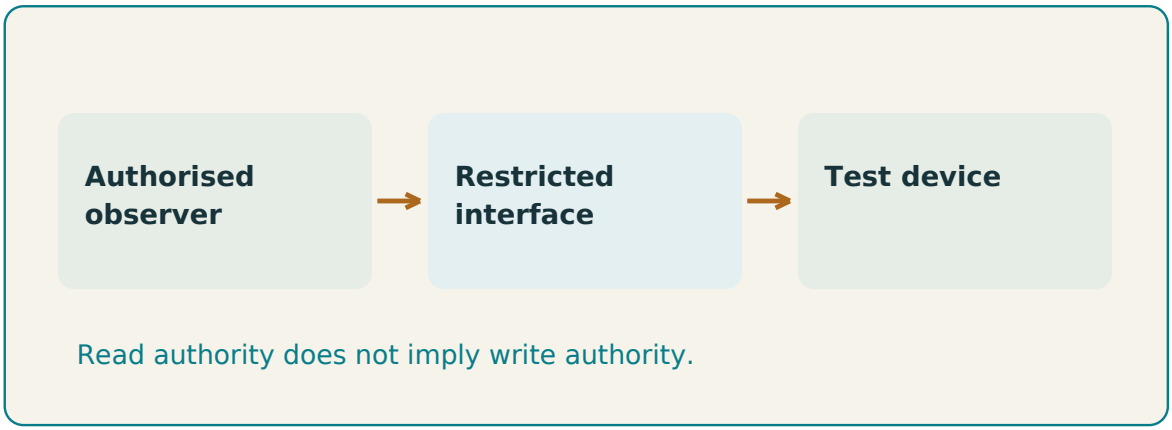
CHECK YOUR UNDERSTANDING

What should happen when a sample timestamp lies far in the future?

Answer: 137 | Sources: B1, B2

Secure the control boundary

Operational technology controls physical equipment. Security failures can therefore become physical failures.



Observation and control need explicit, limited authority.

HOW IT WORKS

Identify assets, authorised users, allowed communication and consequences before connecting a controller. Use least privilege: a read-only observer does not need write authority. Separate the learning network from unrelated or production equipment. Protect credentials and avoid exposing plain control protocols to the public internet.

USE IT CORRECTLY

Define authentication, update, backup and recovery procedures appropriate to the system. Restrict changes and keep an auditable configuration history. Security and safety interact: an update or isolation action must consider process consequences. An encrypted channel does not make an unsafe command safe or a poor sensor accurate.

WORKED EXAMPLE / HYPOTHETICAL

For an early integration, allow only the needed read operations on an authorised test device. Record the mapping and configuration. Prove the observer cannot issue writes before using it to gather evidence.

CHECK YOUR UNDERSTANDING

Does a private IP address alone make a Modbus controller secure?

Answer: 137 | Sources: S20

Build Lupine with evidence

Carry uncertainty honestly, test real failure modes and earn the next stage with evidence.

L Execution, reconciliation and discovery

EIGHT IDEAS TO UNDERSTAND

89 Rail needs an honest lifecycle 107

90 Retry safety depends on semantics 108

91 Reconcile without inventing certainty 109

92 Durable state survives a restart 110

93 Cloked records evidence, not physical truth 111

94 Test the failure sequence, not only the row 112

95 Advance by evidence, not by confidence 113

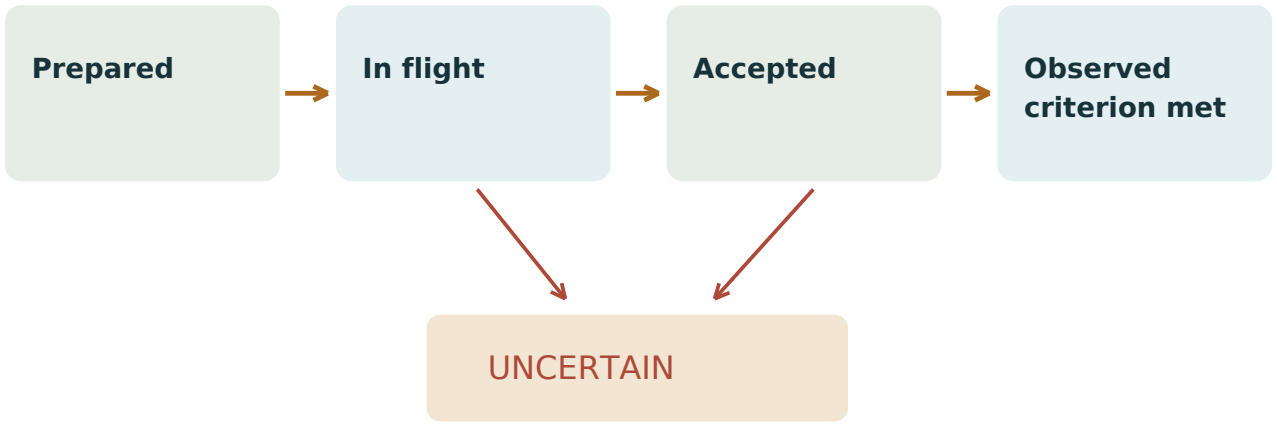
96 Build a company around a real gap 114

Before you move on

Explain one mechanism without looking, solve one example with units and identify one limit of the model. Use the answer section to correct your reasoning, then keep your corrected sketch.

Rail needs an honest lifecycle

Sending, acceptance and observed physical result are separate events.



Uncertainty is a legitimate state, not an exception to hide.

HOW IT WORKS

Represent a command as an explicit state machine. Prepared means it has not yet been sent. In-flight means an attempt may be underway. Accepted describes a device response. Completed is a scoped evidence verdict, not a claim of perfect knowledge. Uncertain preserves an unresolved outcome.

USE IT CORRECTLY

Define allowed transitions and required evidence for each. A timeout must not silently become failed. A mismatch means the relevant observed result does not meet the criterion after the allowed response interval. A terminal verdict can later need review if new contradictory evidence appears; keep an audit trail rather than rewriting history invisibly.

WORKED EXAMPLE / HYPOTHETICAL

Accepted plus suitable fresh post-settling evidence in the agreed band can support “observed target reached within stated measurement limits.” It does not prove that the command caused the change or that every part of the equipment is healthy.

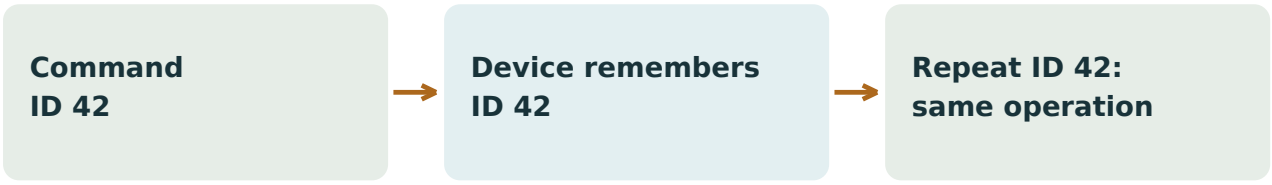
CHECK YOUR UNDERSTANDING

Which state fits a lost reply after a command may have executed?

Answer: 138 | Sources: B1, B2

Retry safety depends on semantics

An operation’s identity and effect determine whether repeating it is safe.



Only valid if the device actually supports the stated durable semantics.

The retry must refer to the original operation, with defined device support.

HOW IT WORKS

An operation is idempotent under defined semantics if repeating it has no additional effect. Setting an absolute value can be idempotent in some designs; “increase by 10” is usually not. Even absolute setpoints can have side effects, races or mode-dependent behaviour. Specify rather than assume.

USE IT CORRECTLY

A stable command ID supports deduplication only if the device or responsible service preserves the association durably for the required period. A new ID on each retry can defeat deduplication. Reconnection, expiry and concurrent commands need explicit rules. Blocking every repeat forever also is not a complete recovery policy.

WORKED EXAMPLE / HYPOTHETICAL

If a device executed “increase by 1” but its reply vanished, another identical increase can produce 2. An absolute “set to 1” has a different effect model, but still needs capability, mode and concurrency checks.

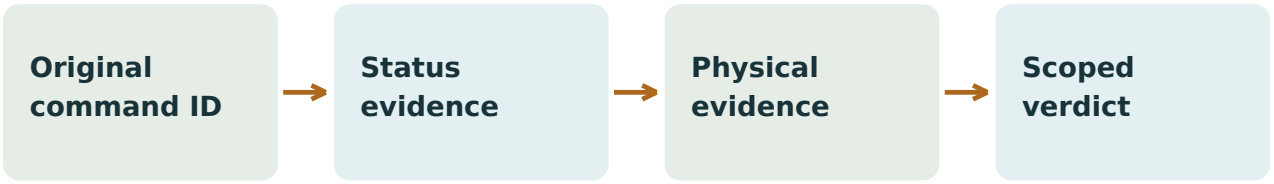
CHECK YOUR UNDERSTANDING

Is a transport transaction ID automatically sufficient for safe retry after a device restart?

Answer: 138 | Sources: B1, B2

Reconcile without inventing certainty

Resolve a pending outcome using further authoritative status and relevant physical evidence.



Status and measurement must be interpreted together, including their limits.

HOW IT WORKS

Query the original command where supported, read device mode and capabilities, and obtain suitable new observations. Determine what each response can actually establish. “Not found” may mean never received, expired history, a restarted device or the wrong identity; a delayed original request may still be in flight.

USE IT CORRECTLY

A refusal can establish rejection of one request without explaining why the physical output is already at target. Keep contradictions visible. Permit another potentially duplicating action only after the original has been resolved or a device-specific safe-repeat policy explicitly handles the uncertainty. Cancellation itself also needs defined semantics.

WORKED EXAMPLE / HYPOTHETICAL

Reply lost, device history later says accepted, and fresh output is at target: the evidence can support a scoped completion verdict. Reply lost, history unavailable, output already at target beforehand: command execution remains uncertain.

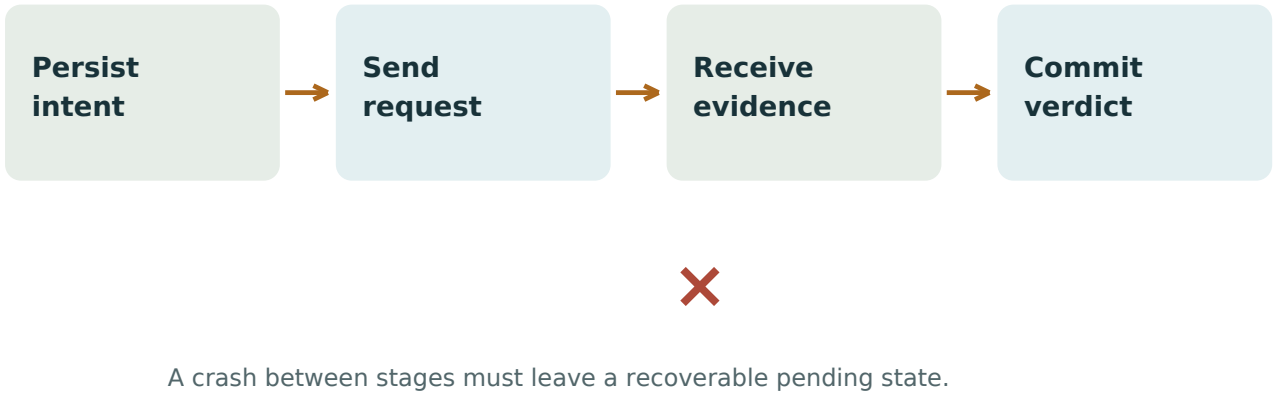
CHECK YOUR UNDERSTANDING

Why is a matching physical reading alone insufficient to resolve every lost-command case?

Answer: 138 | Sources: B1, B2

Durable state survives a restart

A process crash must not erase the fact that a physical action may already have occurred.



A durable pending record preserves the unresolved physical possibility.

HOW IT WORKS

Persist command identity, intent and lifecycle evidence using a defined transactional strategy. A database transaction groups changes so they commit together under the database's guarantees. The exact durability depends on configuration, filesystem and hardware. An in-memory dictionary does not provide restart recovery.

USE IT CORRECTLY

There is still a boundary between committing a local record and acting on remote equipment. A crash can occur between them. Design recovery around that ambiguity, using stable IDs, an outbox or pending-intent record and reconciliation. "Exactly once" is not created by a local transaction alone.

WORKED EXAMPLE / HYPOTHETICAL

Persist intent, send a command, then crash before storing the reply. On restart, the durable pending record tells Rail to reconcile. Treating the absent reply as proof of failure would invite duplicate action.

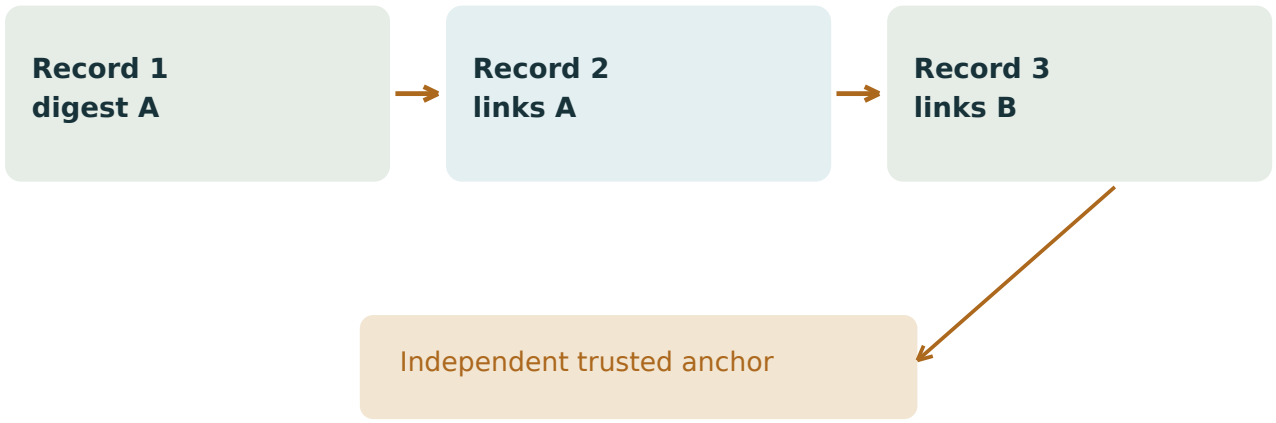
CHECK YOUR UNDERSTANDING

Why can a database transaction not atomically guarantee a remote physical action by itself?

Answer: 138 | Sources: S21

Cloked records evidence, not physical truth

Integrity, origin and measurement validity need different mechanisms.



An external trust anchor and a valid physical measurement solve different problems.

HOW IT WORKS

A hash maps data to a fixed-length digest. In a hash chain, each record includes a link to the prior digest, so changes break internal consistency unless later hashes are rebuilt. Canonical serialisation ensures the same intended data produces the same byte representation for hashing.

USE IT CORRECTLY

A trusted external anchor can reveal a rewritten history relative to that anchor. Signatures can help authenticate the holder of a key. Neither establishes sensor accuracy, correct mounting or causation. Protect keys and define trust boundaries. Store enough raw evidence and method information for the physical claim to be evaluated separately.

WORKED EXAMPLE / HYPOTHETICAL

An attacker rewrites a record and recomputes every subsequent hash. An internal chain check can pass. Comparing with an independently protected earlier final digest can reveal disagreement, provided that anchor is trustworthy.

CHECK YOUR UNDERSTANDING

Does a valid signature prove that a temperature reading was physically accurate?

Answer: 138 | Sources: B1, B2

Test the failure sequence, not only the row

Many execution defects appear only after a particular order of events.

Lost reply	uncertain
Stale sample	uncertain
Accepted + mismatch	mismatch
Restart + pending ID	reconcile

Combine communication, time, persistence and evidence failures deliberately.

HOW IT WORKS

Create tests for lost instruction, lost reply, refusal, stale data, conflicting evidence, restart and delayed delivery. Then combine events into sequences. Verify invariants such as “no unsupported completion” and “no unsafe repeat while an operation may still execute.” Include the device’s documented semantics in the test model.

USE IT CORRECTLY

A mock device tests software against invented behaviour; a hardware test adds a real communication and measurement boundary. Neither alone proves production reliability. Fault injection must remain inside a safe, authorised setup. Repeat selected cases and preserve failures, software versions, timing and actual device configuration.

WORKED EXAMPLE / HYPOTHETICAL

Sequence: send → device acts → reply lost → supervisor restarts → old sample arrives. The correct behaviour should not depend on the restart making the command “new” or the old sample appearing fresh.

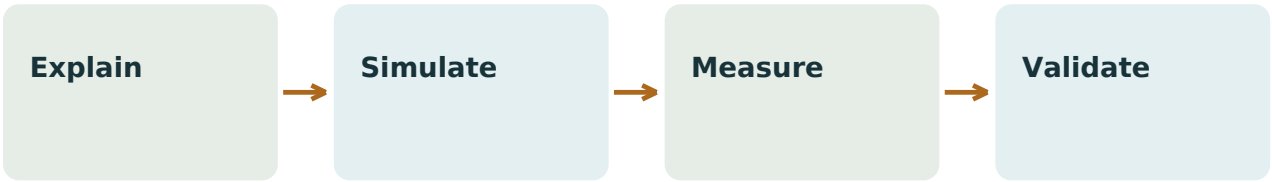
CHECK YOUR UNDERSTANDING

What does passing 30 software tests establish?

Answer: 138 | Sources: B1, B2

Advance by evidence, not by confidence

The next stage should answer a specific uncertainty that matters to the project.



Each stage earns a more specific claim; none guarantees a company.

HOW IT WORKS

Start by explaining the rule, then reproduce it in a model, exercise it on a small reviewed bench and compare with a real device interface. Before a field pilot, identify an equipment user, the actual failure, current handling, permitted access and measurable acceptance criteria.

USE IT CORRECTLY

A model-only result is useful preparation. A measured bench result is evidence about that bench. A successful authorised pilot adds site-specific evidence. Each stage has a different claim boundary. Do not call a small fan an energy-infrastructure deployment or interpret expert interest as a purchase commitment.

WORKED EXAMPLE / HYPOTHETICAL

Gate: “Can Rail detect a stale post-command report on the named controller without issuing an unintended repeat?” This is testable. “Is Lupine the future of the grid?” is not an experiment specification.

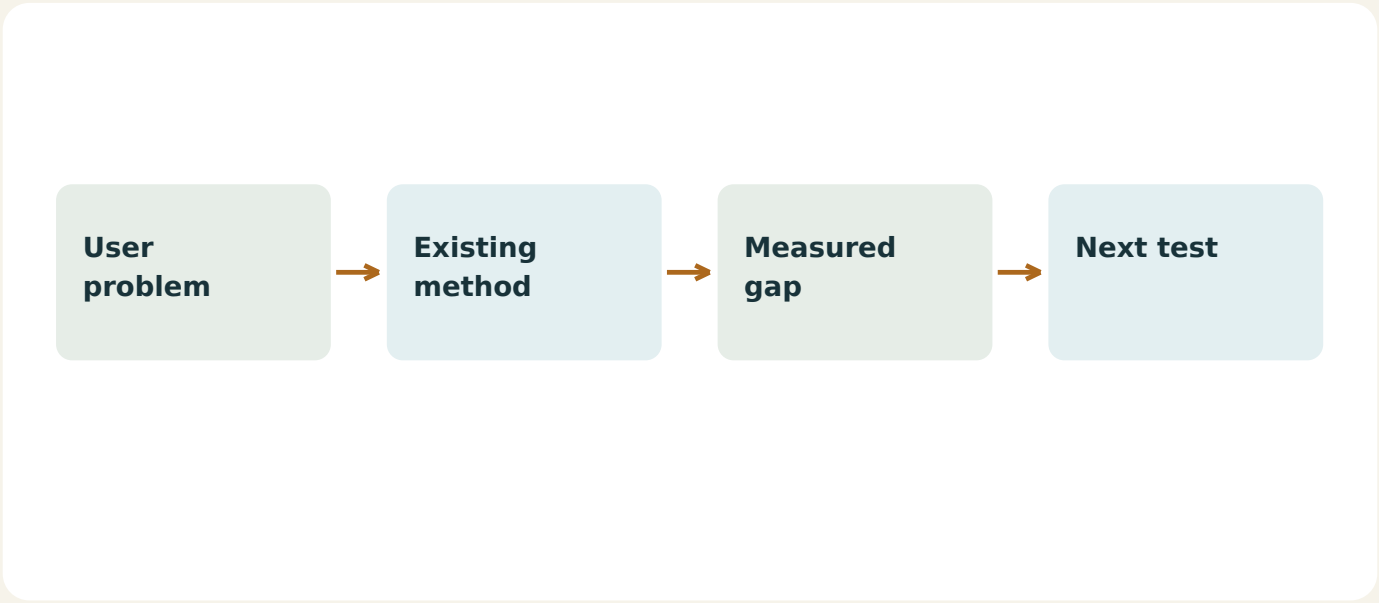
CHECK YOUR UNDERSTANDING

What should determine whether you add a more complex model or buy another instrument?

Answer: 138 | Sources: B1, B2

Build a company around a real gap

Your direction can be long term while your first validated problem stays narrow.



A useful company needs a validated problem as well as working software.

HOW IT WORKS

Keep Lupine’s layers as working hypotheses: Aiva selects among feasible actions, Rail manages execution and reconciliation, and Cloked records evidence within stated trust limits. The energy domain gives these layers physical meaning, but architecture transfer alone is not a reason for adoption.

USE IT CORRECTLY

Find one equipment user with a repeated, consequential failure their current method handles poorly. Study existing tools and ask why the present method is insufficient. Compare a baseline with a narrow improvement. Your industrial-engineering strength is understanding the complete process, interfaces and constraints - not claiming originality before investigation.

WORKED EXAMPLE / HYPOTHETICAL

Write: “For this user, under this failure, the present method loses this evidence. We will test this capability against this baseline, with these limits.” Keep the brief one page until the underlying evidence supports expansion.

CHECK YOUR UNDERSTANDING

What is the strongest early output you can produce?

Answer: 138 | Sources: B1, B2

Project 1 / Start with evidence on paper

Calculate energy and classify readings without buying or connecting anything.

SETUP AND BOUNDARY

Paper or a spreadsheet, a calculator and the supplied four-row example. Treat every value as invented teaching data.

01

Predict

For a 5 V, 100 Ω ideal resistor, predict I and P. Write the assumptions and the measurement point.

02

Calculate

Use 2 W for 30 min followed by 1 W for 30 min. Calculate energy before reading the answer.

03

Classify

Target: 1.00 W ±0.05 W after settling. Classify a fresh 0.99 W, fresh 0.70 W, missing value and stale 1.00 W.

04

Explain

Separate numerical in-band status from sensor accuracy and command causation. State what further evidence is missing.

WHAT COUNTS AS A RESULT

Predicted resistor values: 0.050 A and 0.25 W. Energy: 1.5 Wh. Numeric classifications: in band, outside band, unknown and stale/unknown. No physical measurement or command execution has been established.

Record: Keep the calculation, units, assumptions and your explanation of why absence is not zero.

Project 2 / Measure a small DC load

Compare a predicted resistor load with real low-voltage measurements.

SETUP AND BOUNDARY

A reviewed, current-limited 5 V DC source suitable for this task; a 100 Ω resistor rated at least 1 W; suitable insulated connections; a multimeter. Set a suitable source limit, for example 0.10 A for this 0.05 A nominal load. Do not alter a mains supply.

01

Plan unpowered

Draw source → resistor → return. Mark the voltmeter across the resistor and the ammeter in the branch. Confirm meter sockets, mode, fuse and range.

02

Predict

Use the resistor tolerance and source specification to predict a plausible range. Nominal values are 0.05 A and 0.25 W.

03

Measure briefly

With the circuit reviewed, power under supervision. Record actual voltage and current, method and time. Stop for unexpected heating or behaviour.

04

Compare

Calculate VI and V^2/R with their limitations. Disconnect power before changing connections. Do not bridge a supply with a current-mode meter.

WHAT COUNTS AS A RESULT

A coherent, safe measurement and an explained difference between prediction and observation. Agreement within broad bounds is a cross-check, not proof of traceable calibration.

Record: Photograph the unpowered arrangement. Record source current limit, meter identity, resistor rating and tolerance, raw readings and limitations.

Project 3 / Observe temperature response

Understand location, lag and repeatability without creating a hot test system.

SETUP AND BOUNDARY

A suitable low-voltage temperature probe/logger, an independent reference if available, a clock and an ordinary room. No heaters, boiling water, opened appliances or deliberate component overheating.

01

Stabilise

Place the probe in one location away from direct sun and drafts. Record the environment and wait for readings to settle.

02

Change gently

Move it to a second ordinary room location, or hold only a probe designed for skin contact gently in your hand. Keep electronics and connectors dry.

03

Log

Capture both sample and receipt times if the system supports them. Record each value and quality flag. Do not pretend logging frequency is sensor bandwidth.

04

Repeat

Return to the first location and repeat. Compare the approach to steady conditions, mounting effects and any apparent offset against the independent instrument.

WHAT COUNTS AS A RESULT

You can explain which temperature was sensed, why it lagged and why two sensors might disagree. Do not infer a component junction temperature from an ambient probe.

Record: Plot temperature against elapsed time. Mark movement times and state whether timing comes from a local monotonic clock or aligned wall clocks.

Project 4 / Characterise a guarded fan

Measure the response of a small controllable load before involving Rail.

SETUP AND BOUNDARY

A guarded low-power 5 V fan and a suitable reviewed driver/control arrangement; current-limited source; power measurement; speed measurement if speed is the target. Exact wiring depends on the identified fan and driver. Check their datasheets before connection.

01

Define the target

Choose electrical input power or rotational speed. Write the sensor, unit, band and relevant limits. Do not call current a speed measurement.

02

Establish a baseline

Record initial command, input power, speed where available and ambient conditions. Keep fingers and objects away from the guarded moving part.

03

Apply one permitted step

Change between two allowed operating points. Record the actual command and response without overriding local protection or the fan’s control-interface requirements.

04

Estimate dynamics

Identify delay, overshoot and time to remain inside a stated band. Repeat both directions. If response is inconsistent, investigate rather than choosing an easier tolerance.

WHAT COUNTS AS A RESULT

A repeatable response trace at the chosen measurement boundary, with a justified provisional settle/hold criterion. This is a fan-bench result, not a data-centre cooling qualification.

Record: Exact device IDs, driver, supply limit, firmware, sensor, capture timing, raw traces and every failed run.

Project 5 / Rehearse lost outcomes

Own the execution rules before asking code or hardware to enforce them.

SETUP AND BOUNDARY

Six command cards and six evidence cards. No physical equipment is needed. Use one stable command ID for each intended operation.

01

Healthy case

Accepted; a fresh relevant post-settling sample meets the band. State a scoped completion verdict and its measurement limits.

02

Mismatch case

Accepted; a fresh relevant sample is outside the band after settling. Record mismatch rather than success.

03

Uncertain cases

Reply lost; missing sensor; stale sample; refusal with conflicting output evidence. Preserve uncertainty and specify the evidence needed next.

04

Sequence case

Device acts, reply is lost, supervisor restarts and an old sample arrives. Explain what must have been stored and why another operation may duplicate the effect.

WHAT COUNTS AS A RESULT

You can explain each verdict without looking it up. You can distinguish observed target from causation, and know why timeout or “not found” may be insufficient to permit a repeat.

Record: A small decision table including current state, evidence quality, verdict, next query and retry policy.

Project 6 / Inject one communication fault

Test whether the same rules survive a real but controlled bench communication failure.

SETUP AND BOUNDARY

Only after Project 4 is understood: the supervised guarded fan bench, durable command records, an independent measurement path and a defined local fallback. Do not use the fan to protect a heat-producing system during this test.

01

Predefine behaviour

Specify which command may be lost, which reply may be suppressed and how capture times will be preserved. State the independent stop method.

02

Lose a reply

Let the local controller apply an allowed target, then deliberately suppress only its response in the test adapter. Avoid accidental duplicate sends.

03

Reconcile

Retain the same command ID. Query supported device status and gather fresh physical evidence through the independent path.

04

Repeat safely

Test a lost instruction, frozen sample and supervisor restart separately. Restore the known baseline between cases. Stop if behaviour differs from the reviewed plan.

WHAT COUNTS AS A RESULT

No unsupported success and no potentially duplicating repeat during unresolved execution. A hardware finding must identify exactly which controller, adapter and measurement path were tested.

Record: Raw commands, replies, capture times, independent readings, restarts, verdicts and the firmware/software versions.

Project 7 / Read a local Modbus mock

Understand a documented register exchange before connecting any external equipment.

SETUP AND BOUNDARY

Python 3 and two terminal windows on your own computer. Use the server and client printed on pages 125-126. They bind/connect only to 127.0.0.1:15020. No packages, public network or real controller are required.

01

Predict

The invented map has holding-register offset 0 = 1,234 and offset 1 = signed scale factor -1. Predict the engineering value.

02

Run the server

Save the printed server code as mock_modbus.py and run python3 mock_modbus.py. Keep this terminal open.

03

Run the client

Save the printed client as read_modbus.py and run python3 read_modbus.py. Check transaction identity, function, byte count and decoded words.

04

Probe the limits

Change only the requested offset to 2 and observe the exception response. Restore the original request. Stop the local server with Ctrl+C when finished.

WHAT COUNTS AS A RESULT

The normal request decodes 123.4 W from the invented map. The invalid range produces an explicit exception. Neither result is a measurement of equipment power.

Record: The request, response bytes, map, scale factor, result and your explanation of why a protocol reply is not a physical-outcome proof.

Project 8 / Design the first real observer

Prepare a technically useful read-only integration with an authorised equipment user.

SETUP AND BOUNDARY

A named willing equipment user, an identified test interface, written access boundaries and a specific observation question. Do not assume permission to scan, connect to or command production infrastructure.

01

Ask about one failure

Request an anonymised account of a lost reply, stale reading or unresolved command. Understand how the current system handles it and why that is insufficient, if it is.

02

Map the interface

Record exact device/model, firmware, quantities, units, update cadence, clock quality, status semantics and read permissions.

03

Specify an observer

Collect only necessary data through a reviewed read-only path. Define retention, redaction, credential handling and how to verify writes are impossible.

04

Compare and decide

Compare what the observer can distinguish with the present method. Agree what result would justify a later controlled test. Keep unproven novelty and commercial claims out of the findings.

WHAT COUNTS AS A RESULT

The equipment user confirms the question matters and the observation is technically interpretable. No product or startup conclusion follows from access alone.

Record: A one-page problem brief, interface boundary, annotated evidence example, baseline comparison and the next decision.

A small calculation you can own

Save the code as first_calculation.py. Run it using Python 3. Expected result: 0.25 W, 0.5 Wh and a message preserving the missing value.

```
# Save as first_calculation.py; run: python3 first_calculation.py
import math

voltage_v = 5.0
current_a = 0.05
duration_h = 2.0

def power_w(voltage_v, current_a):
    if voltage_v is None or current_a is None:
        raise ValueError("missing measurement")
    if not math.isfinite(voltage_v):
        raise ValueError("invalid voltage")
    if not math.isfinite(current_a):
        raise ValueError("invalid current")
    return voltage_v * current_a

p_w = power_w(voltage_v, current_a)
energy_wh = p_w * duration_h
print("Power:", p_w, "W")
print("Energy:", energy_wh, "Wh")

sample = {"value": None, "unit": "W", "quality": "missing"}
if sample["value"] is None:
    print("No physical zero has been observed.")
```

EXPLAIN BEFORE YOU CHANGE IT

Identify the inputs, validation, state and result. Predict one changed input. Run it and explain any difference. Passing these examples does not validate real equipment, settle command causation or implement a production execution engine.

READ THE PYTHON

import loads a module. **def** defines a function; indented lines belong to it. **return** supplies its result. **if** chooses a branch. **raise** reports an exception. **None** marks absence. A dictionary groups named fields; square brackets retrieve one. **print** displays a value. Do not remove a validation branch merely to avoid its error.

Classify a sample before execution

This toy function checks a measurement against a band. It deliberately does not decide command execution. All times use one simulated clock.

```python
# Save as classify_sample.py; run: python3 classify_sample.py
import math

def classify(value_w, quality, captured_s, now_s,
             command_s, max_age_s, target_w, band_w):
    # Times share one simulated clock; no real device is involved.
    if quality != "valid" or value_w is None:
        return "uncertain: no valid measurement"
    numbers = [value_w, captured_s, now_s, command_s,
               max_age_s, target_w, band_w]
    if not all(math.isfinite(x) for x in numbers):
        return "uncertain: invalid numeric input"
    if max_age_s < 0 or band_w < 0:
        return "uncertain: invalid policy"
    if not command_s <= captured_s <= now_s:
        return "uncertain: outside observation window"
    if now_s - captured_s > max_age_s:
        return "uncertain: stale measurement"
    if abs(value_w - target_w) <= band_w:
        return "observed in band; not an execution verdict"
    return "observed outside band"

print(classify(0.99, "valid", 10, 11, 5, 3, 1, 0.05))
print(classify(None, "missing", 10, 11, 5, 3, 0, 0.05))
print(classify(1.0, "valid", 6, 11, 5, 3, 1, 0.05))
```

EXPLAIN BEFORE YOU CHANGE IT

Identify the inputs, validation, state and result. Predict one changed input. Run it and explain any difference. Passing these examples does not validate real equipment, settle command causation or implement a production execution engine.

READ THE LOGIC

!= means not equal; **<=** means less than or equal. **and** requires both conditions. A list holds several values; **all(...)** requires each test to pass. The function returns at the first failed gate. No communication, acknowledgement or settling-hold logic is present: this is only a sample classifier.

A read-only mock on your computer

Save as mock_modbus.py. The invented register map is raw = 1,234 and signed scale factor = -1. This local teaching service is not a production controller.

```
# Save as mock_modbus.py; run: python3 mock_modbus.py
# Read-only teaching mock, NOT an industrial Modbus server.
import socket
import struct

def recv_exact(sock, count):
    data = b""
    while len(data) < count:
        chunk = sock.recv(count - len(data))
        if not chunk:
            raise ConnectionError("peer closed")
        data += chunk
    return data

registers = [1234, 65535]  # scale factor word = signed -1
with socket.socket() as server:
    server.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
    server.bind(("127.0.0.1", 15020))
    server.listen(1)
    print("Local mock ready on 127.0.0.1:15020")
    while True:
        conn, _ = server.accept()
        with conn:
            conn.settimeout(3)
            try:
                head = recv_exact(conn, 7)
                tid, pid, length, uid = struct.unpack(">HHHB", head)
                if pid != 0 or not 2 <= length <= 254:
                    continue
                pdu = recv_exact(conn, length - 1)
                fc = pdu[0]
                if fc != 3:
                    reply = bytes([fc | 128, 1])
                elif len(pdu) != 5:
                    reply = bytes([131, 3])
                else:
                    start, count = struct.unpack(">HH", pdu[1:])
                    if not 1 <= count <= 125:
                        reply = bytes([131, 3])
                    elif start + count > len(registers):
                        reply = bytes([131, 2])
                    else:
                        words = registers[start:start + count]
                        reply = bytes([3, 2 * count])
                        reply += struct.pack(">" + "H" * count, *words)
                conn.sendall(struct.pack(">HHHB", tid, 0,
                                         len(reply) + 1, uid) + reply)
            except (OSError, ConnectionError, struct.error):
                pass
```

Read and decode the invented map

Start the server first. Save this as read_modbus.py and run it in another terminal. Expected engineering value: 123.4 W. This is simulated data.

```
# Save as read_modbus.py; run: python3 read_modbus.py
# Connects ONLY to the local invented read-only mock.
import socket
import struct

def recv_exact(sock, count):
    data = b""
    while len(data) < count:
        chunk = sock.recv(count - len(data))
        if not chunk:
            raise ConnectionError("peer closed")
        data += chunk
    return data

tid, uid = 1, 1
start, count = 0, 2
pdu = struct.pack(">BHH", 3, start, count)
request = struct.pack(">HHHB", tid, 0, len(pdu) + 1, uid) + pdu
with socket.create_connection(("127.0.0.1", 15020), 3) as sock:
    sock.sendall(request)
    head = recv_exact(sock, 7)
    rt, pid, length, ru = struct.unpack(">HHHB", head)
    if (rt, pid, ru) != (tid, 0, uid) or not 2 <= length <= 254:
        raise ValueError("unexpected response header")
    reply = recv_exact(sock, length - 1)

print("Response PDU:", reply.hex(" "))
if reply[0] == 131:
    raise ValueError("Modbus exception code " + str(reply[1]))
if reply[:2] != bytes([3, 4]) or len(reply) != 6:
    raise ValueError("unexpected function or byte count")
raw = struct.unpack(">H", reply[2:4])[0]
scale = struct.unpack(">h", reply[4:6])[0]
# This scale meaning belongs to our invented map, not Modbus itself.
power_w = raw * (10 ** scale)
print("Invented map value:", power_w, "W")
```

READ THE CONNECTION

socket provides the local connection. **bytes** holds raw data. **struct** packs/unpacks numbers; “>” selects big-endian representation. **while** repeats until enough bytes arrive; **with** closes resources. A short recv() is not necessarily a whole message.

Think in systems

Compare the reasoning and the limits, not only the final number. Lesson page numbers let you return directly to the explanation.

01 The physical system comes first / page 8

Only that communication failed. The fan could be on, off or changing. Obtain suitable physical evidence before claiming an outcome.

02 Units are part of the answer / page 9

800 Wh; 800 × 3,600 = 2,880,000 J = 2.88 MJ.

03 Use algebra to ask better questions / page 10

Input = output/efficiency = 40/0.80 = 50 W. Loss = 10 W.

04 Power is a rate; energy accumulates / page 11

3 × (20/60) = 1 Wh, or 3,600 J.

05 Efficiency needs a boundary / page 12

Yes. It moves heat as well as consuming work. Heat moved divided by electrical work is not the same metric as conversion efficiency.

06 Read a graph before trusting it / page 13

Their peaks, rates of change, durations and timing can differ. Equipment limits may depend on these features rather than the mean.

07 Keep claims at the level of evidence / page 14

No. It is a modelled estimate using assumed inputs. Only appropriate plant measurements could establish observed output.

08 Set a sensible home-lab boundary / page 15

Available current, stored energy, component ratings, wiring, exposed moving parts and temperature also matter.

Understand circuits

Compare the reasoning and the limits, not only the final number. Lesson page numbers let you return directly to the explanation.

09 Charge and current / page 17

Not in the steady one-path model. Current is equal around the loop; voltage changes across components.

10 Voltage is a difference / page 18

Not safely. Establish the relationship between the reference nodes before comparing the voltages.

11 Ohm’s law and a real resistor / page 19

Power becomes four times larger because $P = V^2/R$.

12 Series and parallel paths / page 20

The 200 Ω branch carries 5/200 = 25 mA. The other carries 50 mA; total is 75 mA.

13 Kirchhoff keeps the accounts balanced / page 21

A missing branch, a sign or measurement error, or an unmodelled accumulation/transient. Conservation has not stopped working.

14 Electrical power becomes something else / page 22

No. A stalled or faulty device may still consume power. Measure motion if the claim concerns motion.

15 Capacitors store electric-field energy / page 23

Because $E = \frac{1}{2}CV^2$, so energy depends on voltage squared for a fixed capacitance.

16 Inductors resist changes in current / page 24

Current cannot change arbitrarily fast without voltage. The stored energy needs a specified safe path and rated components.

Meet the electronics

Compare the reasoning and the limits, not only the final number. Lesson page numbers let you return directly to the explanation.

17 Diodes and LEDs are directional / page 26

Its voltage-current relationship is nonlinear and temperature-dependent; use an appropriate diode model and operating limits.

18 A transistor lets a signal control power / page 27

No. Check guaranteed behaviour at 3.3 V or a lower specified gate drive, including current and thermal conditions.

19 A microcontroller is a small computer / page 28

They may use different chips, pin mappings, peripherals and supporting circuitry. Identify the exact hardware.

20 Digital inputs are not exact voltmeters / page 29

No. It identifies the state of that input or output representation. Independent contact or process feedback is needed for a physical claim.

21 PWM changes time, not the supply label / page 30

Period = 1 ms. On-time = 0.25 ms.

22 ADCs turn analogue signals into codes / page 31

No. It may reduce some random variation but cannot automatically remove bias, reference error, nonlinearity or aliasing.

23 Sensors convert one quantity to another / page 32

A suitable tachometer or rotational-speed signal. Input current or ambient temperature alone is insufficient.

24 Read a datasheet as an engineer / page 33

Start-up, fault and worst-case operating currents may be higher. Use the relevant guaranteed limits and an appropriate design margin.

Make measurements

Compare the reasoning and the limits, not only the final number. Lesson page numbers let you return directly to the explanation.

25 Use a multimeter for the right task / page 35

The connection topology and often the lead socket change. Voltage is measured across; current is measured through a path.

26 Accuracy, precision and resolution differ / page 36

They show repeatability at the display resolution in those conditions. They do not alone establish accuracy or calibration.

27 Calibration connects indication to reference / page 37

No. Both can share errors. Traceability requires a documented calibration chain with associated uncertainty.

28 Build an uncertainty budget / page 38

Not on that evidence alone. Improve the measurement or define a justified decision rule that preserves ambiguous cases.

29 Sampling can conceal a signal / page 39

No. The time resolution and likely averaging cannot resolve that behaviour. Use a suitable high-rate measurement system.

30 Use statistics without hiding failure / page 40

Raw results, count, conditions, maximum and useful summaries, with limitations. Avoid an unsupported high-confidence tail claim.

31 Timestamps describe different events / page 41

It measures elapsed duration without being distorted by a wall-clock correction. It does not itself align clocks on different devices.

32 Design a test that could disprove you / page 42

To check that the system rejects unsupported success, rather than merely succeeding in the easiest case.

Understand power systems

Compare the reasoning and the limits, not only the final number. Lesson page numbers let you return directly to the explanation.

33 AC repeats; DC keeps its polarity / page 44

It halves, from 20 ms to 10 ms.

34 RMS and phase explain AC behaviour / page 45

No. Its RMS is about 7.07 V; at equal resistance its mean power is half that of 10 V DC.

35 Real, reactive and apparent power / page 46

P = 500 × 0.9 = 450 W, using compatible definitions and measurements.

36 Three phases share the work / page 47

They can be represented as 0°, -120° and +120° relative to a chosen reference.

37 Generators and motors exchange energy / page 48

The mechanical driver supplies the energy; delivering electrical output requires mechanical work against electromagnetic torque.

38 Transformers trade voltage for current / page 49

No. Current capability changes in the ideal model; energy conservation still applies, with losses in a real device.

39 Networks have thermal and voltage limits / page 50

Energy availability does not guarantee local transfer capability, voltage support, thermal capacity or acceptable performance after a fault.

40 Protection must act without your software / page 51

They must remain effective through the local protection design. Supervision failing must not disable the required protection.

Ground it in Tasmania

Compare the reasoning and the limits, not only the final number. Lesson page numbers let you return directly to the explanation.

41 Tasmania is your engineering setting / page 53

The equipment manufacturer or relevant site/equipment team, not automatically the generator or transmission-network operator.

42 Hydro power comes from head and flow / page 54

Twice the power. A real plant may not keep head and efficiency fixed and may reach another limit.

43 Water and turbines have dynamics / page 55

Commands, load changes and generator controls act at different speeds. A model that ignores those speeds can predict a misleading outcome.

44 Pumped hydro shifts energy in time / page 56

No. You also need usable energy capacity, operating conditions and relevant constraints.

45 Wind and solar need interfaces / page 57

It can shift some energy and supply or absorb power within its limits. It does not eliminate every network or resource constraint.

46 Frequency is a system balance signal / page 58

Its power rating, controls, response delay, state of charge, connection and operating permissions also matter.

47 System strength is not inertia / page 59

Appropriate site data, validated device/network models, relevant disturbances and specialist assessment; normal averages alone are insufficient.

48 A connection is a technical interface / page 60

No. It aggregates many effects and lacks the device-specific, time-aligned evidence needed for that claim.

Storage and continuity

Compare the reasoning and the limits, not only the final number. Lesson page numbers let you return directly to the explanation.

49 A battery has both W and Wh limits / page 62

No. The arithmetic ignores its discharge power/current, thermal and protection limits.

50 State of charge is an estimate / page 63

Integration accumulates it. A persistent offset produces a growing error in counted charge.

51 A BMS enforces battery boundaries / page 64

No. Reconcile the reason, current capability and device state. Repetition can conceal or worsen a fault.

52 Converters change electrical form / page 65

It may not be bidirectional or designed to sink current. The receiving energy path and protection must be specified.

53 Grid-following and grid-forming differ / page 66

No. Its response remains limited by hardware, current, energy, control settings and the connected system.

54 A microgrid needs a defined boundary / page 67

Synchronisation, voltage, phase, frequency, protection, operating permissions and the equipment’s reconnection procedure need suitable assessment.

55 A UPS is a designed continuity path / page 68

No. Topology, switching, limits, faults and which load is actually connected all matter.

56 Redundancy can share a common failure / page 69

One cause affects both, such as shared power loss, common mounting error, shared reference bias or identical software failure.

Follow the heat

Compare the reasoning and the limits, not only the final number. Lesson page numbers let you return directly to the explanation.

57 Heat and temperature are different / page 71

Yes. Temperature differences have equal numerical size in these units. Absolute temperatures use different zero points.

58 Thermal resistance links heat to rise / page 72

The steady temperature rise doubles. This can happen if an interface or cooling condition deteriorates.

59 Flow can carry thermal power / page 73

It doubles in this steady model, assuming the other properties remain unchanged.

60 Pumps pay for pressure and flow / page 74

No. You need the actual flow path and suitable flow evidence, plus relevant thermal measurements.

61 Air cooling needs a managed path / page 75

Local recirculation, uneven supply, blocked paths or a failed fan can create an inlet or component hot spot.

62 Liquid cooling still needs heat rejection / page 76

No. It changes the transport path. The heat still needs a suitable sink or a useful heat consumer.

63 PUE measures a specific ratio / page 77

No. The 20% is total included non-IT overhead, not only cooling.

64 Heat reuse needs the right temperature / page 78

Transport losses, pipe cost, temperature requirements and matching demand affect whether the available heat is practically useful.

Understand the AI factory

Compare the reasoning and the limits, not only the final number. Lesson page numbers let you return directly to the explanation.

65 Compute hardware has a physical job / page 80

It does not show whether the required useful work was completed correctly, at the required quality and latency.

66 Draw the complete rack power path / page 81

A suitable rack-input energy measurement, with matching workload, time interval and operating conditions.

67 Workloads can change power quickly / page 82

Time-aligned high-rate measurements, relevant boundaries, representative load behaviour and an assessment of competing causes and system dynamics.

68 The network can be the bottleneck / page 83

No. Communication, memory, storage, software and synchronisation may remain the limiting resources.

69 A workload is more than a power trace / page 84

The fixed task, quality, completion or latency requirement, recovery behaviour, measurement boundary and allowed operating limits.

70 Choose metrics that match the claim / page 85

No. Quality, latency, reliability and other hard constraints must still be met.

71 Flexibility has a feasible envelope / page 86

It records those events under their measurement and trust limits. It does not by itself prove the counterfactual reduction.

72 Trace failure through the dependencies / page 87

Its value can remain in range. Detection needs capture-time, sequence, plausibility or independent evidence, not only numeric range checks.

Control the system

Compare the reasoning and the limits, not only the final number. Lesson page numbers let you return directly to the explanation.

73 Feedback compares target and observation / page 89

Not without a justified relationship or airflow measurement. It directly regulates the variable in its feedback path.

74 Dynamics determine settling time / page 90

No. The simulator can test software logic. Real timing needs device evidence or a validated model.

75 Hysteresis prevents constant switching / page 91

It stays on under the stated hysteresis rule; switch-off occurs only below the lower threshold.

76 PID has three different memories / page 92

The limit must be enforced, and the controller design should handle saturation appropriately rather than silently assuming the request occurred.

77 Feasibility comes before optimisation / page 93

A hard temperature limit is not a negotiable preference. Exclude the action or preserve uncertainty if feasibility is unknown.

78 Separate control time scales / page 94

The appropriate local control/protection design, independently of the supervisory connection.

79 SCADA observes and commands / page 95

No. It indicates some communication condition. Physical outcome needs relevant fresh evidence.

80 A model is not automatically a digital twin / page 96

It tests software against specified artificial behaviours; agreement does not establish the behaviour or reliability of real equipment.

Connect the equipment

Compare the reasoning and the limits, not only the final number. Lesson page numbers let you return directly to the explanation.

81 Code should preserve physical meaning / page 98

It can turn absence into a plausible physical zero. The system can then falsely confirm a zero-output command.

82 Give each record a schema / page 99

The device identifies the equipment; the command ID identifies one intended operation and supports correlation, reconciliation and deduplication.

83 TCP delivers bytes, not outcomes / page 100

Only that the expected communication outcome was not observed in time. It does not establish physical failure.

84 Modbus exposes a data model / page 101

No. Confirm device semantics, acceptance status and fresh relevant physical evidence.

85 A register map is part of the interface / page 102

The parser may apply the wrong sign, scaling, register order, units, sentinel rule or mapping version.

86 An adapter isolates device differences / page 103

An explicit unsupported/unknown capability, with available evidence. Do not fabricate an authoritative status.

87 Quality gates precede verdicts / page 104

Treat it as a clock/quality problem. Do not accept a negative age as evidence of exceptional freshness.

88 Secure the control boundary / page 105

No. Access paths, segmentation, credentials, host security and the wider system configuration still matter.

Build Lupine with evidence

Compare the reasoning and the limits, not only the final number. Lesson page numbers let you return directly to the explanation.

89 Rail needs an honest lifecycle / page 107

Uncertain, pending reconciliation. Neither completed nor failed is established merely by the timeout.

90 Retry safety depends on semantics / page 108

No. It may be connection-local and not durably tied to a physical operation or deduplication record.

91 Reconcile without inventing certainty / page 109

The state may predate the command or have another cause, and the command’s pending status may remain unknown.

92 Durable state survives a restart / page 110

The remote device is outside that local transaction. Communication and physical execution create additional failure windows.

93 Cloked records evidence, not physical truth / page 111

No. It can support an origin/integrity claim under its key-trust assumptions, not measurement validity.

94 Test the failure sequence, not only the row / page 112

Those specified cases passed on that implementation and environment. It does not establish untested hardware behaviour or universal safety.

95 Advance by evidence, not by confidence / page 113

A specific unresolved question that the additional model or measurement can answer, with a planned decision based on the result.

96 Build a company around a real gap / page 114

A clear technical problem, reproducible evidence, an honest comparison and an equipment user who confirms why the difference matters.

Relationships, units and limits

A formula is useful only when its model and boundary apply. Keep the unit beside every intermediate result.

Current

$$I = \Delta Q/\Delta t$$

A = C/s; average charge-flow rate.

Ohmic resistor

$$V = IR$$

V, A, Ω. Use an appropriate component model.

Steady DC power

$$P = VI$$

W = V × A.

Resistor heating

$$P = I^2R = V^2/R$$

W. The resistance and operating conditions must apply.

Constant-power energy

$$E = Pt$$

W × h = Wh; W × s = J.

Variable-power energy

$$E \approx \Sigma P_i \Delta t_i$$

State sample/interval assumptions and units.

Efficiency

$$\eta = \text{useful output / input}$$

Use matching boundaries and periods.

Series resistance

$$R = R_1 + R_2 + \ldots$$

Same current in a steady series path.

Parallel resistance

$$1/R = 1/R_1 + 1/R_2 + \ldots$$

Same voltage across the parallel branches.

Capacitor

$$Q = CV;\ E = \tfrac{1}{2}CV^2$$

C in farads; energy in joules.

Inductor

$$v = L\,di/dt;\ E = \tfrac{1}{2}LI^2$$

L in henries; changing current induces voltage.

Period

$$T = 1/f$$

Seconds when frequency is in Hz.

Relationships, units and limits

A formula is useful only when its model and boundary apply. Keep the unit beside every intermediate result.

Sine RMS

$$V_{rms} = V_{pk}/\sqrt{2}$$

For an ideal sine, not every waveform.

Single-phase AC

$$P = V_{rms}I_{rms}PF$$

Real power in W; apparent power S = VI in VA.

Balanced three-phase

$$P = \sqrt{3}\ V_LL\ I_L\ PF$$

Line-to-line voltage and line current; stated assumptions.

Ideal transformer

$$V_2/V_1 = N_2/N_1$$

Changing flux; real losses and limits remain.

Mechanical rotation

$$P = \tau\omega$$

N·m × rad/s = W.

Hydro estimate

$$P = \rho gQH\eta$$

kg/m³, m/s², m³/s and m give watts.

Sensible heating

$$E = mc\Delta T$$

kg × J/(kg·K) × K = J.

Thermal resistance

$$R\theta = \Delta T/P$$

K/W; specify path and steady conditions.

Fluid heat transport

$$P = \dot{m}c_p\Delta T$$

kg/s × J/(kg·K) × K = W.

Hydraulic power

$$P = \Delta pQ$$

Pa × m³/s = W; input must include losses.

Idealised storage duration

$$t = \text{usable output energy} / \text{power}$$

Use consistent output boundary and power limits.

PUE

$$\text{Facility energy} / \text{IT energy}$$

Same period and specified measurement boundary.

Words with physical meaning

Use the definition as a starting point, then return to the relevant lesson and example.

AC Alternating current; periodically reverses direction.

Actuator Device that changes a physical input to a process.

ADC Analogue-to-digital converter; produces discrete codes.

Aiva Lupine's proposed feasible-action selection layer.

Alias A false apparent signal caused by unsuitable sampling.

Ampere Current unit: one coulomb per second.

Apparent power RMS voltage × RMS current, in VA.

Bandwidth Range of signal frequencies or a stated data-transfer capability; specify context.

Baseline Defined comparison behaviour without the tested intervention.

BMS Battery-management system; monitors and manages pack operation.

Boundary The named limit around a system or measurement claim.

Calibration Characterisation against references under stated conditions.

Capture time When the reported physical sample was taken.

Cloked Lupine's proposed evidence-recording layer.

Closed loop Control using observed output as feedback.

COP Heat moved/delivered divided by work input for a defined mode.

Current Rate of charge flow, in amperes.

DC Current or voltage with one polarity/direction; magnitude may vary.

Dead time Delay before an observable response begins.

Deduplication Recognising repeat requests as the same defined operation.

Dew point Temperature at which air becomes saturated for its water-vapour content.

Digital twin A stated model linked to a specific physical system; define its scope.

Driver Interface that provides suitable electrical control or load drive.

Efficiency Useful output divided by input over matching boundaries.

Energy Transferred or stored amount, measured in J or Wh.

Fault An abnormal condition requiring defined handling or protection.

Firmware Program running on embedded hardware.

Words with physical meaning

Use the definition as a starting point, then return to the relevant lesson and example.

Flow Volume or mass transferred per time; state which.

Frequency Cycles per second, in hertz.

GPIO General-purpose digital input/output pin.

GPU Processor designed for large-scale parallel computation.

Grid-forming Inverter control providing a controlled voltage reference, within limits.

Grid-following Inverter control synchronising to an existing voltage waveform.

Hash Digest of data; not proof of physical validity.

Head Usable height difference in a hydraulic-energy calculation.

HMI Human-machine interface for observing and controlling equipment.

Hysteresis Different transition thresholds depending on prior state.

Idempotent Repetition has no additional effect under specified semantics.

Inertia Resistance to changes in motion; relevant stored energy affects frequency response.

Input register A 16-bit data object in the Modbus model.

Inverter Power-electronic converter producing AC from DC.

Joule Energy unit; one watt-second.

Kilowatt-hour 1,000 Wh = 3.6 million J.

Latency Delay for a specified communication or operation.

Modbus Protocol for accessing documented bit/register data objects.

Monotonic clock Clock used for local elapsed time without wall-clock adjustments.

MOSFET Transistor often used to control electrical current.

NEM National Electricity Market and its associated interconnected system context.

Ohm Resistance unit: volt per ampere.

Overshoot Response exceeding the desired final value during a transient.

PID Proportional-integral-derivative control family.

PLC Programmable logic controller used in industrial control.

Power Rate of energy transfer, in watts.

Power factor Real power/apparent power using compatible definitions.

Words with physical meaning

Use the definition as a starting point, then return to the relevant lesson and example.

PUE Facility energy divided by IT equipment energy.

PWM Pulse-width modulation; control through switched on-time fraction.

Rail Lupine's proposed execution and reconciliation layer.

Reactive power Sinusoidal AC reactive-exchange quantity, in var.

Receive time When a report arrived at the observing system.

Reconciliation Resolving pending execution using additional evidence.

Register map Documented addresses, types, scaling and semantics of device data.

Resolution Smallest distinguishable indicated change.

RMS Root mean square; effective magnitude for specified use.

SCADA Supervisory control and data acquisition.

Setpoint Requested operating target.

Settling time Time to meet a defined band and hold criterion.

SOC Estimated state of charge relative to a defined battery capacity.

Stale Too old for the decision's stated evidence requirement.

System strength Ability to maintain/control the local voltage waveform under relevant conditions.

Telemetry Reported measurements and status with context.

Thermal resistance Temperature difference per heat-flow rate for a stated path.

Throughput Valid work completed per unit time under stated conditions.

Timeout Expected communication did not complete within a defined duration.

Traceability Documented reference/calibration chain with associated uncertainty.

Transaction Grouped database changes with defined commit guarantees.

Uncertainty Dispersion of values reasonably attributed to a measured quantity.

UPS Uninterruptible power supply with topology-specific continuity behaviour.

Verification Check of a specified requirement using stated evidence.

Volt Potential-difference unit: joule per coulomb.

Watt Power unit: joule per second.

Watt-hour Energy: one watt sustained for one hour; 3,600 J.

A result you can revisit

Use one sheet for one question. Keep the raw evidence alongside this summary.

QUESTION AND CRITERION

What exactly are you testing? What would count as a failure?

SYSTEM AND BOUNDARY

Device, firmware, source, load, sensor, location and permitted limits.

METHOD AND PRECONDITIONS

Baseline, planned change, time window, repeats and stop condition.

RAW EVIDENCE AND QUALITY

File names, command IDs, sample timing, units, missing/stale values.

RESULT AND LIMITS

Measured result, decision rule, uncertainty and competing explanations.

NEXT QUESTION OR CORRECTION

What did you learn? What changes next? Date any correction.

One problem, one next test

Use this sheet before adding hardware, complexity or a commercial claim.

Specific equipment user

Who experiences the problem, and in which equipment?

Concrete failure

What happened, under which conditions, and how often is known?

Current method and gap

What already handles this? What specific evidence or capability is missing?

Proposed comparison

What will your method do differently, and what baseline will you compare?

Acceptance and stop criteria

Which measured result justifies proceeding? What would make you change direction?

MASTERY CHECK BEFORE YOU ADVANCE

I can draw the system and name the quantities.
I can solve the example with units and assumptions.
I can explain missing data, limits and a contrary result.
I can identify what my result does not establish.
I can propose the next measurement without relying on a slogan.

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Subject index

Alphabetical lesson titles with page numbers. The PDF bookmarks also provide section and lesson navigation.

AC repeats; DC keeps its polarity 44
Accuracy, precision and resolution differ 36
adapter isolates device differences 103
ADCs turn analogue signals into codes 31
Advance by evidence, not by confidence 113
Air cooling needs a managed path 75
battery has both W and Wh limits 62
BMS enforces battery boundaries 64
Build a company around a real gap 114
Build an uncertainty budget 38
Calibration connects indication to reference 37
Capacitors store electric-field energy 23
Charge and current 17
Choose metrics that match the claim 85
Cloked records evidence, not physical truth 111
Code should preserve physical meaning 98
Compute hardware has a physical job 80
connection is a technical interface 60
Converters change electrical form 65
Design a test that could disprove you 42
Digital inputs are not exact voltmeters 29
Diodes and LEDs are directional 26
Draw the complete rack power path 81
Durable state survives a restart 110
Dynamics determine settling time 90
Efficiency needs a boundary 12
Electrical power becomes something else 22
Feasibility comes before optimisation 93
Feedback compares target and observation 89
Flexibility has a feasible envelope 86
Flow can carry thermal power 73
Frequency is a system balance signal 58
Generators and motors exchange energy 48
Give each record a schema 99
Grid-following and grid-forming differ 66
Heat and temperature are different 71
Heat reuse needs the right temperature 78
Hydro power comes from head and flow 54
Hysteresis prevents constant switching 91
Inductors resist changes in current 24
Keep claims at the level of evidence 14
Kirchhoff keeps the accounts balanced 21
Liquid cooling still needs heat rejection 76
microcontroller is a small computer 28
microgrid needs a defined boundary 67
Modbus exposes a data model 101
model is not automatically a digital twin 96
network can be the bottleneck 83

Subject index

Alphabetical lesson titles with page numbers. The PDF bookmarks also provide section and lesson navigation.

Networks have thermal and voltage limits 50
Ohm’s law and a real resistor 19
physical system comes first 8
PID has three different memories 92
Power is a rate; energy accumulates 11
Protection must act without your software 51
PUE measures a specific ratio 77
Pumped hydro shifts energy in time 56
Pumps pay for pressure and flow 74
PWM changes time, not the supply label 30
Quality gates precede verdicts 104
Rail needs an honest lifecycle 107
Read a datasheet as an engineer 33
Read a graph before trusting it 13
Real, reactive and apparent power 46
Reconcile without inventing certainty 109
Redundancy can share a common failure 69
register map is part of the interface 102
Retry safety depends on semantics 108
RMS and phase explain AC behaviour 45
Sampling can conceal a signal 39
SCADA observes and commands 95
Secure the control boundary 105
Sensors convert one quantity to another 32
Separate control time scales 94
Series and parallel paths 20
Set a sensible home-lab boundary 15
State of charge is an estimate 63
System strength is not inertia 59
Tasmania is your engineering setting 53
TCP delivers bytes, not outcomes 100
Test the failure sequence, not only the row 112
Thermal resistance links heat to rise 72
Three phases share the work 47
Timestamps describe different events 41
Trace failure through the dependencies 87
Transformers trade voltage for current 49
transistor lets a signal control power 27
Units are part of the answer 9
UPS is a designed continuity path 68
Use a multimeter for the right task 35
Use algebra to ask better questions 10
Use statistics without hiding failure 40
Voltage is a difference 18
Water and turbines have dynamics 55
Wind and solar need interfaces 57
workload is more than a power trace 84
Workloads can change power quickly 82